Supplementary Material for: Identifiability by Stationarity: Blind Gain–Object Separation in Self-Calibrating Ghost Imaging

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***Abstract*—This document supplements the main manuscript. Sections S1–S5 report the supplementary verification experiments promised in Sec. 9 of the main text: the permutation-whitening / factorized-randomization ablation (S1), the coherent-residual validation of the master identity (S2), the C_0 audit and the oracle-to-blind baselines (S3), and the extended flip-boundary detail with the low-photon drift-floor probe and replication tables (S4), and the shared simulation protocol (S5). Appendices A–F contain the complete proofs of every theorem, corollary, and proposition quoted in the main text. Citation numbers, section numbers (Sec. 1–10), figure numbers (Fig. 1–6), theorem labels, and equation tags (e.g. (F.10)) referring to the main manuscript retain their main-text meaning; figures and tables introduced here *continue* the main numbering (Figs. 7–12, Tables 2–3), and only the supplementary *sections* keep an S-prefix (S1–S5). All numerical values are descriptive results of the specified simulation grids unless a resampled interval is stated.**

S1. Permutation Whitening and the Factorized Randomization Ablation

This section carries the multi-permutation whitening-power experiment and the factorized randomization ablation summarized in Sec. 9.6 of the main text (data: `results/perm_ablation_r1/`; 32 draws per randomized arm × 10 objects under the Fig. 2 protocol — $K = 4096$, 64×64 objects, 8192 paired frames, noiseless carrier, Brown–Forsythe/Levene screen [29], [30] at 8 chunks, reject at $p < 10^{-3}$). Two randomization semantics must be kept separate (Appendix E.6): the *pixel-level* randomization of the SRHT theory — pixel permutation $P_{\rm col}$ and pixel signs D, whose Lemma E.1 covariance is invariant to acquisition order — and the *acquisition-order* reordering $P_{\rm row}$, which controls temporal stationarity of the acquired series. Figure 7(a) reports the factorized attribution: either permutation alone substantially reduces the temporal-stationarity screen's rejection rate — from 70% (ordered) to 5.6% ($P_{\rm col}$) and 12.2% ($P_{\rm row}$), all still far above the $\sim 0.1\%$ nominal rate of the $p < 10^{-3}$ test — the theory's own randomization $P_{\rm col}{+}D$ reduces it to 11.9%, the code's SRHT arm ($P_{\rm row}{+}D$) to 5.9%, and the historical combined arm ($P_{\rm col}{+}P_{\rm row}$) reproduces its $19/320 = 5.9\%$ headline — whereas pixel signs D alone barely move it (43.4%), because sign flips whiten carrier amplitudes but leave the coarse-to-fine sequency-energy ordering of natural-Hadamard rows intact. Figure 7(b) shows the honest per-draw certificate null quoted in the main text: on the historical arm, the maximum non-DC Walsh coefficient of the permuted T^2 does not predict which draws reject (Spearman 0.092, $p = 0.10$), consistent with the probabilistic — not per-draw-certifiable — nature of permutation whitening (Sec. 7).

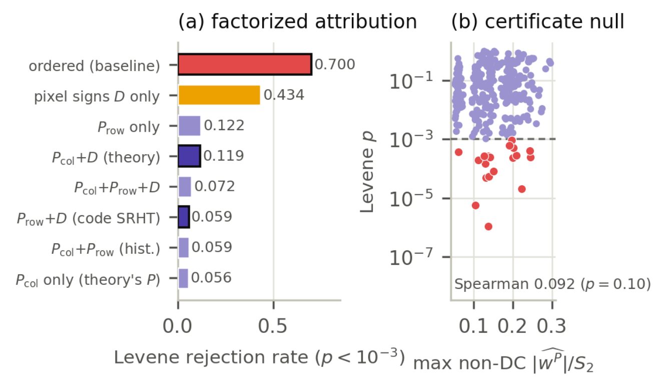

Fig. 7. Permutation whitening and factorized randomization ablation (32 draws per randomized arm × 10 objects; Levene screen at $p < 10^{-3}$). (a) Rejection rate per arm: either permutation alone substantially reduces the rejection rate (70% → 5.6–12.2%, all $\gg$ the $\sim 0.1\%$ nominal rate); pixel signs alone barely move it; outlined bars mark the theory's randomization ($P_{\rm col}{+}D$) and the code's SRHT arm ($P_{\rm row}{+}D$). (b) Per-draw certificate null on the historical $P_{\rm col}{+}P_{\rm row}$ arm (320 draws): max non-DC $|\widehat{w^P}|/S_2$ of the permuted squared object versus the Levene p-value (violet: accepted draws; red: rejected; dashed line: the $p = 10^{-3}$ gate). Walsh flatness of the realized draw does not predict rejection (Spearman 0.092, $p = 0.10$).

The acquisition-time ablation of the main text's Sec. 9.4 (moved here from the main document for space) is reproduced as Fig. 8: at the single above-floor drift point $\rho = 10^{-3}$, acquisition-order permutation alone gains +5.39 dB over ordered Hadamard, pixel sign flips alone +5.47 dB, and the code's full SRHT arm +5.45 dB — each ingredient by itself recovers essentially the full advantage, and combining them is not additive. All Fig. 8 arms randomize acquisition time only; the pixel-permutation ($P_{\rm col}$) attribution is the subject of Fig. 7 above.

S2. Validation of the Master Identity: Reconstruction Bridge and Coherent Residuals (E4)

This section carries the two validations of Theorem 1. Figure 9 (moved from the main document for space) shows the reconstruction-bridge test of main-text Sec. 9.3: predicted

Draft rendering; author list and affiliations to be finalized before submission. Extended supplementary material; length subject to editorial approval.

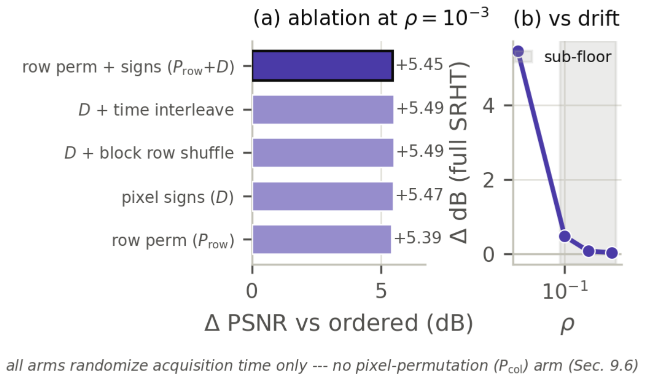

Fig. 8. Ablation isolating the acquisition-time randomization mechanism (moved from the main document). (a) PSNR gain over ordered Hadamard at the single above-floor drift point $\rho = 10^{-3}$ for five arms: acquisition-order (row) permutation P_{row} alone gains $+5.39$ dB, pixel sign flips D alone $+5.47$ dB, and the code's full SRHT arm ($P_{\text{row}}+D$) $+5.45$ dB, with the block-shuffle and time-interleave sign variants in the same $+5.5$ dB band — each ingredient by itself recovers essentially the full advantage, and combining them is not additive. All five arms randomize acquisition time only: none applies the pixel/column permutation P_{col} of the Appendix E SRHT analysis, whose factorized attribution is Fig. 7. (b) The full-SRHT advantage versus drift rate ρ: the gain collapses toward 0 dB as every blind variant approaches the noise floor (shaded: sub-floor cells). What matters is that the chronology is randomized, not the particular form of the randomization. Descriptive values at single grid points (no resampled intervals).

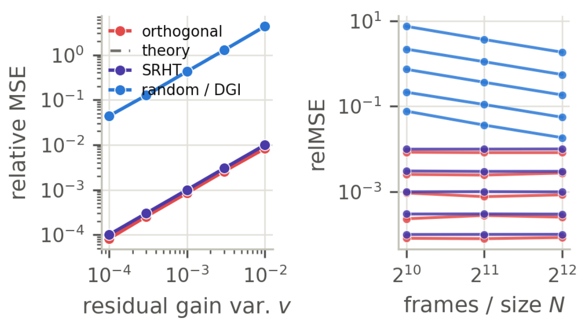

Fig. 9. Reconstruction bridge (Theorem 1 tested as an identity, on a balanced 8100-row design: three reconstructors, $N \in \{1024, 2048, 4096\}$, six injected residual-gain variances v, ten objects, fifteen seeds; moved from the main document). Predicted versus measured scale-aligned relMSE: the deterministic inverses are flat in N, as $B_L = 1$ demands, while the random/DGI arm decays as $1/N$ at every v with a quantitative leverage law. Slopes, leverage values, and the DC-dominated alignment effect: main-text Sec. 9.3.

versus measured scale-aligned relMSE on the balanced 8100-row design; the quantitative slopes and leverage values are in main-text Sec. 9.3. The remainder of the section carries experiment E4: the direct, raw-metric validation of Theorem 1 under *injected* (exogenous) coherent residuals, the regime in which hypotheses (H1)–(H3) hold by construction.

Five residual-covariance structures — i.i.d., AR(1), sinusoidal, low-pass OU, blockwise — are injected at fixed $v = 10^{-3}$ across three reconstructors and ten objects (150 cells, $K = N = 1024$; data: `results/coherent_residual_e4_r1/`), and the measurement is made in the raw-residual (identity-verification) metric rather than the scale-aligned image metric of the bridge

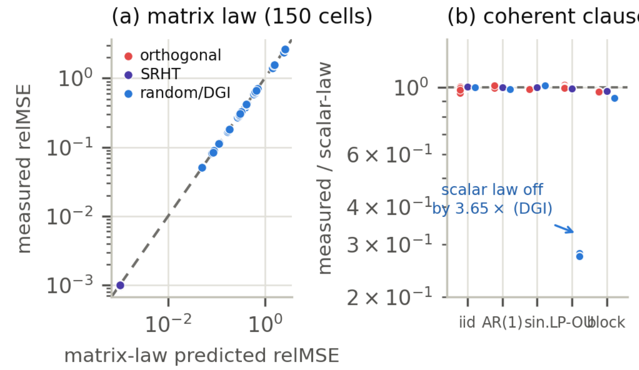

Fig. 10. Coherent-residual validation of Theorem 1 (E4; five injected residual structures $\times$ three reconstructors $\times$ ten objects at $v = 10^{-3}$, raw identity-verification metric). (a) Matrix-law prediction versus measured relMSE: agreement within 4.2% across all 150 cells. (b) Ratio of measured relMSE to the scalar $v\,B_L$ law by residual structure: exact for orthogonal and SRHT inversions, but up to $3.65\times$ off for random/DGI under low-pass OU residuals — a direct verification of Theorem 1's coherent (full-matrix) clause.

experiment (Fig. 9). Figure 10(a) shows the full matrix law of Theorem 1 tracking the measured relMSE within 4.2% across all 150 cells. Figure 10(b) isolates the coherent clause: the scalar $v\,B_L$ shortcut is exact for the complete orthogonal and SRHT inversions (for which $LL^\top \propto I/N$ analytically, so the gain term feels only $\operatorname{diag}(V_\delta)$) but fails by up to $3.65\times$ for random/DGI under low-pass-coherent residuals — exactly the failure mode Theorem 1's full matrix form exists to capture. E4 validates the matrix law under injected residuals; it does not probe the same-record δ–ξ dependence flagged after Theorem 1 in the main text.

S3. C_0 AUDIT AND ORACLE-TO-BLIND BASELINES

C_0 audit (E9). The one-sample DGI correlator constant $C_0 = \mathbb{E}\|Z\|^2/S_2 - 1$ is measured from 2×10^4 single-frame pattern draws per ensemble, for four pattern ensembles $\times$ five objects (data: `results/c0_audit_e9_r1/`). Figure 11(a) compares the measurement with the moment formula of Appendix F: mean relative error 0.35%, worst 2.01%, across values of C_0 spanning 10^3 to 5.7×10^6. The strong ensemble dependence — $O(K)$ for signed Rademacher patterns versus $O(K_{\text{eff}}K)$ for the nonnegative offset ensembles — confirms the main text's bookkeeping rule that C_0 is pipeline-specific, as Theorem 1's DGI specialization insists: there is no universal correlator constant to tabulate.

Oracle, metered, and blind baselines (E11). On four held-out objects with OU drift ($\rho = 10^{-3}$, $s = 0.1$; five gain treatments $\times$ five seeds; data: `results/oracle_baselines_e11_r1/`), Fig. 11(b) shows the accountability ordering static $\geq$ oracle $\geq$ blind-AGC $\geq$ no-correction holding in CNR for 4 of 4 objects, with an oracle$-$blind CNR gap of only 0.035–0.147: at this drift point the blind windowed estimator of Theorem B concedes little to a gain oracle, while forgoing correction altogether costs the most. This supports the main text's reading that

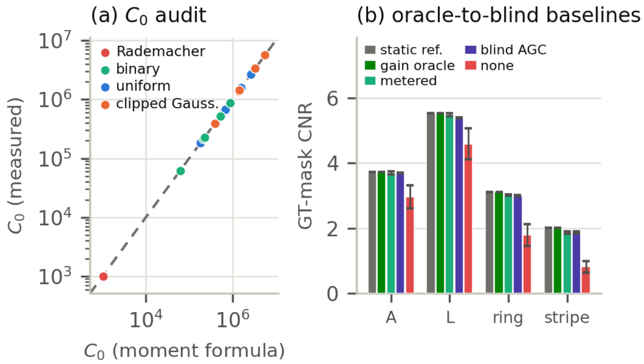

Fig. 11. C_0 audit and oracle-to-blind baselines. (a) Measured one-sample correlator constant C_0 versus the Appendix-F moment formula for four pattern ensembles × five objects (2×10^4 draws each): mean relative error 0.35% (worst 2.01%); the $O(K)$ Rademacher versus $O(K_{\mathrm{eff}}K)$ offset-ensemble spread confirms that C_0 is pipeline-specific. (b) GT-mask CNR under five gain treatments on four held-out objects (OU drift, $\rho = 10^{-3}$, $s = 0.1$; error bars: s.d. over five seeds): the ordering static $\geq$ oracle $\geq$ blind-AGC $\geq$ none holds for 4 of 4 objects, with oracle−blind gaps of 0.035–0.147 CNR.

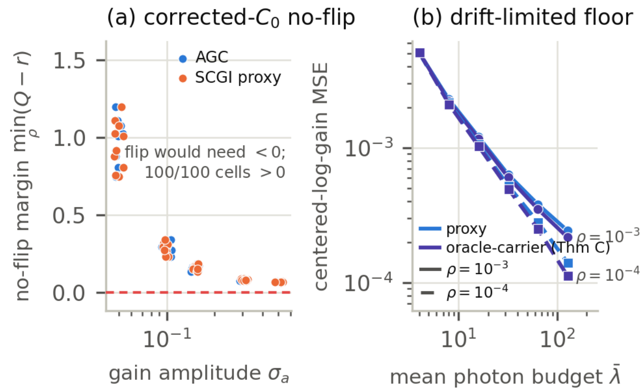

Fig. 12. Extended flip-boundary detail and the low-photon floor probe. (a) Corrected-C_0 no-flip margins for the blind random+DGI arm: per-cell minimum over the drift grid of $Q - r$ (the flip criterion of Prop 3 with drift-leverage constant $1 + C_0^{\mathrm{raw}}$), for the AGC and SCGI-proxy corrections; a flip would require a negative value, and all 100/100 cells are positive, matching the no-flip outcome of Sec. 9.4. (b) High-photon floor probe: log-domain gain MSE versus mean photon budget $\bar{\lambda}$ for the uncalibrated proxy and the oracle-known-carrier calibrated soft-log (Theorem C) at $\rho = 10^{-3}$ (solid) and $\rho = 10^{-4}$ (dashed); the two drift rates share the photon-limited regime and separate only at the floor (2.43×10^{-4} versus 1.41×10^{-4} for the proxy, 2.17×10^{-4} versus 1.13×10^{-4} for the oracle-carrier arm, $\sim 11\%$ lower), which is therefore drift-limited, not photon-limited.

the practical price of blindness is small once the stationarity anchor is available.

S4. Extended Quantitative Detail: Flip Boundary, Low-Photon Floor, and Replication Tables

This section collects the extended flip-boundary detail, the low-photon floor probe, and the per-arm replication values behind main-text Fig. 3 and Table 1.

Corrected-C_0 no-flip consistency. Section 9.4 of the main text reports that, once the drift-leverage constant of the mean-subtracted blind random+DGI pipeline is corrected to the raw-correlator value $1 + C_0^{\mathrm{raw}}$ of (F.10), Prop 3 predicts *no flip* anywhere on the simulated grid — and the data confirm it in 100/100 cells. Figure 12(a) shows the per-cell margin behind that count (data: `results/prop3_nofreeparam_r1/`): for both blind corrections (AGC and SCGI proxy), the minimum over the drift grid of the flip criterion $Q - r$ stays strictly positive in every one of the 2×50 (correction × cell) checks, with the margin narrowing as σ_a grows — consistent with the boundary being pushed beyond the simulated grid rather than absent in principle.

Low-photon drift-limited floor. Figure 12(b) carries the floor probe behind the closing claim of Sec. 9's low-photon experiment (data: `results/paper_fig7_lowphoton_r5_final/`): at the high-photon end the log-domain gain-MSE floor is drift-limited rather than photon-limited — the floor of 2.43×10^{-4} at $\rho = 10^{-3}$ falls to 1.41×10^{-4} when the drift slows to $\rho = 10^{-4}$ for the proxy arms, and from 2.17×10^{-4} to 1.13×10^{-4} for the oracle-known-carrier estimator of Theorem C, whose floor sits $\sim 11\%$ *below* the proxies because it deconvolves the known carrier exactly; the arms coincide through the photon-limited part of the probed $\bar{\lambda} \geq 4$ range and separate only at the floor.

Table II collects the secondary measured constants and grid conventions behind main-text Fig. 4 and Table 1 (moved here from the main document's Table 1 for space; the headline constants C_0, D_H, v_{blind}, and the Prop 3 no-free-parameter agreement remain in the main text).

Per-arm replication values (main-text Fig. 3 and Sec. 9.2). Table III tabulates the best-window blind gain errors and the slow-drift variance-segment slopes per arm from the 20-seed uncertainty replication (fixed $W \leq 16$ variance-segment convention; 31,600 rows; medians with two-way seeds-and-objects crossed cluster bootstrap 95% CIs), together with the raw-arm single-pass diagnostics and the exploratory 5-seed adaptive-segment slopes.

S5. Shared Simulation Protocol: Per-Figure Enumerables

Bucket-signal conventions (full statement; main-text Sec. 5): (i) *signed coefficient model* — B_n is a signed transform coefficient (e.g. a bare Walsh–Hadamard coefficient); it can vanish or change sign, so $\log B_n$ is undefined and no log-domain gain estimator is available without further structure; (ii) *physical nonnegative offset model* — the optical bucket is a nonnegative intensity, with signed patterns realized with a bias/offset so that $B_n \geq B_{\min} > 0$; Theorem B and the main-text Sec. 6 estimators are stated in this model; (iii) $\pm$ *pairwise model* — each signed pattern is acquired as a complementary pair of nonnegative patterns whose buckets are differenced; the difference is signed again, but the drift enters through adjacent-pair increments controlled by ρ_{pair} (main-text Sec. 8). The raw signed arms of main-text Sec. 9 follow convention (i) and are diagnostic only.

Notation (consolidated from main-text Sec. 2): K pixels, N frames, drift class $S \ni \mathbf{1}$ with $p = \dim S$; gauge $(a, T) \mapsto$

TABLE II

SECONDARY MEASURED CONSTANTS AND CONVENTIONS BEHIND MAIN-TEXT FIG. 4 (COMPANION TO MAIN-TEXT TABLE 1). EVERY CONSTANT IS PIPELINE-SPECIFIC; CONFIDENCE CONVENTIONS AS IN THE MAIN TEXT (TWO-WAY SEEDS-AND-OBJECTS CROSSED CLUSTER BOOTSTRAP WHERE STATED).

Quantity	Value in this pipeline	Uncertainty
K_{eff} (object-panel range)	59.8–864.3	—
Drift / noise / flux conventions	OU log-gain, $\rho \in [10^{-3}, 10]$, drift s.d. 0.030–0.104; noise $\sigma \in [0.05, 0.5]$; equal frames and equal total flux per arm	—
Above-floor gate	relMSE < 0.5; 29 of 45 cells above floor (16 at noise floor)	cell count, not resampled; no bootstrap CI computed
Empirical SRHT-paired vs. ordered-Hadamard crossover, two-parameter power-law fit	$R^2 = 0.989$–0.9995, fitted exponents -0.94 to -1.93	descriptive fits of a comparison Prop 3 does not rank; survival-style CIs left to future work

TABLE III

PER-ARM BEST-WINDOW BLIND GAIN ERRORS AND $\log(\mathrm{err})$-VS-$\log W$ SLOPES BEHIND MAIN-TEXT FIG. 3 ($\rho_{\mathrm{pair}} = 10^{-3}$, $s = 0.1$, 2048-FRAME BUDGET; MEDIANS OVER 20 SEEDS $\times$ 10 OBJECTS WITH TWO-WAY CLUSTERED 95% CIS; EXPLORATORY SLOPES FROM THE EARLIER 5-SEED ADAPTIVE-SEGMENT CONVENTION). THE RAW SIGNED-WALSH ARMS ARE DIAGNOSTIC-ONLY (BUCKET CONVENTION (I)); THEIR SINGLE-PASS 1024-FRAME MEDIANS ARE GIVEN FOR THE BUDGET CONTROL.

Arm	Best-window error (ratio AGC)	Best-window error (log estimator)	Slope, ratio AGC [95% CI]	Explor. slope
random uniform	— (bundle 0.007–0.017)	agrees to $\sim 2\%$	-0.39 $[-0.42, -0.31]$	—
random binary	0.0150 $[0.0130, 0.0165]$	agrees to $\sim 2\%$	-0.41 $[-0.44, -0.37]$	-0.39
SRHT-paired	0.0077 $[0.0066, 0.0083]$	agrees to $\sim 2\%$	-0.58 $[-0.66, -0.41]$	-0.60
randomly permuted paired Hadamard	0.0073 $[0.0063, 0.0079]$	0.0119 $[0.0090, 0.0133]$	-0.62 $[-0.67, -0.44]$	-0.61
naturally ordered paired Hadamard	—	shallow decay -0.222 $[-0.306, -0.054]$ (slope)	$+0.03$ $[-0.06, +0.17]$ (no decay)	-0.24
raw ordered (2048; single-pass 0.896)	0.899 $[0.892, 0.903]$, 0.950 at W=512	0.708 $[0.563, 0.840]$, ≥ 0.71, non-decaying	non-decaying	—
raw shuffled (2048; single-pass 0.788)	0.794 $[0.775, 0.816]$	0.136 $[0.083, 0.196]$	—	—

$(c\,a,\, T/c)$; bandwidth ρ_{bw} ($p \approx \rho_{\mathrm{bw}} N$) versus adjacent-pair decorrelation ρ_{pair}; object functionals $K_{\mathrm{eff}} = S_1^2/S_2$, $K_\infty = S_2/\|T\|_\infty^2$, $K_4 = S_2^2/\sum_j T_j^4$; pipeline constants $C_0 = \mathbb{E}\|Z\|^2/S_2 - 1$, D_H, and the reconstructor leverage $B_L = (1/S_2)\sum_n b_n^2 \|Le_n\|^2$.

All supplementary and main-text experiments share the protocol of Sec. 9 of the main text (matched acquisition budgets, shared OU drift traces within each seed, a single sliding-window-mean AGC estimator for all arms with no per-arm tuning, the ten-object panel spanning $K_{\mathrm{eff}} = 59.8$–864.3, base seed 20240708). Per-figure enumerables (main-text figure numbers):

- **Fig. 2 (carrier stationarity):** 8192 frames per arm at 64×64, split into 8 chunks with a 128-frame transient cutoff, AGC window $W = 64$, five arms.
- **Fig. 3 (blind gain identifiability):** 32×32 objects ($K = 1024$), 2048 frames per arm, nine arms — the raw signed-Walsh arms at both the frame-matched 2048 budget (sequence acquired twice) and the single-pass 1024 diagnostic — $\rho_{\mathrm{pair}} \in \{10^{-3}, 10^{-2}\}$, drift scale $s = 0.1$, W swept over a log-spaced grid from 4 to 1024; every row is scored under both the ratio AGC and the log-domain estimator of Theorem B; the final uncertainty replication (Table III) uses 20 drift seeds per (object, ρ) pair — 31,600 rows — while the exploratory R^2 analysis used 5 seeds.
- **Reconstruction bridge (Fig. 9):** injected $v \in \{0, 10^{-4}, 3\times 10^{-4}, 10^{-3}, 3\times 10^{-3}, 10^{-2}\}$, $N \in \{1024, 2048, 4096\}$, 15 seeds, three reconstruction arms.
- **Fig. 5 (low-photon robustness):** mean photon budgets $\bar{\lambda} \in \{0.25, \ldots, 128\}$ on a $\times 2$ grid, soft-log $\alpha = 0.5$, $W = 64$, $\rho_{\mathrm{pair}} = 10^{-3}$, $s = 0.1$, 5 seeds, with four estimator arms — naive clipped log, Anscombe, the uncalibrated soft-log proxy, and the oracle-known-carrier calibrated soft-log of Theorem C (per-window carrier-calibration root, carriers supplied from the simulation truth as a flat-field-calibrated benchmark; m_α tabulated exactly by truncated Poisson summation, tail bound $\leq 3.5 \times 10^{-34}$; the mean-Poisson proxy $m_\alpha^{-1}(\mathrm{mean}_W\, \psi_\alpha)$ retained for continuity) — compared on the same paired Poisson draw at each budget; all windowed arms use truncated, renormalized windows at the record boundaries (no replicate padding).
- **Fig. 6 (threshold scan):** rank/solver scan at $K = 64$ (20 seeds, 1300 cells) and $K = 128$ (30 seeds, 1950 cells), $p \in \{3, 5, 9, 17, 33\}$, margins $N - K - p \in [-8, +16]$ (lower end truncated at $N > K$ for small p).
- **Pattern conventions:** i.i.d. uniform $[0, 1]$; i.i.d. binary Bernoulli(0.5); paired Hadamard as complementary $(1 \pm H)/2$ frames (2 frames per coefficient); SRHT as $H \cdot \mathrm{diag}(\pm 1)$ with an optional row permutation, paired the same way; raw signed Walsh arms are diagnostic-only (main-

text Sec. 5, bucket convention (i)) and not a valid physical estimator without offset or pairing.

- **Metrics:** blind gain error is the scale-aligned relative error (both gain traces normalized to unit mean, optimal scalar alignment); mask-based contrast uses GT-mask CNR at threshold 0.5; reconstruction error is the scale-aligned relMSE of main-text Sec. 8. The OU drift trace is generated with `normalize_mean=1`.

Appendix A supports Sec. 4 (Theorem A and Corollaries 1–2); Appendix B supports Sec. 4 (Theorem A′, parametric-transversality proof); Appendix C supports Sec. 5 (Propositions 1–2, condition (★), Theorem B); Appendix D supports Sec. 6 (Theorems C–D, Fisher geometry, low-photon); Appendix E supports Sec. 7 (SRHT Walsh condition, permutation bounds); Appendix F supports Sec. 8 (Theorem 1, specializations, Prop 3). Heuristic or genericity steps are labeled as such inside each proof; every appendix ends its theorems with a "What this does not claim" paragraph.

APPENDIX A
SQUARE-DESIGN NON-IDENTIFIABILITY

This appendix proves Theorem A of Sec. 4 together with its two corollaries, in the notation of the main text: noiseless bucket record $R_n = a_n B_n$, carrier $B_n = (MT)_n$, log-gain $\ell = \log a$ (entrywise), drift class S a linear subspace of $\mathbb{R}^N$ with $\operatorname{span}\{\mathbf{1}\} \subseteq S$ and $\dim S = p$. This appendix is the archival form of these results under the manuscript labels Theorem A, Corollary 1, Corollary 2.

A. A.0 Standing assumptions

- **(A1) Model.** Noiseless record $R_n = a_n(MT)_n$, $n = 1, \dots, N$, with $N = K$ and $M \in \mathbb{R}^{K\times K}$ **invertible** — Hadamard, Fourier, random square, or otherwise. Write $c = MT$ for the carrier coefficient vector, so $R = e^{\ell} \odot c$ ($\odot$ = entrywise product).
- **(A2) Gain class.** $a_n > 0$ for all n; the admissible log-gains form (or contain a neighborhood inside) the linear subspace $S \ni \mathbf{1}$, $\dim S = p \geq 2$; write $p - 1 = \dim(S/\operatorname{span}\{\mathbf{1}\})$ for the nonconstant drift dimension.
- **(A3) Object class.** $T \in \mathbb{R}^K$ unconstrained (Theorem A, Corollary 1); constrained classes are treated in Corollary 2.
- **(A4) Carrier support.** For the clean dimension statements we assume full carrier support, $c_n = (MT)_n \neq 0$ for all n; this holds for generic T (each $\{c_n = 0\}$ is a hyperplane). Zero buckets are discussed in Remark A.2.
- **(A5) Gauge.** The global-scale gauge orbit of a pair (ℓ, T) is $\mathcal{G}(\ell, T) = \{(\ell + t\mathbf{1},\, e^{-t}T) : t \in \mathbb{R}\}$; identifiability is always meant modulo $\mathcal{G}$.

B. A.1 Theorem A (square non-identifiability, any invertible design)

Theorem A. *Assume (A1)–(A5). For every $s \in S$ the pair*

$$\ell' = \ell + s, \qquad c' = e^{-s} \odot c, \qquad T'(s) = M^{-1}\big(e^{-s} \odot MT\big)$$

reproduces the bucket record exactly: $e^{\ell'_n} c'_n = R_n$ for all n. If s is nonconstant on $\operatorname{supp}(MT)$ — under (A4), simply nonconstant — then $(\ell', T'(s)) \notin \mathcal{G}(\ell, T)$ and $T'(s) \neq \gamma T$ for every $\gamma \in \mathbb{R}$. The set of admissible pairs consistent with the record therefore contains a real-analytic family of dimension $\dim S = p$, of which the gauge orbit accounts for only $\dim(S \cap \operatorname{span}\{\mathbf{1}\}) = 1$; the non-gauge ambiguity dimension is at least $p - 1 \geq 1$. In particular, gain–object separation from a square design is not identifiable modulo gauge whenever S contains a nonconstant element.

Proof. **(Data equality.)** $e^{\ell'_n} c'_n = e^{\ell_n + s_n} e^{-s_n} c_n = e^{\ell_n} c_n = R_n$, and $T'(s) = M^{-1}c'$ is well defined because M is invertible. Both steps are exact; no smallness of s is used.

(Distinctness of the object.) $T'(s) = \gamma T$ for some γ iff $MT'(s) = \gamma MT$, i.e. $e^{-s_n} c_n = \gamma c_n$ for every n, i.e. $e^{-s_n} = \gamma$ for every $n \in \operatorname{supp}(c)$. Since $x \mapsto e^{-x}$ is injective, this forces s constant on $\operatorname{supp}(c)$. Contrapositively, s nonconstant on $\operatorname{supp}(MT)$ implies $T'(s)$ is not proportional to T.

(Distinctness modulo gauge.) $(\ell + s, T'(s)) \in \mathcal{G}(\ell, T)$ requires $\ell + s = \ell + t\mathbf{1}$ in the gain coordinate, i.e. $s = t\mathbf{1}$ (log is injective on positive gains). A nonconstant s therefore leaves the orbit; note that when $s = t\mathbf{1}$ the construction reduces exactly to the gauge, $T'(t\mathbf{1}) = e^{-t}T$, so the family contains the orbit and nothing of the orbit is double-counted.

(Dimension.) The map $s \mapsto (\ell + s, T'(s))$ is injective (read off the first coordinate) and real-analytic in s, hence its image is a p-dimensional analytically parametrized family of exact solutions. Intersecting with the gauge orbit uses up $\dim(S \cap \operatorname{span}\{\mathbf{1}\}) = 1$ dimension, leaving non-gauge ambiguity dimension $\geq p - 1$. ■

The display in the main text (Sec. 4) is the instance $s = \bar{\ell}\,\mathbf{1} - \ell$: the drift is absorbed *in one step* into the relabelled object $T' = M^{-1}\operatorname{diag}(a)MT$ (up to the gauge constant), leaving a constant gain. The hypothesis "a nonconstant on the support of MT" in the main-text statement is exactly the distinctness condition established above.

a) What this does not claim.: Theorem A is a statement about the square case $N = K$ and an unconstrained object; for $N > K$ the construction fails unless $e^{-s} \odot MT$ remains in $\operatorname{range}(M)$, and the correct analysis is the tall-design identifiability theory of Theorem A′ (see Appendix B). It does not say the record is uninformative about the gain in a statistical model: under random illumination, the *relative* gain is consistently estimable (Theorem B; see Appendix C), a different — statistical, asymptotic — notion of identifiability which Theorem A neither contradicts nor implies. Finally, Theorem A is about uniqueness, not conditioning: it makes no claim about how well- or ill-posed any estimator is.

C. A.2 Corollary 1 (slowness does not help)

Corollary 1. *Assume (A1)–(A5).*

*(i) If the admissible log-gain class is the linear space S itself, the entire ambiguity family of Theorem A lies inside the class: $\ell \in S,\ s \in S \Rightarrow \ell + s \in S$. In particular, choosing $s = \bar{\ell}\,\mathbf{1} - \ell \in S$ produces a **constant-gain** exact explanation of the record with the static object $T' = M^{-1}(e^{\ell - \bar{\ell}} \odot MT)$: a slowly drifting gain over the true object is indistinguishable from no drift at all over a relabelled object.*

*(ii) If the admissible class is a slowness ball $\mathcal{B}_L = \{\ell \in S : \|\ell\|_* \leq L\}$ for any seminorm $\|\cdot\|_*$ vanishing on constants (e.g. a Hölder or low-pass-energy seminorm), and ℓ is interior ($\|\ell\|_* < L$), then every nonconstant $s \in S$ with $\|s\|_*$ small enough keeps $\ell + s \in \mathcal{B}_L$. The local non-gauge ambiguity dimension remains $p - 1$: slowness bounds the **size** of the ambiguity, never its **dimension**.*

Proof. (i) is linearity of S plus Theorem A applied with the stated s; nonconstancy of ℓ on $\mathrm{supp}(c)$ makes the constant-gain alternative distinct modulo gauge. (ii) Subadditivity of the seminorm gives $\|\ell+s\|_* \le \|\ell\|_* + \|s\|_* \le L$ once $\|s\|_* \le L - \|\ell\|_*$; the set of such s is a neighborhood of 0 in S and Theorem A applies to each nonconstant member. ■

a) What this does not claim.: Slowness is not useless in general — in the tall-design regime the whole point is that slow drift means small p, making the threshold $N \ge K + p$ cheap to satisfy (Sec. 4). The corollary only asserts that at $N = K$ a slowness prior *by itself* removes nothing from the ambiguity, because the ambiguity directions live in the slow class itself.

D. A.3 Corollary 2 (nonnegativity is insufficient; known support zeros can restore local identifiability)

Throughout this corollary write H for the square invertible design ($H = M$; the letter matches the fixed-basis usage, e.g. Hadamard), and define the gain-absorption operator

$$M_s := H^{-1}\,\mathrm{diag}(e^{-s})H, \qquad T'(s) = M_sT,$$

so that the ambiguity set at T under an object prior class $\mathcal{T}$ is $V_{\mathcal{T}}(T) = \{s \in S : M_sT \in \mathcal{T}\}$, and identifiability modulo gauge is the assertion $V_{\mathcal{T}}(T) = S \cap \mathrm{span}\{\mathbf{1}\}$.

Corollary 2. *Assume (A1)–(A5).*

(i) (Positivity insufficient.) Let $\mathcal{T}_+ = \{T : T_j \ge 0\ \forall j\}$ and let T have full support, $T_j > 0$ for all j. Then there is $r > 0$ such that every $s \in S$ with $\|s\|_\infty < r$ satisfies $M_sT > 0$; explicitly, it suffices that

$$\|H^{-1}\|_2\,\|HT\|_2\,(e^r - 1) < \min_j T_j.$$

Hence $V_{\mathcal{T}_+}(T)$ contains a full neighborhood of 0 in S and the local non-gauge ambiguity dimension is still $p - 1$.

(ii) (Known support zeros; local criterion.) Let $\Omega \subset \{1, \dots, K\}$, $|\Omega| = q$, and $\mathcal{T}_\Omega = \{T : T_{\Omega^c} = 0,\ T \ge 0\}$, with the true $T \in \mathcal{T}_\Omega$, $T_\Omega > 0$. Then local identifiability modulo gauge holds at T if and only if (at the first-order level)

$$\ker\big(P_{\Omega^c}H^{-1}\,\mathrm{diag}(HT)\,\big|_S\big) \;=\; S \cap \mathrm{span}\{\mathbf{1}\},$$

where P_{Ω^c} is coordinate projection onto Ω^c. The gauge direction always lies in the kernel; when the kernel is exactly the gauge, no ambiguity exists in a quantifiable ball: with γ the smallest singular value of the linearized map on a complement $S_\perp$ of $\mathrm{span}\{\mathbf{1}\}$ in S and C_H a second-order remainder constant (below), every s with $0 < \|s_\perp\|_2 < \gamma/C_H$ violates the support constraint.

(iii) (Generic counts — dimension-counting arguments, labeled as such.) The linearized map in (ii) sends the $(p-1)$-dimensional space $S/\mathrm{span}\{\mathbf{1}\}$ into $\mathbb{R}^{K-q}$; injectivity is possible only if $K_0 := K - q \ge p - 1$, i.e. ***at least one known zero per nonconstant drift dimension****. For generic $(H, \mathcal{T}_\Omega)$ this necessary count is expected to be sufficient. For unknown support of size $\le q$, dimension counting predicts the threshold $K \ge 2q + (p-1)$, up to endpoint conventions.*

Proof. **(i)** $M_0T = T$ and $\|M_sT - T\|_2 = \|H^{-1}\big((e^{-s} - \mathbf{1}) \odot HT\big)\|_2 \le \|H^{-1}\|_2\,\|HT\|_2 \max_n |e^{-s_n} - 1| \le \|H^{-1}\|_2\,\|HT\|_2\,(e^{\|s\|_\infty} - 1)$. If this is below $\min_j T_j$ then $M_sT > 0$ entrywise. Since the positive orthant is open and $s \mapsto M_sT$ is continuous, positivity excludes nothing locally; Theorem A supplies the distinctness. ■ (i)

(ii) Define the support-constraint map $F : S \to \mathbb{R}^{K-q}$, $F(s) = P_{\Omega^c}M_sT$. It is real-analytic, $F(0) = P_{\Omega^c}T = 0$, and its differential at 0 is

$$DF_0\,u = -\,P_{\Omega^c}H^{-1}\,\mathrm{diag}(HT)\,u,$$

by differentiating $e^{-tu} = \mathbf{1} - tu + O(t^2)$ inside M_{tu}. The gauge is always in the kernel: $H^{-1}\,\mathrm{diag}(HT)\mathbf{1} = H^{-1}(HT) = T$ and $P_{\Omega^c}T = 0$; consistently, $F(t\mathbf{1}) = e^{-t}F(0) = 0$ exactly, for all t.

Necessity of the criterion (first-order sense). If the kernel strictly exceeds the gauge, some nonconstant $u \in S$ satisfies the linearized support constraint. This is a first-order ambiguity direction: it makes T a critical point of the constrained model map and hence a Fisher-information null direction for the corresponding functional (cf. Remark A.1), so local *differential* identifiability fails. Whether the direction integrates to an exact ambiguity curve requires a further constant-rank or analyticity argument and can fail at second order; we claim only the first-order (differential) statement, which is what the criterion tests.

Sufficiency in a ball. Suppose the kernel equals the gauge and let $\gamma = \sigma_{\min}(DF_0|_{S_\perp}) > 0$. Entrywise Taylor with remainder, $|e^{-x} - 1 + x| \le \frac{x^2}{2}e^{|x|}$, gives for $\|s\|_\infty \le 1$

$$\|F(s) - DF_0\,s\|_2 \;\le\; C_H\,\|s\|_2^2, \qquad C_H := \tfrac{e}{2}\,\|H^{-1}\|_2\,\|HT\|_2.$$

For gauge-fixed $s = s_\perp \in S_\perp$ (any s may be reduced to this form using $F(s + t\mathbf{1}) = e^{-t}F(s)$, which rescales but never annihilates F),

$$\|F(s_\perp)\|_2 \;\ge\; \gamma\,\|s_\perp\|_2 - C_H\,\|s_\perp\|_2^2 \;>\; 0 \quad \text{for } 0 < \|s_\perp\|_2 < \gamma/C_H.$$

A nonzero $F(s)$ means $M_sT \notin \mathcal{T}_\Omega$ (support constraint violated), so no non-gauge ambiguity exists in that ball. This is the quantitative inverse-function-theorem argument in explicit constants. ■ (ii)

(iii) *These are genericity/dimension-count arguments, not proofs, and we state them as such.* The necessity direction is rigorous: a linear map from a $(p-1)$-dimensional space cannot be injective into $\mathbb{R}^{K-q}$ if $K - q < p - 1$, so at least $p - 1$ known zeros are required. The sufficiency direction is generic: the entries of $P_{\Omega^c}H^{-1}\,\mathrm{diag}(HT)|_{S_\perp}$ are polynomial in (H, T), so full column rank holds off a proper algebraic subvariety *provided a single witness exists*; the dimension count $K - q \ge p - 1$ makes a witness expected but, for a *fixed* named basis (e.g. Hadamard with a specific low-pass S), the rank must be verified directly — the correct fixed-basis statement is rank-based, not dimensional. The unknown-support count $K \ge 2q + (p-1)$ arises from demanding that the q-sparse model avoid all of its gain-transformed copies $M_s\Sigma_q$, in analogy with the generic sparsity thresholds of Kech–Krahmer [13] for generic bilinear maps; it is a heuristic prediction of the generic threshold, unverified for structured designs, and the uniform no-support-switching condition it summarizes is strictly stronger than (ii). ■ (heuristic scope) *a) What this does not claim.:* Part (i) says nonnegativity fails to restore *identifiability*; it may still improve estimator conditioning and is routinely useful as regularization — no claim is made either way. Part (ii) is local: it excludes ambiguities in a ball around T and is silent about isolated global aliases, which can exist even with a nonsingular linearization; global uniqueness needs a separate compactness/non-vanishing argument outside the ball. The criterion is also stated at fixed, *known* support; for unknown support only the heuristic count of part (iii) is offered, and part (iii) makes no claim for any specific fixed basis without an explicit rank check.

E. A.4 Remarks

a) Remark A.1 (Fisher-theoretic reading).: Because the ambiguity of Theorem A is an explicit real-analytic curve $t \mapsto (\ell + ts, T'(ts))$ along which the noiseless data are constant, every nonconstant $s \in S/\operatorname{span}\{\mathbf{1}\}$ yields an exact null direction of the (gauge-quotient) Fisher information in any regular dominated noise model built on this record: the Cramér–Rao bound is infinite for any functional that varies along the curve. Here the implication "explicit ambiguity curve $\Rightarrow$ Fisher null direction" is exact, not merely first-order.

b) Remark A.2 (zero buckets).: If some $c_n = (MT)_n = 0$, two separate pathologies arise, which the statements above deliberately quarantine via (A4): the gain values a_n at zero-carrier frames are entirely unobservable (a trivial extra ambiguity, of dimension $\dim(S \cap \{s : s = 0 \text{ on } \operatorname{supp}(c)\})$ beyond the count of Theorem A), and log-domain processing of $R_n = 0$ is undefined, an estimation-side issue treated by the calibrated soft-log machinery of the low-photon analysis (Sec. 6; see Appendix D), not an identifiability issue.

APPENDIX B
TALL-DESIGN IDENTIFIABILITY: THE PARAMETRIC-TRANSVERSALITY PROOF

This appendix proves the tall-design identifiability results quoted as **Theorem A′** in Sec. 4, restated in full as Theorem A′(i)–(iv) in Sec. B.8: the generic local threshold with a genuine *iff* at $N = K + p - 1$ for $K \geq 2$, genuine (exact, positive-dimensional) failure below that wall, generic exact identifiability at $N \geq K + p$, uniform identifiability over the nonvanishing-carrier set at $N \geq 2K + p - 1$, and the degenerate $K = 1$ stratum. The proof route is **parametric transversality** (a rowwise design-perturbation witness plus Fubini), verified adversarially in review round R11; it holds for **every fixed drift subspace** $S \ni \mathbf{1}$ — no genericity of S is assumed anywhere, so the physical Fourier low-pass space S_{LP} is covered directly, with no transfer lemmas. Sharpness (necessity) of the two global thresholds is **not** claimed as a theorem; its status is stated precisely after Theorem A′. This appendix **supersedes** the earlier incidence-variety/stratum-counting route (see Sec. B.9); the superseded "sharp iff" statements at $N = K + p$ and $N = 2K + p - 1$ are withdrawn.

A. B.0 Setting, notation, and the reduction

The bucket record is $R_n = a_n B_n$ with $B = MT$: in vector form the data map is

$$\Phi_M(T, s) \;=\; D_{e^s}\, M\, T, \qquad M \in \mathbb{R}^{N\times K},\; N \geq K,$$
$$s \in S,\; T \in \mathbb{R}^K,$$

where $D_x = \operatorname{diag}(x)$, e^s is componentwise, and the log-gain trajectory is confined to the fixed, known drift class $S \subset \mathbb{R}^N$ with $\dim S = p \geq 2$ and $\mathbf{1} \in S$. Write $y = MT$ for the carrier and define

$$\mathcal{U} = \{(M, T) : (MT)_n \neq 0 \;\forall n\},$$
$$\mathcal{U}_M = \{T \in \mathbb{R}^K : (MT)_n \neq 0 \;\forall n\},$$
$$\mathcal{R} = \{M \in \mathbb{R}^{N\times K} : \operatorname{rank} M < K\}.$$

$\mathcal{U}$ is open with null complement (each $\{m_n^\top T = 0\}$ is a proper algebraic subset); $\mathcal{R}$ is a proper algebraic subset (vanishing of all $K \times K$ minors), hence Lebesgue-null. **Every exceptional set below explicitly includes $\mathcal{R}$ (or $\mathcal{R} \times \mathbb{R}^K$).** The gauge is $(T, s) \sim (cT,\, s - \log c\,\mathbf{1})$, $c > 0$; identifiability is always modulo this gauge. "For a.e." always refers to Lebesgue measure on the stated parameter space.

Lemma B.0 (reduction). *Assume* $\operatorname{rank} M = K$ *and* $(M, T) \in \mathcal{U}$*, and fix any* $s \in S$*. Then* $(T', s') \in \mathbb{R}^K \times S$ *is an alias of* (T, s) *— i.e.* $\Phi_M(T', s') = \Phi_M(T, s)$ *— iff, with* $d := s - s' \in S$,

$$M\, T' \;=\; D_{e^d}\, M\, T. \tag{B.1}$$

If $d = a\mathbf{1}$ is constant, then (B.1) reads $MT' = e^a MT$ and full column rank gives $T' = e^a T$ — exactly the gauge orbit with $c = e^a$; conversely every gauge point solves (B.1) with constant d. Hence (M, T) is identifiable iff (B.1) has no solution (T', d) with $d \in S \setminus \mathbb{R}\mathbf{1}$. Moreover, fixing a complement S_0 of $\mathbb{R}\mathbf{1}$ in S ($\dim S_0 = p - 1$) and writing $d = d_0 + a\mathbf{1}$, the normalization $(T', d) \mapsto (e^{-a}T', d_0)$ maps solutions to solutions; so (M, T) is identifiable iff (B.1) has no solution with $d \in S_0 \setminus \{0\}$.

Proof. D_{e^s} is invertible, so record equality is equivalent to (B.1) with $d = s - s'$. Constant $d = a\mathbf{1}$: row n reads $m_n^\top T' = e^a y_n$, i.e. $MT' = e^a MT$, and $\operatorname{rank} M = K$ forces $T' = e^a T$, which is the gauge point $(cT,\, s - \log c\,\mathbf{1})$ with $c = e^a > 0$. The equivariance $(T', d) \mapsto (e^a T', d + a\mathbf{1})$ is a direct substitution in (B.1). ∎

Two structural remarks. *(a) The reduction eliminates s:* identifiability is a property of (M, T, S) alone, uniform over the true drift — this is the source of the "for every $s \in S$ simultaneously" quantifier in every theorem below. *(b) Full column rank is necessary:* for rank-deficient M the equivalence "constant $d \Leftrightarrow$ gauge-trivial" is **false** even on $\mathcal{U}$. Counterexample ($\operatorname{rank} M = 1$): $N = K = 2$, both rows of M equal to $(0, 1)$, $T = (0, 1)^\top$, $T' = (1, 1)^\top$: $MT = MT' = (1, 1)^\top$, so $(T', d = 0)$ solves (B.1) with constant d, yet $T' \notin \mathbb{R}T$ — an object-kernel ambiguity invisible to the normalized $d \neq 0$ incidence set. This is why $\mathcal{R}$ is carried explicitly in every exceptional set.

a) What this does not claim.: The reduction is exact but presupposes $\operatorname{rank} M = K$ and a nonvanishing carrier; neither is a restriction for the a.e. statements (both failure sets are null), but both must be — and are — carried explicitly.

B. B.1 Lemma B.1 (row dichotomy)

Lemma B.1. *Let $(M,T) \in \mathcal{U}$ and let (T', d) solve (B.1) with d nonconstant ($d \notin \mathbb{R}\mathbf{1}$). Then T' is not a scalar multiple of T; consequently $T' - e^{d_n}T \neq 0$ for every $n = 1, \dots, N$.*

Proof. Suppose $T' = \beta T$. Row n of (B.1): $\beta y_n = e^{d_n} y_n$; since $y_n \neq 0$ on $\mathcal{U}$, $e^{d_n} = \beta$ for all n, so d is constant — contradiction. For the display: if $T' = e^{d_n}T$ for a single n, then $T' \parallel T$ ($T \neq 0$ because $y = MT \neq 0$). ■

Only *global* nonconstancy of d is used — partial constancy (some equal components) is harmless — and the only division is by y_n, legitimate on $\mathcal{U}$.

C. B.2 Lemma B.2 (joint submersivity)

Define $F : \mathbb{R}^{N\times K} \times \mathbb{R}^K \times \mathbb{R}^K \times S_0 \to \mathbb{R}^N$,

$$\begin{aligned} F(M,T,T',d) \;&=\; MT' - D_{e^d}MT, \\ F_n &= m_n^\top \big(T' - e^{d_n}T\big). \end{aligned} \tag{B.2}$$

Lemma B.2. *At every zero of F with $(M,T) \in \mathcal{U}$ and $d \in S_0 \setminus \{0\}$, the differential of F with respect to M alone is surjective onto $\mathbb{R}^N$.*

Proof. Perturb row n only: $\delta M = e_n\, \delta m_n^\top$ gives $\delta F = e_n \cdot \delta m_n^\top(T' - e^{d_n}T)$, which spans $\mathbb{R}e_n$ as δm_n ranges over $\mathbb{R}^K$, provided $T' - e^{d_n}T \neq 0$ — which holds at every such zero for every n by Lemma B.1. Rows are decoupled, so the image contains every e_n. ■

Only δM is used: no genericity of S, no structure of d beyond nonconstancy, and the object is untouched. This rowwise witness is uniform over the whole zero set — the ingredient the superseded route lacked.

D. B.3 A measure lemma, and generic exact identifiability at $N \geq K + p$

Lemma B.3 (C^1 image of a lower-dimensional manifold is null). *Let X be a second-countable C^1 manifold with $\dim X = m < q$, and $f : X \to \mathbb{R}^q$ a C^1 map. Then $f(X)$ has q-dimensional Lebesgue measure zero.*

Proof. Second countability gives a countable chart cover $\varphi_i : U_i \to \mathbb{R}^m$, and each chart domain is exhausted by countably many compact cubes; by countable subadditivity it suffices to treat one compact cube Q of side ℓ. On Q, $g = f \circ \varphi_i^{-1}$ is C^1, hence Lipschitz with some constant L. Subdividing Q into k^m subcubes of side ℓ/k, each image lies in a ball of radius $L\sqrt{m}\,\ell/k$ in $\mathbb{R}^q$, so the outer measure of $g(Q)$ is at most $k^m\, c_q(2L\sqrt{m}\,\ell/k)^q = C\,k^{m-q} \to 0$ since $m < q$. ■

No properness or compactness of f is needed; the zero manifolds below are embedded in Euclidean space, hence second countable and σ-compact automatically. The statement is **false** for merely continuous maps (space-filling curves) — the C^1 regularity supplied by Lemma B.2 is load-bearing.

Theorem B.4 (generic exact identifiability; Theorem A$'$(ii)). *Fix $S \ni \mathbf{1}$, $\dim S = p \geq 2$. If $N \geq K+p$, there is a Lebesgue-null set $E_{\text{gen}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ such that every $(M,T) \notin E_{\text{gen}}$ is identifiable: for every $s \in S$, the only $(T', s') \in \mathbb{R}^K \times S$ with $\Phi_M(T', s') = \Phi_M(T, s)$ are the gauge orbit of (T, s).*

Proof. Work on the open set $\Omega = \{(M,T,T',d) : (M,T) \in \mathcal{U},\; T' \in \mathbb{R}^K,\; d \in S_0 \setminus \{0\}\}$. By Lemma B.2, 0 is a regular value of F on Ω, so $Z = F^{-1}(0) \cap \Omega$ is an embedded real-analytic submanifold of codimension N:

$$\dim Z = (NK + K) + K + (p-1) - N.$$

The projection $\pi : Z \to \mathbb{R}^{N\times K} \times \mathbb{R}^K$ onto (M,T) is C^1 and

$$\dim Z - \dim(\mathbb{R}^{N\times K} \times \mathbb{R}^K) = K + p - 1 - N \leq -1$$

when $N \geq K + p$, so $\pi(Z)$ is Lebesgue-null by Lemma B.3. By the reduction (Lemma B.0, which requires $\operatorname{rank} M = K$),

$$\{\text{non-identifiable pairs in } \mathcal{U}\} \;\subset\; (\mathcal{R} \times \mathbb{R}^K) \cup \pi(Z),$$

and the full non-identifiable set is contained in $(\mathcal{R} \times \mathbb{R}^K) \cup \pi(Z) \cup \mathcal{U}^c$ — three null sets. Set E_{gen} to their union. ■

a) Remark B.4.1 (quantifier order, proved).: For a.e. (M,T): for **every** $s \in S$, no non-gauge alias — the null set E_{gen} is independent of s, because the reduction eliminated s before any measure-theoretic step. This does **not** imply that one design works for every object; that is Theorem B.5 at its own larger threshold. ($T' = 0$ needs no special handling: $T' = 0$ forces $D_{e^d}y = 0$, impossible on $\mathcal{U}$.)

b) Remark B.4.2 (fixed-object slice).: For each **fixed** $T \neq 0$: a.e. M identifies that particular T when $N \geq K + p$ (the exceptional design set may depend on T). Proof: freeze T, run the same argument with $F_T(M, T', d) = MT' - D_{e^d}MT$ on $\{M : (M,T) \in \mathcal{U}\} \times \mathbb{R}^K \times (S_0 \setminus \{0\})$; Lemma B.2's witness uses only δM, the zero set has dimension $NK + K + (p - 1) - N \leq NK - 1$ when $N \geq K + p$, and Lemma B.3 applies; add $\mathcal{R}$ and the proper algebraic set $\{M : (M,T) \notin \mathcal{U}\}$. This slice form is what covers the fixed structured objects of the Fig. 6 protocol.

c) What this does not claim.: No claim of uniformity over objects for a fixed design (see Theorem B.5 and its counterexample); no claim at the boundary $N = K + p - 1$ (necessity is open — Sec. B.8, sharpness paragraph); nothing about conditioning, noise, or estimators; deterministic structured designs are not covered by any a.e. statement.

E. B.4 Uniform identifiability on the nonvanishing-carrier set at $N \geq 2K + p - 1$

Theorem B.5 (uniform sufficiency on $\mathcal{U}_M$; Theorem A$'$(iii)). *Fix $S \ni \mathbf{1}$, $\dim S = p \geq 2$. If $N \geq 2K + p - 1$, there is a Lebesgue-null set $E_{\text{unif}} \subset \mathbb{R}^{N\times K}$ such that for every $M \notin E_{\text{unif}}$: every $T \in \mathcal{U}_M$ is identifiable, for every $s \in S$ simultaneously.*

Proof. The solution set of (B.1) is scaling-equivariant — (T, T', d) solves iff (cT, cT', d) does, $c \neq 0$ — and nonvanishing carrier and nonconstancy are scale-invariant, so it suffices to rule out solutions with $|T| = 1$. Let

$$\begin{aligned} \Omega' = \{(M,T,T',d) : |T| = 1,\; (M,T) \in \mathcal{U}, \\ T' \in \mathbb{R}^K,\; d \in S_0 \setminus \{0\}\}, \end{aligned}$$

an open subset of $\mathbb{R}^{N\times K} \times S^{K-1} \times \mathbb{R}^K \times S_0$. Lemma B.2 holds verbatim on Ω' (the sphere constrains T, not M), so $Z' = F^{-1}(0) \cap \Omega'$ is an embedded real-analytic manifold of codimension N with

$$\begin{aligned}\dim Z' &= NK + (K-1) + K + (p-1) - N \\ &= NK + (2K + p - 2 - N),\end{aligned}$$

and the projection $\pi' : Z' \to \mathbb{R}^{N\times K}$ onto M has deficit $2K + p - 2 - N \le -1$ when $N \ge 2K + p - 1$; $\pi'(Z')$ is null by Lemma B.3. Set $E_{\mathrm{unif}} = \mathcal{R} \cup \pi'(Z')$ (rank-deficient designs added explicitly: the reduction needs $\operatorname{rank} M = K$). If $M \notin E_{\mathrm{unif}}$ and (T', d) solved (B.1) for some $T \in \mathcal{U}_M$, $d \in S_0 \setminus \{0\}$, then rescaling by $c = 1/|T|$ would put $(M, cT, cT', d) \in Z'$, i.e. $M \in \pi'(Z')$ — contradiction. Lemma B.0 concludes. ■

a) Remark B.5.1 (why $\mathcal{U}_M$ cannot be dropped: a zero-carrier counterexample at the threshold).: "Every nonzero object" is **false** even at $N = 2K + p - 1$. Take $K = 2$, $p = 2$, $N = 5 = 2K + p - 1$, $S = \operatorname{span}\{\mathbf{1}, e_1\}$, $T = (1, 1)^\top$, and M with rows $(1, -1), (1, 0), (0, 1), (1, 1), (1, 2)$, so $MT = (0, 1, 1, 2, 3)^\top$ — full column rank, first carrier coordinate zero. For every $t \neq 0$, $d = t\,e_1 \in S$ is nonconstant and $(T', d) = (T, t\,e_1)$ solves (B.1): row 1 reads $m_1^\top T' = e^t \cdot 0 = 0$, rows 2–5 are unchanged. So (T, s) and $(T, s - t\,e_1)$ produce identical records for every t — a one-parameter family of non-gauge aliases. This mechanism is available for **any** design: every object on the hyperplane $\{m_1^\top T = 0\}$ acquires such aliases whenever S contains a vector supported on the zero-carrier row. The correct conclusion is the stated one: for a.e. M, every $T \in \mathcal{U}_M$, every $s \in S$.

b) What this does not claim.: Uniformity is over nonvanishing-carrier objects and drifts, not over designs: an adversarially chosen M can defeat the bound. Necessity of $2K + p - 1$ is not claimed (Sec. B.8). No claim about recovery algorithms or noise.

F. B.5 Lemma W: position of the twisted column space

For $(M, T) \in \mathcal{U}$ with $\operatorname{rank} M = K$ define

$$W(M,T) \;:=\; D_y^{-1}\operatorname{col}(M) \subset \mathbb{R}^N, \qquad y = MT,$$

a K-dimensional subspace with $\mathbf{1} \in W$ (since $D_y \mathbf{1} = y \in \operatorname{col}(M)$). Aliases relate to W by: $d \in S$ *solves (B.1) for some* T' *iff* $e^d \in W(M, T)$ — indeed $D_{e^d} y = D_y e^d$, so (B.1) reads $MT' = D_y e^d$, i.e. $e^d = D_y^{-1} M T' \in W$; conversely $e^d = D_y^{-1} M u$ gives the (unique, by full column rank) solution $T' = u$.

Lemma W. *Fix any subspace $S \ni \mathbf{1}$ with $\dim S = p \le N$, and $K \ge 1$. For a.e. $(M, T) \in \mathcal{U}$, with $W = W(M, T)$:*

$$\begin{aligned}\dim(S \cap W) &= 1 + \max\{0,\ (p-1) + (K-1) - (N-1)\} \\ &= 1 + \max\{0,\ K + p - N - 1\}.\end{aligned} \tag{B.3}$$

No genericity of S is assumed or used — only $\mathbf{1} \in S$.

Proof. *(Step 1: basis completion; fixed T.)* Fix $T \neq 0$ and complete it to a basis $B_T = [T, b_2, \dots, b_K] \in GL_K$. The map $M \mapsto MB_T = (y, v_2, \dots, v_K)$ is a linear isomorphism of $\mathbb{R}^{N\times K}$, hence maps null sets to null sets both ways: a.e. M $\Leftrightarrow$ a.e. $(y, v_2, \dots, v_K)$.

(Step 2: fixed-y isomorphism.) For fixed $y \in (\mathbb{R}^*)^N$ (a co-null condition), set $w_j := D_y^{-1} v_j$. For fixed y this is a linear isomorphism of $\mathbb{R}^{N(K-1)}$ in $(v_2, \dots, v_K)$ — it preserves "a.e." and introduces no correlation between the w_j and the fixed S. Then $W = D_y^{-1}\operatorname{span}(y, v_2, \dots, v_K) = \operatorname{span}(\mathbf{1}, w_2, \dots, w_K)$.

(Step 3: quotient.) Let $q : \mathbb{R}^N \to \mathbb{R}^N/\mathbb{R}\mathbf{1}$ (dimension $N - 1$). $A := q(S)$ has dimension $p - 1$ (the kernel $\mathbb{R}\mathbf{1}$ lies in S); $q(W) = \operatorname{span}(qw_2, \dots, qw_K)$.

(Step 4: general position against the fixed A.) For a.e. $(w_2, \dots, w_K)$: the $q(w_j)$ are independent and $\dim(A \cap q(W)) = \max\{0, (p-1) + (K-1) - (N-1)\}$. Failure is the vanishing of suitable maximal minors of the matrix $[A\text{-basis} \mid qw_2 \cdots qw_K]$ — a polynomial condition in the w_j, not identically zero (a witness exists: against one fixed subspace, vectors can be chosen stepwise outside a finite union of proper subspaces), hence a proper algebraic — null — subset. This is general position with respect to the fixed A only; no genericity of S.

(Step 5: $q(S \cap W) = q(S) \cap q(W)$, via $\mathbf{1} \in W$.) "$\subseteq$" is trivial. "$\supseteq$": if $x \in S$, $x' \in W$ and $q(x) = q(x')$, then $x - x' \in \mathbb{R}\mathbf{1} \subseteq W$, so $x \in W$, i.e. $x \in S \cap W$. The kernel of q on $S \cap W$ is exactly $\mathbb{R}\mathbf{1}$ (as $\mathbf{1} \in S \cap W$), whence $\dim(S \cap W) = 1 + \dim(q(S) \cap q(W))$.

(Step 6: Fubini twice, with Borel measurability.) The quantifier order so far: for every fixed $T \neq 0$, for a.e. y, for a.e. (v_j) — hence, by Steps 1–2 and Fubini on $\mathbb{R}^N \times \mathbb{R}^{N(K-1)}$, for every fixed $T \neq 0$, for a.e. M, formula (B.3) holds. The failure set $B = \{(M, T) \in \mathcal{U} : \text{(B.3) fails}\}$ is Borel: the dimension condition is a Boolean combination of rank conditions (minor vanishing/nonvanishing) on matrices whose entries are rational in (M, T) on $\mathcal{U}$ (the w_j are entries of M divided by coordinates of MT). Measurability licenses a second Fubini, over T: every T-slice of B is M-null, so B is null in $\mathbb{R}^{N\times K} \times \mathbb{R}^K$. ■

a) Remark W.1 ($K = 1$).: For $K = 1$ the lemma **holds**: $W = \mathbb{R}\mathbf{1}$ identically, $\dim(S \cap W) = 1$, and (B.3) also gives $1 + \max\{0, p - N\} = 1$ since $p \le N$. What degenerates at $K = 1$ is the necessity of the global thresholds (Sec. B.7), not Lemma W.

b) What this does not claim.: (B.3) is an a.e. statement over (M, T); it says nothing about any *particular* design (e.g. structured M), and nothing about the conditioning of the intersection, only its dimension.

G. B.6 The generic local threshold ($K \ge 2$) and genuine failure below the wall

a) Kernel derivation (independent; no import from the superseded route).: Fix $(M, T) \in \mathcal{U}$ with $\operatorname{rank} M = K$ and any $s \in S$. The differential of Φ_M at (T, s) is

$$D\Phi_M(T,s)[\delta T, \delta s] \;=\; D_{e^s}\big(M\,\delta T + D_y\,\delta s\big), \qquad y = MT. \tag{B.4}$$

The diagonal factor is invertible, so $(\delta T, \delta s) \in \ker \Leftrightarrow M\delta T = -D_y \delta s$; the right side lies in $\operatorname{col}(M)$ iff $\delta s \in D_y^{-1}\operatorname{col}(M) =$

$W(M,T)$, and then $\delta T = -(M^\top M)^{-1}M^\top D_y \delta s$ is unique. Hence $\ker D\Phi_M(T,s) \cong S \cap W$ via $(\delta T, \delta s) \mapsto \delta s$, and

$$\dim\ker D\Phi_M(T,s) = \dim(S\cap W),$$
$$\operatorname{rank} D\Phi_M(T,s) = K + p - \dim(S\cap W),$$

independently of s. The gauge tangent is the line spanned by $(-T, \mathbf{1})$ (from $c \mapsto (cT, s - \log c\,\mathbf{1})$), and it lies in the kernel since $MT = D_y\mathbf{1}$; as $\mathbf{1} \in S \cap W$ always, the maximal gauge-compatible rank is $K + p - 1$ and the defect from it is $\dim(S\cap W) - 1$. This derivation is self-contained: nothing is imported from the superseded route, and no genericity of S is used. By Lemma W, for a.e. $(M,T) \in \mathcal{U}$ and every s:

$$\operatorname{rank} D\Phi_M(T,s) = \min\{N,\ K + p - 1\}.$$

Theorem B.6 (generic local iff; Theorem A′(i), rank part). *Fix $S \ni \mathbf{1}$, $\dim S = p \geq 2$, and $K \geq 2$. There is a Lebesgue-null set $E_{\text{loc}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ — take $E_{\text{loc}} = (\mathcal{R} \times \mathbb{R}^K) \cup \mathcal{U}^c \cup B$ with B the failure set of Lemma W — such that for every $(M,T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: $\operatorname{rank} D\Phi_M(T,s) = \min\{N, K+p-1\}$, and (T,s) is locally identifiable modulo gauge* ***iff*** *$N \geq K + p - 1$.*

Proof. The rank formula is the kernel derivation plus Lemma W. *Sufficiency at $N \geq K + p - 1$ (ordinary local injectivity modulo gauge):* write $S = \mathbb{R}\mathbf{1} \oplus S_0$; the bookkeeping

$$S \cap W = \mathbb{R}\mathbf{1} \oplus (S_0 \cap W)$$

holds because for $x \in S\cap W$, $x = a\mathbf{1} + x_0$ with $x_0 \in S_0$ gives $x_0 = x - a\mathbf{1} \in W$. At maximal rank $\dim(S \cap W) = 1$, so $S_0 \cap W = 0$. Restrict Φ_M to the gauge slice $\mathbb{R}^K \times (s + S_0)$, of dimension $K + p - 1$: by (B.4) its differential at (T,s) has kernel $\cong S_0 \cap W = 0$, so the restricted map is an immersion there, hence (constant-rank/immersion theorem) a local embedding — locally injective on the slice. Any nearby alias is gauge-normalized into the slice by a factor c near 1, so ordinary local identifiability modulo gauge holds. *Necessity at $N \leq K + p - 2$:* Theorem B.7 below supplies a positive-dimensional manifold of exact aliases through (T,s) — genuine failure, not merely a rank diagnostic. ∎

Theorem B.7 (genuine failure below the wall; Theorem A′(i), below-wall part). *Fix $S \ni \mathbf{1}$, $K \geq 2$, $N \leq K + p - 2$. For every $(M,T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: through (T,s) passes a real-analytic manifold of exact aliases of dimension $K + p - N - 1 \geq 1$; every point with $d \neq 0$ is gauge-inequivalent to (T,s), and the aliased objects satisfy $T'(d) \to T$ as $d \to 0$.*

Proof. Let $P = \Pi_{W^\perp}$ be the orthogonal projection onto $W^\perp$ ($\dim W^\perp = N - K$ by full column rank) and define the real-analytic map $\varphi : S_0 \to W^\perp$, $\varphi(d) = P(e^d)$. Then $e^d \in W \Leftrightarrow \varphi(d) = 0$ and $\varphi(0) = P\mathbf{1} = 0$. The differential is $D\varphi_0 = P|_{S_0}$ ($\frac{d}{dt}\big|_0 e^{td} = d$ componentwise; P linear). By the bookkeeping $S \cap W = \mathbb{R}\mathbf{1} \oplus (S_0 \cap W)$ and Lemma W with $N \leq K + p - 2$ (the max in (B.3) is attained), $\dim(S_0 \cap W) = K + p - N - 1$, so

$$\operatorname{rank} D\varphi_0 = (p-1) - (K+p-N-1) = N - K = \dim W^\perp :$$

$D\varphi_0$ is surjective. By the (real-analytic) submersion/implicit-function theorem, $\varphi^{-1}(0)$ is, near 0, a real-analytic manifold of dimension $(p-1) - (N-K) = K + p - N - 1 \geq 1$. For each d in it, $e^d \in W$; with the left inverse $L = (M^\top M)^{-1}M^\top$ set

$$T'(d) \;:=\; L\,D_y\,e^d,$$

so that $MT'(d) = D_y e^d$ ($D_y e^d = D_{e^d} y \in \operatorname{col}(M)$ precisely when $e^d \in W$): $(T'(d), d)$ solves (B.1), $T'(0) = Ly = T$, and $T'(d) \to T$ by continuity. Every $d \neq 0$ lies in $S_0 \setminus \{0\}$, hence $d \notin \mathbb{R}\mathbf{1}$: a genuine non-gauge alias — (T,s) and $(T'(d), s-d)$ produce identical records but are gauge-inequivalent. ∎

b) Consistency check at $N = K$.: At $N = K$ (square, $\operatorname{rank} M = K$): $W = \mathbb{R}^N$, $W^\perp = 0$, $\varphi \equiv 0$, and the alias manifold is a full neighborhood of 0 in S_0 — ambiguity dimension $p - 1$, exactly Theorem A. There is no hidden $N \geq K + 1$ assumption, and no contradiction ($p \geq 2$ gives $K < K + p \leq 2K + p - 1$, so the square case lies strictly below both global thresholds).

c) What this does not claim.: The local iff is generic in (M,T) — a particular structured design must be checked separately. Local identifiability at $N \geq K + p - 1$ does not imply global uniqueness (far aliases at the boundary are exactly the open sharpness question, Sec. B.8). Nothing about the singular values of the Jacobian, i.e. conditioning.

H. B.7 The $K = 1$ stratum

Theorem B.8 ($K = 1$**: global identifiability; Theorem A′(iv)).** *Let $K = 1$. For every M, every $T \in \mathcal{U}_M$, and every $s \in S$, the pair (T,s) is globally identifiable modulo gauge, for every admissible N.*

Proof. $T \in \mathcal{U}_M$ means $T \neq 0$ and $m_n \neq 0$ for all n (scalars $m_n T \neq 0$). Any solution of (B.1) satisfies $e^{d_n} = m_n T'/(m_n T) = T'/T$ for every n, so d is constant and $c := T'/T = e^{d_1} > 0$ — the gauge. ∎

The local threshold at $K = 1$ reads $N \geq K + p - 1 = p$, which is **automatically satisfied** ($S \subseteq \mathbb{R}^N$ forces $p \leq N$) — the threshold is vacuous, not "inapplicable". Consequently neither $N \geq K + p$ nor $N \geq 2K + p - 1$ is necessary at $K = 1$: corner strata break naive universal-necessity claims, which is why all necessity statements in this appendix are restricted to $K \geq 2$.

I. B.8 Theorem A′ — the manuscript statement, and sharpness status

Theorem A′ (proved tall-design identifiability results). *Let $K \geq 1$, $N \geq K$, and let $S \leq \mathbb{R}^N$ be a fixed, known linear subspace of dimension $p \geq 2$ with $\mathbf{1} \in S$. For $M \in \mathbb{R}^{N\times K}$ define $\Phi_M(T,s) = D_{e^s}MT$ and $\mathcal{U}_M = \{T \in \mathbb{R}^K : (MT)_n \neq 0 \text{ for every } n\}$. Parameters are identified modulo the positive global-scale gauge $(T,s) \sim (cT,\ s - \log c\,\mathbf{1})$, $c > 0$.*

(i) ***(Generic local threshold and genuine failure below it,*** *$K \geq 2$.**)** There is a Lebesgue-null set $E_{\text{loc}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ such that for every $(M,T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: $\operatorname{rank} D\Phi_M(T,s) = \min\{N, K+p-1\}$. Consequently (T,s) is locally identifiable modulo gauge iff $N \geq K+p-1$. If $N \leq K + p - 2$ then a gauge-normalized neighborhood of (T,s) contains a real-analytic manifold of exact aliases of*

dimension $K + p - N - 1 \geq 1$, and every other sufficiently nearby point of it is gauge-inequivalent.

(ii) ***(Generic exact-identifiability sufficiency.)*** *If $N \geq K + p$, there is a Lebesgue-null set $E_{\mathrm{gen}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ such that for every $(M, T) \notin E_{\mathrm{gen}}$ and every $s \in S$: $D_{e^{s'}} MT' = D_{e^s} MT$ with $(T', s') \in \mathbb{R}^K \times S$ implies $T' = cT$, $s' = s - \log c\,\mathbf{1}$ for a unique $c > 0$. Exact identifiability holds simultaneously for every possible true drift $s \in S$. This statement is* ***not*** *uniform over all objects for one fixed M.*

(iii) ***(Uniform exact-identifiability sufficiency on the nonvanishing-carrier set.)*** *If $N \geq 2K + p - 1$, there is a Lebesgue-null set $E_{\mathrm{unif}} \subset \mathbb{R}^{N\times K}$ such that for every $M \notin E_{\mathrm{unif}}$, every $T \in \mathcal{U}_M$, every $s \in S$, and every $(T', s') \in \mathbb{R}^K \times S$: $D_{e^{s'}} MT' = D_{e^s} MT$ implies $T' = cT$, $s' = s - \log c\,\mathbf{1}$ for a unique $c > 0$. The same generic design works simultaneously for all nonvanishing-carrier objects and all true drifts in S.*

(iv) ***(Degenerate one-dimensional object stratum.)*** *If $K = 1$ then for every M and every $T \in \mathcal{U}_M$ and every $s \in S$, the pair (T, s) is globally identifiable modulo gauge, for every admissible N. Neither $N \geq K + p$ nor $N \geq 2K + p - 1$ is necessary when $K = 1$.*

Proof. (i) = Theorems B.6–B.7 (E_{loc} from Lemma W); (ii) = Theorem B.4; (iii) = Theorem B.5; (iv) = Theorem B.8 (which holds for every M, not just a.e.). ■

Sharpness status. Parts (ii) and (iii) are sufficient conditions, not proved if-and-only-if thresholds. For $K \geq 2$, the below-wall part of (i) proves generic exact failure whenever $N \leq K + p - 2$. The remaining necessity assertion for the $K + p$ bound is the boundary case $N = K + p - 1$: dimension counting predicts nonlocal aliases there, but dominance of the bad-pair projection has not been proved. Likewise, dimension counting predicts that a generic design fails uniform identifiability for some object when $N \leq 2K + p - 2$, but this necessity direction has not been proved in the intermediate regime. Thus $K + p$ and $2K + p - 1$ are conjecturally sharp for $K \geq 2$ under generic/nondegenerate drift geometry, not universal sharp thresholds for every fixed S. The $K = 1$ stratum explicitly rules out any universal necessity claim.

a) What this does not claim.: Necessity of the two global thresholds (see the sharpness paragraph); anything about deterministic structured designs (the ordered-Hadamard obstruction of Theorem A is unaffected, and no a.e. statement covers a fixed named design); anything about conditioning, noise robustness, or estimators — uniqueness at a threshold is compatible with arbitrarily bad stability; and no coverage of objects constrained to lower-dimensional sets (e.g. exactly sparse) — open constraint sets (e.g. positivity with nonvanishing carrier) inherit all a.e. statements by restriction. The Fig. 6 experiment (Sec. 9) is evidence for the *local rank wall* of part (i) — with the fixed-object slice covered by Remark B.4.2 — and probes algorithmic recovery; it neither tests nor proves the sharpness predictions.

J. B.9 Relation to the superseded incidence-variety route

Earlier versions of this appendix proved Theorem A′ through a collision incidence variety: a stratum-counting lemma over the gain-ratio manifold, a dominant-stratum rank hypothesis $\mathrm{rank}[M, -D_b M] = \min(N, 2K)$, a determinantal-transversality argument for the rank-drop locus, and two low-pass lemmas (a level-multiplicity bound for S_{LP} and a generic-rank lemma for $[M, -DM]$) feeding a three-level transfer proposition. **That route is superseded (R11), and its claims are withdrawn or replaced as follows.**

- The former "sharp iff" statements at $N \geq K+p$ (generic) and $N \geq 2K+p-1$ (uniform) are **withdrawn**: their necessity directions rested on unproved dominant-projection and determinantal-transversality assertions. The proved statements are the sufficiency directions, now established by Theorems B.4–B.5 above, together with the genuinely sharp *local* iff at $N = K+p-1$ (Theorem B.6) and genuine below-wall failure (Theorem B.7). Sharpness of the two global thresholds is a labeled prediction (Sec. B.8).
- The former uniform claim quantified over "every nonzero object"; that is **false** at the stated threshold (Remark B.5.1) and is replaced by uniformity over the nonvanishing-carrier set $\mathcal{U}_M$.
- The former route needed genericity of the drift family (or the low-pass transfer lemmas to emulate it). The parametric-transversality proof holds for **every fixed** $S \ni \mathbf{1}$, so the low-pass transfer machinery (level-multiplicity bound, generic-rank lemma, three-level transfer hierarchy, and the L_S multiplicity check) is no longer needed for Theorem A′ and is not restated here; the withdrawn material remains available in the archived drafts for provenance.
- The $K = 1$, $N = p = 2$ counterexample that broke the old claimed generic threshold is absorbed as the degenerate stratum, Theorem A′(iv).

a) What this section does not claim.: Nothing in the superseded route is relied upon anywhere in Secs. B.0–B.8; conversely, no assertion of the superseded route beyond the ones re-proved above should be quoted from earlier drafts.

Appendix C
Carrier statistics and the stationarity anchor

This appendix proves Proposition 1, formalizes condition (★) quantitatively, proves Proposition 2 (with counterexamples in both directions), and proves Theorem B. Throughout, $R_n = a_n B_n$ is the bucket record, $B_n = \langle I_n, T\rangle = \sum_{j=1}^{K} I_{n,j} T_j$ the carrier, $\ell_n = \log a_n$ the log-gain, S the drift class with $p = \dim S$, $S_1 = \sum_j T_j$, $S_2 = \|T\|_2^2$, $K_{\mathrm{eff}} = S_1^2/S_2$, $K_\infty = S_2/\|T\|_\infty^2$, $K_4 = S_2^2/\sum_j T_j^4$, and W denotes the window length of the blind estimator $\hat{g}_n = W^{-1}\sum_{k\in W_n} Y_k$ applied to $Y_n = \log R_n$. Asymptotic statements are for a triangular array indexed by (K, N): the object $T = T^{(K)}$ and pattern law may vary with K, and limits are taken along $K, N \to \infty$.

A. C.1 Proposition 1 — random-basis carrier statistics

Proposition 1 (carrier moments, stationarity, CLT, Berry–Esseen). *Assume:*

- ***(P1)*** *$T_j \geq 0$ for all j, $S_1 > 0$, $S_2 > 0$.*
- ***(P2)*** *The pattern entries $\{I_{n,j}\}$ are i.i.d. across n and j with mean $\mu_I > 0$ and variance $\sigma_I^2 > 0$; the entries are*

nonnegative (so $B_n \geq 0$ a.s., and $B_n > 0$ a.s. whenever $P(I=0)^{|\operatorname{supp} T|} = 0$).

- ***(P3)*** *(for the CLT) Spikiness $K_\infty \to \infty$ along the array, with the family $\{(I-\mu_I)^2\}$ uniformly integrable (automatic for a fixed pattern law).*
- ***(P4)*** *(for Berry–Esseen) $\nu_3 := \mathbb{E}|I-\mu_I|^3/\sigma_I^3 < \infty$.*

Then:

(i) Exact moments and stationarity. *$\mathbb{E}B_n = \mu_I S_1$, $\operatorname{Var} B_n = \sigma_I^2 S_2$, hence $\mathrm{CV}_B = (\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$. The sequence $\{B_n\}_{n\geq 1}$ is i.i.d., hence strictly stationary and white.*

(ii) CLT. *Under (P1)–(P3), $(B_n - \mu_I S_1)/(\sigma_I\sqrt{S_2}) \Rightarrow \mathcal{N}(0,1)$.*

(iii) Berry–Esseen. *Under (P4), for a universal constant C,*

$$\sup_x \left| P\Big(\tfrac{B_n-\mu_I S_1}{\sigma_I\sqrt{S_2}} \leq x\Big) - \Phi(x)\right| \;\leq\; C\,\nu_3 \frac{\sum_j |T_j|^3}{S_2^{3/2}} \;\leq\; C\,\nu_3\, K_\infty^{-1/2}.$$

The correct CLT small parameter is $K_\infty^{-1/2}$, ***not*** *K_{eff}^{-1}.*

(iv) Higher cumulants and the log carrier. *For $r \geq 2$ the cumulants are $\kappa_r(B_n) = \kappa_r(I)\sum_j T_j^r$, so the standardized law of B_n depends on T through* all *ratios $\sum_j T_j^r/S_2^{r/2}$, $r \geq 3$; these vanish as $K_\infty \to \infty$ but are not controlled by K_{eff} alone. Writing $B_n = \mu_I S_1(1+u_n)$ with $\mathbb{E}u_n = 0$, $\operatorname{Var} u_n = (\sigma_I/\mu_I)^2/K_{\text{eff}}$, a third-order Taylor expansion gives*

$$\mathbb{E}\log B_n = \log(\mu_I S_1) - \tfrac{1}{2}\operatorname{Var}(u_n) + O(\mathbb{E}|u_n|^3),$$
$$\operatorname{Var}(\log B_n) = \frac{(\sigma_I/\mu_I)^2}{K_{\text{eff}}} + O(\mathbb{E}|u_n|^3).$$

Proof. (i) Linearity of expectation and independence across j give the mean and (Bienaymé) the variance; independence across n gives the i.i.d. claim. (ii) Apply the Lindeberg–Feller CLT [Billingsley, *Probability and Measure*, Thm. 27.2] to the triangular array $X_j = T_j(I_{n,j} - \mu_I)$ with $s^2 = \sigma_I^2 S_2$. Since $|X_j| \leq \|T\|_\infty |I-\mu_I|$, the event $\{|X_j| > \varepsilon s\}$ is contained in $\{|I-\mu_I| > \varepsilon\sigma_I\sqrt{K_\infty}\}$, so the Lindeberg fraction is

$$\frac{1}{\sigma_I^2 S_2}\sum_j T_j^2\,\mathbb{E}\big[(I-\mu_I)^2\mathbf{1}\{|X_j|>\varepsilon s\}\big] \leq\; \frac{1}{\sigma_I^2}\,\mathbb{E}\big[(I-\mu_I)^2\mathbf{1}\{|I-\mu_I| > \varepsilon\sigma_I\sqrt{K_\infty}\}\big] \;\longrightarrow\; 0$$

by uniform integrability as $K_\infty \to \infty$. (iii) Esseen's inequality for sums of independent, non-identically distributed variables [Petrov, *Sums of Independent Random Variables*, Ch. V] gives the bound with third-moment ratio $\nu_3 \sum_j |T_j|^3/S_2^{3/2}$, and $\sum_j |T_j|^3 \leq \|T\|_\infty S_2$ yields the $K_\infty^{-1/2}$ form. (iv) Cumulant homogeneity ($\kappa_r(cX) = c^r\kappa_r(X)$) and additivity over independent summands give the cumulant formula. The log expansion is Taylor's theorem with Lagrange remainder applied to $\log(1+u)$ on the event $\{|u_n| \leq \frac{1}{2}\}$. ∎

Status of the log expansion (flagged). Step (iv) is a **controlled moment expansion, not a distribution-free theorem**: making the $O(\mathbb{E}|u_n|^3)$ remainder rigorous, and ensuring integrability of $\log B_n$ near $B_n = 0$, requires a high-probability bound $P(|u_n| > \frac{1}{2})$ decaying fast enough — e.g. sub-exponential pattern entries plus K_∞-driven concentration, or a strictly positive pattern floor $I \geq I_{\min} > 0$. This step is a moment calculation; it enters only through the variance scale $\sigma_{\text{abs}}^2 = \sigma_{\text{LR}}^2 \asymp 1/K_{\text{eff}}$ (per-frame, summable-absolute, and long-run variances all coincide for the i.i.d. carrier) entering rate constants, never in identifiability statements. Zero and near-zero buckets are excluded here; the calibrated soft-log $\psi_\alpha(c) = \log(c+\alpha)$ of Appendix D removes this restriction with a photon-dependent sensitivity factor.

a) What this does not claim.: The proposition does not say the carrier law is determined by (μ_B, σ_B) — that reduction holds only at the Gaussian-approximation level; the exact law feels every weighted cumulant of T. For gain recovery this residual object dependence enters $m_T = \mathbb{E}\log B_n$ only as a *time-constant* scalar, absorbed by the global-scale gauge. It also does not assert cross-row decorrelation or window-energy whitening for structured (SRHT) carriers — those require the Walsh-spectrum/K_4 conditions of Appendix E, where K_{eff} alone is again insufficient.

B. C.2 The quantitative stationarity condition (★)

Definition (★)(W,ϵ,δ). Let $\{B_n\}_{n\leq N}$ be the carrier and W_n length-W windows covering $\{1,\dots,N\}$. The carrier satisfies the *quantitative stationarity anchor* with tolerance ϵ and confidence $1-\delta$ if there exists a scalar $m_T \in \mathbb{R}$ (allowed to depend on T, not on n) such that

$$P\Big(\max_{1\leq n\leq N}\Big|W^{-1}\sum_{k\in W_n}\log B_k - m_T\Big| \leq \epsilon\Big) \;\geq\; 1-\delta.$$

We say **(★) holds asymptotically** if for every $\delta > 0$ there are windows $W = W_N \to \infty$, $W/N \to 0$, with $\epsilon = \epsilon_{W,N,\delta} \to 0$. The scalar m_T is exactly the quantity absorbed by the global-scale gauge $a \mapsto ca$, $T \mapsto T/c$ (which shifts $m_T \mapsto m_T - \log c$, $\ell \mapsto \ell + \log c$, leaving Y invariant).

C. C.3 Proposition 2 — (★) is a window-estimator criterion, not a universal one

Proposition 2. *Assume the drift is slow in the sense of hypothesis (B3) below, with $C_\beta L\,(W/N)^\beta \to 0$. Then:*

(a) Equivalence for the windowed corrector. *The blind windowed estimator $\hat{g}_n = W^{-1}\sum_{k\in W_n} Y_k$ is uniformly consistent for $\ell_n + m_T$ (in probability, uniformly over n)* ***if and only if*** *(★) holds asymptotically.*

(b) (★) is not necessary for identifiability. *There are designs violating (★) that are nonetheless identifiable: a tall design with $N \geq K + p$ is exactly algebraically identifiable modulo scale for a.e. design–object pair (M,T), simultaneously for every drift $s \in S$ (Theorem A′(ii); see Appendix B.4 and Remark B.4.1), with* no *stationarity whatsoever — the a.e.-(M,T) quantifier does not cover deterministic structured (e.g. ordered) designs, which must be checked individually (Sec. B.8); likewise, $K_0 \geq p-1$ known support zeros restore local identifiability under ordered Hadamard (see Appendix A).*

(c) (★) is not sufficient for algebraic identifiability. *There are settings where (★) holds yet exact separation of (a,T) is impossible: (c1) any* square *random design $N = K$ — Proposition 1 gives (★), yet Theorem A applies to every invertible realization, so a nonconstant $s \in S$ still maps (ℓ,T) to an exact alias; (c2) a uniform random re-ordering π of* *the* deterministic *ordered Hadamard record makes (★) hold without changing the coefficient multiset — hence without adding any algebraic information. Quantitatively, Serfling's inequality for sampling without replacement [31] gives, for the finite population $g_k = \log B_k$ with variance τ_g^2 and range R_g,*

$$P\Big(\Big|W^{-1}\sum_{i=n}^{n+W-1} g_{\pi(i)} - \bar{g}\Big| \geq t\Big) \;\leq\; 2\exp\Big[-\frac{Wt^2}{2\tau_g^2 + \frac{2}{3}R_g t}\Big],$$

so a union bound over the $\leq N$ windows yields (★)(W,ϵ,δ) whenever $\log N \ll W$ (and W remains below the drift coherence time). This device fails when some g_k are undefined (zero buckets) or the population log-variance/range is uncontrolled, and it does ***not*** *deliver the small carrier variance $\sigma^2_{\mathrm{LR}} \asymp 1/K_{\mathrm{eff}}$ of genuinely random nonnegative patterns.*

Proof. (a) Decompose exactly

$$\hat{g}_n - (\ell_n + m_T) \;=\; \underbrace{\Big(W^{-1}\sum_{k\in W_n} \ell_k - \ell_n\Big)}_{\text{drift bias}} + \underbrace{\Big(W^{-1}\sum_{k\in W_n} \log B_k - m_T\Big)}_{\text{carrier deviation}}.$$

The drift bias is $\leq C_\beta L\,(W/N)^\beta \to 0$ deterministically by (B3). Hence $\max_n |\hat{g}_n - (\ell_n + m_T)| \to 0$ in probability iff the maximal carrier deviation does — which is verbatim the asymptotic form of (★). Both implications are immediate from this identity. (b) Theorem A′ (see Appendix B) is proved without any distributional assumption on the carrier order; the support-zero criterion is Corollary 2 (see Appendix A). (c1) Theorem A (see Appendix A) holds for *any* invertible M, in particular a random draw, conditionally on the realization; (★), a statement about the temporal law, is unaffected. (c2) Serfling's bound plus the union over windows; the multiset $\{g_k\}$, hence every order-independent (algebraic) functional of the data, is invariant under π. ∎

a) What this does not claim.: Part (a) is an equivalence for *one estimator family* (windowed local averages of $\log R_n$; the same proof covers kernel weights of matching order). It is not a characterization of identifiability by all mechanisms — (b) and (c) are precisely the two failure modes of that over-reading. Nor does (★) by itself bound the *rate*: rates require the variance model of Theorem B, and reconstruction quality after gain correction additionally requires the corrected linear inverse problem to be well posed (see Appendix F).

D. C.4 Theorem B — statistical relative-gain identifiability

Theorem B. *Assume:*

- ***(B1) Signal model.*** *$Y_n = \log R_n = \ell_n + m_T + z_n$, $n = 1,\dots,N$, where $m_T = \mathbb{E}\log B_n$ is a time-constant scalar (finite by $B_n > 0$ a.s. and $\mathbb{E}|\log B_n| < \infty$), and $z_n = \log B_n - m_T$.*
- ***(B2) Carrier noise.*** *$\{z_n\}$ is centered and strictly stationary, and* either *(B2a) α-mixing with $\mathbb{E}|z_0|^{2+\eta} < \infty$ and $\sum_{h\geq 1}\alpha(h)^{\eta/(2+\eta)} < \infty$, which by Davydov's covariance inequality [32] gives absolutely summable autocovariances — write $\gamma(h) = \mathrm{Cov}(z_0, z_h)$ and $\sigma^2_{\mathrm{abs}} := \sum_{h\in\mathbb{Z}}|\gamma(h)| < \infty$ for the* ***summable-autocovariance constant****, the quantity entering every non-asymptotic variance bound below — and hence also a finite* ***signed long-run variance*** *$\sigma^2_{\mathrm{LR}} := \gamma(0) + 2\sum_{h\geq 1}\gamma(h)$ $(\leq \sigma^2_{\mathrm{abs}})$, which is the* asymptotic *variance of window means, $W\,\mathrm{Var}(W^{-1}\sum_{k\in W_n} z_k) \to \sigma^2_{\mathrm{LR}}$; the two constants play distinct roles and must not be conflated;* or *(B2b), for high-probability bounds, β-mixing with $\beta(k) \leq \beta_0\exp[-(k/b)^\kappa]$ and $\|z_n\|_{\psi_1} \leq M$.*
- ***(B3) Slow drift (normalized time).*** *$\ell \in S$ embeds in a normalized-time Hölder ball: $|\ell_j - \ell_k| \leq L\,(|j-k|/N)^\beta$, $\beta \in (0,2]$ — equivalently the per-frame constant $L_a = L\,N^{-\beta}$ of Appendix D — matched to the window order so the windowed bias obeys $|W^{-1}\sum_{k\in W_n}\ell_k - \ell_n| \leq C_\beta L\,(W/N)^\beta$ (for $\beta \in (1,2]$ this requires centered symmetric windows so the first-order bias cancels).*
- ***(B4) Window asymptotics.*** *Both $W = W_N \to \infty$* and *$W/N \to 0$; for fixed L the windowed drift bias $C_\beta L\,(W/N)^\beta \to 0$ then holds automatically (for a triangular array with drift constant $L = L_N$ growing with N, require additionally $L\,(W/N)^\beta \to 0$).*

Then:

(i) Pointwise consistency with rate. *Under (B2a), the* non-asymptotic *bound $\mathbb{E}\big[\hat{g}_n - (\ell_n + m_T)\big]^2 \leq C\big(L^2(W/N)^{2\beta} + \sigma^2_{\mathrm{abs}}/W\big) \to 0$ holds, the windowed bias being $O(L\,(W/N)^\beta)$; the* asymptotic *variance of the window mean is σ^2_{LR}/W (signed long-run variance, (B2a)). Under (B2b), with block length $q = \lceil b\,[\log(4W/\delta)]^{1/\kappa}\rceil$, with probability $\geq 1-\delta$,*

$$\big|\hat{g}_n - (\ell_n + m_T)\big| \;\leq C_\beta L\,(W/N)^\beta + C\Big[\sqrt{\tfrac{\nu_q^2\, q\log(4/\delta)}{W}} + \tfrac{Mq\log(4/\delta)}{W}\Big],$$

where ν_q^2 is a block-variance proxy bounded by a multiple of σ^2_{abs}; uniformity over $n \leq N$ follows with $\delta \mapsto \delta/N$.

(ii) Centered-gain recovery. *$\hat{\ell}^c_n := \hat{g}_n - N^{-1}\sum_m \hat{g}_m$ satisfies $\mathbb{E}\big[\hat{\ell}^c_n - (\ell_n - \bar{\ell})\big]^2 \leq 4C\big(L^2(W/N)^{2\beta} + \sigma^2_{\mathrm{abs}}/W\big)$; hence the relative gain $\ell_n - \bar{\ell}$ (equivalently a_n up to one multiplicative constant) is identifiable and consistently estimable.*

(iii) Gauge. *The scalar $m_T + \bar{\ell}$ is* not *identifiable: for every $c > 0$, the substitution $a \mapsto ca$, $T \mapsto T/c$ leaves the law of $\{Y_n\}$ exactly invariant while shifting $\ell \mapsto \ell + \log c$, $m_T \mapsto m_T - \log c$. This is precisely the constant direction $\mathrm{span}\{1\} \subset S$ of Theorem A, and it is the only ambiguity remaining.*

Proof. (i) Bias–variance split as in Proposition 2(a). Bias: (B3). Variance: for a stationary sequence, $\mathrm{Var}\big(W^{-1}\sum_{k\in W_n} z_k\big) = W^{-1}\sum_{|h|<W}(1 - |h|/W)\gamma(h) \leq W^{-1}\sum_h|\gamma(h)| = \sigma^2_{\mathrm{abs}}/W$ — the non-asymptotic bound, with the summable-autocovariance constant; as $W \to \infty$ the same expression converges to $\sigma^2_{\mathrm{LR}}/W\,(1 + o(1))$ with the *signed* long-run variance, which is the correct asymptotic constant and can be strictly smaller. Finiteness of both is (B2a)/Davydov [32]; $(x+y)^2 \leq 2x^2 + 2y^2$ combines bias and variance. For (B2b), partition the window into alternating blocks of length q; the

exponential β-mixing coupling replaces blocks by independent copies up to total-variation cost $W\beta(q)/q \leq \delta/2$ by the choice of q, and Bernstein's inequality for independent sub-exponential block sums (the Merlevède–Peligrad–Rio inequality [33] with the standard blocking scheme) gives the display. (ii) $\hat{\ell}_n^c - (\ell_n - \bar{\ell}) = [\hat{g}_n - (\ell_n + m_T)] - N^{-1}\sum_m[\hat{g}_m - (\ell_m + m_T)]$; Jensen bounds the second term's second moment by the maximal pointwise one, and $(x+y)^2 \leq 2x^2 + 2y^2$ gives the factor 4. (iii) Direct computation: $B_n(T/c) = B_n(T)/c$, so Y_n is unchanged term by term; conversely, consistency in (ii) implies identifiability of the estimand $\ell_n - \bar{\ell}$ (two parameter configurations inducing the same law of $\{Y_n\}$ would force the same limit), so the ambiguity is exactly one dimensional. ■

a) What this does not claim.: Theorem B is a **statistical, asymptotic** statement about *relative*-gain recovery in a triangular array; it is not exact finite-sample algebraic identifiability of (a, T) — Theorem A still holds verbatim for a square random design, and the two results are complementary, not in tension. It presupposes $B_n > 0$ and integrable logs: in photon-counting records with empty bins the theorem must be replaced by its calibrated soft-log variant (see Appendix D), where $\psi_\alpha(c) = \log(c + \alpha)$, the calibration curve m_α is inverted, and the same bias–variance structure holds with sensitivity κ_α and photon-starved variance $\sim 1/(W\bar{\lambda})$. The constants σ^2_{abs} and σ^2_{LR} are model inputs (they coincide for the i.i.d. carrier, where $\gamma(h) = 0$ for $h \neq 0$); identifying them with $(\sigma_I/\mu_I)^2/K_{\text{eff}}$ invokes the flagged expansion of Proposition 1(iv). Finally, consistent gain correction does not by itself guarantee good reconstruction — the corrected inverse problem must be independently well posed, and the propagation of the residual gain variance v through a basis with leverage B_L, floor C_0/N, and the ρ_{pair} flip boundary is the subject of Appendix F.

APPENDIX D
ESTIMATION RATES, MINIMAX LOWER BOUNDS, AND FISHER GEOMETRY

This appendix proves the rate and optimality statements of Sec. 6: the high-probability windowed-estimator bound (Theorem C's engine), its minimax optimality at the level of rates (Theorem D), the quotient-Fisher geometry underlying the identifiability dichotomy, and the low-photon soft-log rate. Notation is locked to the main text: $R_n = a_n B_n$, $\ell = \log a \in S$ with $\dim S = p$ and $\text{span}\{1\} \subset S$, $Y_n = \log R_n = \ell_n + m_T + z_n$ with $m_T = \mathbb{E}\log B_n$ a time-independent scalar, windows W, and the spread functionals K_{eff}, K_∞ as in Appendix C. The Hölder smoothness order is written $\beta \in (0, 2]$ with seminorm L_a (the drift constant of Theorem C); the symbol α is reserved for the soft-log offset $\psi_\alpha(c) = \log(c + \alpha)$.

Throughout, the *discrete Hölder ball* $H_\beta(L_a)$ is defined operationally: ℓ belongs to $H_\beta(L_a)$ relative to a window family $\{W_n\}$ of order β if the deterministic smoothing bias obeys

$$\Big| W^{-1} \sum_{k \in W_n} \ell_k - \ell_n \Big| \;\leq\; C_\beta\, L_a\, W^\beta \qquad \text{for all } n. \tag{D.1}$$

For $\beta \in (0, 1]$ a one-sided moving average of ℓ with modulus $|\ell_k - \ell_n| \leq L_a|k - n|^\beta$ satisfies (D.1). For $\beta \in (1, 2]$, (D.1) requires a *centered symmetric* window (or a kernel of order $\lfloor\beta\rfloor$), so that the first-order bias cancels and only curvature L_2 survives; a one-sided window does not attain $\beta > 1$.

A. D.1 Lemma D.1 (windowed gain-estimation rate under mixing)

a) Statement.: Assume:

1) *(Model)* $Y_n = \ell_n + m_T + z_n$, with $\{z_n\}$ strictly stationary and $\mathbb{E}z_n = 0$.
2) *(Mixing)* $\{z_n\}$ is β-mixing with $\beta(k) \leq \beta_0 \exp[-(k/b)^\kappa]$ for some $\beta_0, b > 0$, $\kappa > 0$.
3) *(Tails)* $\|z_n\|_{\psi_1} \leq M$ (sub-exponential).
4) *(Smoothness)* $\ell \in H_\beta(L_a)$ in the sense (D.1), with the window order matched to β as above.

Let $\hat{g}_n = W^{-1}\sum_{k \in W_n} Y_k$ and set $q = \lceil b\,[\log(4W/\delta)]^{1/\kappa}\rceil$. Then for each fixed n and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\big|\hat{g}_n - (\ell_n + m_T)\big| \;\leq\; C_\beta L_a W^\beta \\ \qquad + \; C\Big[\sqrt{\tfrac{\nu_q^2\, q\, \log(4/\delta)}{W}} \;+\; \tfrac{M\, q\, \log(4/\delta)}{W}\Big], \tag{D.2}$$

where $\nu_q^2 = \text{Var}\big(q^{-1/2}\sum_{i=1}^q z_i\big)$ is the block-variance proxy; when the autocovariances are absolutely summable, $\nu_q^2 \leq C'\sigma^2_{\text{abs}}$ with the summable-autocovariance constant $\sigma^2_{\text{abs}} = \sum_h |\gamma(h)|$ of Appendix C (hypothesis (B2a)) — the signed long-run variance $\sigma^2_{\text{LR}} \leq \sigma^2_{\text{abs}}$ of Appendix C is the asymptotic variance of window means and plays no role in these non-asymptotic bounds. Uniformity over $n = 1, \ldots, N$ follows by replacing δ with δ/N (a $\log N$ inflation inside the brackets).

(Expectation version, weaker hypotheses.) If instead $\{z_n\}$ is only α-mixing with $\sum_{h \geq 1} \alpha(h)^{\eta/(2+\eta)} < \infty$ and $\|z_0\|_{2+\eta} < \infty$ for some $\eta > 0$, then

$$\mathbb{E}\big[\hat{g}_n - (\ell_n + m_T)\big]^2 \;\leq\; C\, L_a^2 W^{2\beta} + C\, \sigma^2_{\text{abs}}/W. \tag{D.3}$$

b) Proof.: Bias. $\hat{g}_n - (\ell_n + m_T) = \big[W^{-1}\sum_k \ell_k - \ell_n\big] + W^{-1}\sum_k z_k$; the first bracket is bounded by $C_\beta L_a W^\beta$ by (D.1).

Fluctuation, high probability. Partition the window W_n into $m = \lceil W/q\rceil$ consecutive blocks of length q and separate them into odd- and even-indexed families. By Berbee's coupling lemma [34] (see also Rio [35]), on an enlarged probability space each family can be replaced by a sequence of *independent* blocks with the same marginals, at total-variation cost at most $m\,\beta(q) \leq \beta_0\, W \exp[-(q/b)^\kappa]$. The choice $q = \lceil b[\log(4W/\delta)]^{1/\kappa}\rceil$ makes this cost at most $\delta/2$ (adjusting β_0 into C). On the coupling event, each family is a sum of independent block sums $S_j = \sum_{i \in \text{block } j} z_i$ with $\|S_j\|_{\psi_1} \leq CqM$ (triangle inequality for the ψ_1 norm) and $\text{Var}(S_j) = q\,\nu_q^2$. Bernstein's inequality for sums of independent sub-exponential variables (e.g. Vershynin [36], Thm. 2.8.1) applied to each family, with a union bound over the two families (whence the $4/\delta$), yields the bracketed term in (D.2) with the characteristic Gaussian-plus-linear tail: a $\sqrt{\nu_q^2 q \log(4/\delta)/W}$ moderate-deviation term and an $Mq\log(4/\delta)/W$ heavy-tail correction.

When $\sum_h |\operatorname{Cov}(z_0, z_h)| < \infty$, $\nu_q^2 = \operatorname{Var}(z_0) + 2\sum_{h=1}^{q-1}(1 - h/q)\operatorname{Cov}(z_0, z_h) \le C'\sigma_{\rm abs}^2$.

Fluctuation, expectation. Davydov's covariance inequality [32] gives $|\operatorname{Cov}(z_0, z_h)| \le 8\,\alpha(h)^{\eta/(2+\eta)}\|z_0\|_{2+\eta}^2$; the summability hypothesis makes the autocovariance series absolutely summable, so $\operatorname{Var}(W^{-1}\sum_{k\in W_n} z_k) \le \bar\sigma^2/W$ with $\bar\sigma^2 \le C\sigma_{\rm abs}^2$. Combining with the squared bias via $(x+y)^2 \le 2x^2 + 2y^2$ gives (D.3). ■

Corollary D.2 (optimized rate = Theorem C's high-photon case). *Minimizing the right side of (D.3) over W gives*

$$W^* \asymp \left(\sigma_{\rm abs}^2/L_a^2\right)^{1/(2\beta+1)},\qquad \mathrm{MSE}^* \asymp L_a^{2/(2\beta+1)}\,\sigma_{\rm abs}^{4\beta/(2\beta+1)}, \tag{D.4}$$

valid when $W^* \in [1, N]$ and W^* is shorter than the gain coherence time; otherwise the boundary window $W = 1$ or $W = N$ applies. For $\beta = 1$ with per-frame drift scale $L_1 \asymp s\rho_{\rm pair}$ (adjacent-frame decorrelation convention — not *$\rho_{\rm bw}$), this is $\mathrm{MSE}^* \asymp \sigma_{\rm abs}^{4/3}(s\rho_{\rm pair})^{2/3}$; for random carriers the carrier is white, $\sigma_{\rm abs}^2 = \sigma_{\rm LR}^2 \approx (\sigma_I/\mu_I)^2/K_{\rm eff}$ under the spikiness condition $K_\infty \to \infty$ of Prop. 1.*

c) What this does not claim.: The constants C, C_β are not tracked and depend on the mixing constants (β_0, b, κ) and M. The bound is for the *relative* gain $\ell_n + m_T$; the scalar m_T + gauge(ℓ) is not estimable (Theorem B). Nothing is claimed for $\beta > 2$ (would need local polynomials), for non-stationary carriers, or about *adaptive* window selection when (β, L_a) are unknown. The step $\nu_q^2 \lesssim \sigma_{\rm abs}^2$ requires summable autocovariances, which the stated mixing plus moment hypotheses supply.

B. D.2 Theorem D (minimax lower bounds; rate level, constants open)

a) Statement.: (Hölder / Assouad–two-point.) Consider the i.i.d. Gaussian submodel $z_n \sim \mathcal{N}(0, \sigma^2)$ — a member of the model class of Lemma D.1. Then for every n interior to the design,

$$\inf_{\hat\ell}\ \sup_{\ell\in H_\beta(L_a)}\ \mathbb{E}\big(\hat\ell_n - \ell_n\big)^2 \ \ge\ c_\beta\, L_a^{2/(2\beta+1)}\,\sigma^{4\beta/(2\beta+1)}, \tag{D.5}$$

the infimum over all measurable estimators of the gauge-fixed log-gain. In normalized continuous time (N samples on $[0, 1]$, bandwidth $h = W/N$) the equivalent form is $\inf\sup \mathbb{E}(\hat\ell(t) - \ell(t))^2 \ge c\,(\sigma^2/N)^{2\beta/(2\beta+1)} L^{2/(2\beta+1)}$.

(Sobolev / van Trees–Pinsker.) In the Gaussian sequence formulation — expand $\ell(t) = \sum_j \theta_j\varphi_j(t)$ in a Fourier basis and impose the Sobolev ellipsoid $\sum_j j^{2\beta}\theta_j^2 \le L^2$ — the integrated risk obeys

$$\inf_{\hat\ell}\ \sup_{\|\ell\|_{H^\beta}\le L} \frac{1}{N}\sum_n \mathbb{E}(\hat\ell_n - \ell_n)^2 \ \ge\ c_\beta\, L^{2/(2\beta+1)}(\sigma^2/N)^{2\beta/(2\beta+1)}, \tag{D.6}$$

with the gauge direction (the constant mode, confounded with m_T) excluded from the risk. Consequently the windowed estimator of Lemma D.1 is **minimax-rate optimal** over $H_\beta(L_a)$ for $\beta \le 1$ with one-sided windows and for $\beta \le 2$ with centered symmetric windows. **Sharp minimax constants are open** (this is Theorem D of the main text).

b) Proof.: Two-point (pointwise). Fix n_0 and let ϕ be a fixed C^∞ bump, $\operatorname{supp}\phi \subset [-1, 1]$, $\phi(0) = 1$, $\|\phi\|_{H_\beta} \le 1$. Take the hypotheses $\ell^{(0)} \equiv 0$ and $\ell_k^{(1)} = L_a W_0^\beta\phi\big((k - n_0)/W_0\big)$; both lie in $H_\beta(L_a)$ (up to an absolute constant absorbed into c_β). Since the noise is i.i.d. Gaussian, the Kullback–Leibler divergence between the two data laws is $\mathrm{KL} = \frac{1}{2\sigma^2}\sum_k(\ell_k^{(1)})^2 \le C\,L_a^2 W_0^{2\beta+1}/\sigma^2$. Choosing $W_0 \asymp (\sigma^2/L_a^2)^{1/(2\beta+1)}$ makes $\mathrm{KL} \le c < \infty$, and the two hypotheses are separated at n_0 by $|\ell_{n_0}^{(1)} - \ell_{n_0}^{(0)}| = L_a W_0^\beta$. Le Cam's two-point lemma [25] then gives (D.5). The nuisance scalar m_T only enlarges the model, so the bound holds a fortiori; the bumps can be taken mean-centered over the record so that the two hypotheses differ in the gauge-invariant component $\ell - \bar\ell$ itself. For the integrated version, tile $[1, N]$ with $\sim N/W_0$ disjoint bumps carrying independent signs and apply Assouad's lemma [25].

van Trees / Pinsker. In the sequence model $y_j = \theta_j + (\sigma/\sqrt{N})\epsilon_j$ (exact for equispaced discrete-Fourier regression with white Gaussian noise), place independent priors $\theta_j \sim \mathcal{N}(0, \tau_j^2)$ truncated inside the ellipsoid. The van Trees inequality [26] bounds the Bayes risk of each coordinate by the reciprocal of (data Fisher + prior Fisher): BayesRisk $\ge \sum_j(N/\sigma^2 + \tau_j^{-2})^{-1}$, the constant mode omitted (gauge). Maximizing the right side over prior spectra supported in the ellipsoid is Pinsker's calculation [27] and yields the scaling (D.6). Since minimax risk dominates Bayes risk, (D.6) follows. ■

c) What this does not claim.: (i) Constants: neither c_β nor a Pinsker-type sharp constant is claimed for our model; the exact Pinsker constant transfers only where the acquisition is genuinely the Gaussian sequence model — for mixing, non-Gaussian carriers the statement is at rate level only (a Brown–Low-type asymptotic-equivalence argument [37] would be needed for more, and is not attempted). (ii) The lower bound is proven inside the i.i.d. Gaussian *submodel*; this suffices for minimaxity of the full class (sup over the class dominates sup over the submodel, while the upper bound (D.3) holds over the full class), but no lower bound is claimed *within* every fixed mixing law. (iii) No adaptivity claim: rate optimality is for known (β, L_a).

C. D.3 Proposition D.4 (quotient Fisher geometry: singularity ⇔ the Theorem-A ambiguity)

a) Statement.: (a) Linear-Gaussian window model. Let $\ell = U\theta$ with $U \in \mathbb{R}^{N\times p'}$, and observe $Y = U\theta + m\mathbf{1} + z$, $z \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$ known, $m \in \mathbb{R}$ the unknown carrier log-mean (confounded with global gain scale). The Fisher information for θ after eliminating the nuisance m is

$$I_\theta \;=\; U^\top\Sigma^{-1/2}P_\perp\Sigma^{-1/2}U,\qquad P_\perp = I - \frac{\Sigma^{-1/2}\mathbf{1}\mathbf{1}^\top\Sigma^{-1/2}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}. \tag{D.7}$$

For a linear functional $L\theta$: if $\operatorname{row}(L) \subseteq \operatorname{range}(I_\theta)$, every unbiased estimator obeys $\operatorname{Cov}(L\hat\theta) \succeq L\,I_\theta^+ L^\top$ (Moore–Penrose CRB for singular information [38], [39]); if L has a nonzero component along $\ker I_\theta$, **no unbiased estimator of $L\theta$ exists**

and the CRB is infinite — the functional is not locally estimable.

(b) Quotient statement. Let Θ be the parameter manifold, G the global-scale group $(a,T) \mapsto (ca,T/c)$, and P_θ a dominated model differentiable in quadratic mean. Define the quotient score operator $S_\theta : T_{[\theta]}(\Theta/G) \to L_0^2(P_\theta)$ and $I_\theta = S_\theta^* S_\theta$. Then: $I_\theta \succ 0$ on $T_{[\theta]}(\Theta/G)$ **iff** $[\theta] \mapsto P_\theta$ is an immersion, i.e. iff the model is *locally differentially identifiable* modulo scale. Any C^1 ambiguity curve $[\theta(t)]$ with $P_{\theta(t)} = P_{\theta(0)}$ satisfies $S_\theta\theta'(0) = 0$, hence produces a Fisher null direction. For the Theorem-A ambiguity this implication is *exact*: the explicit data-preserving path $\ell' = \ell + ts$, $T' = M^{-1}(c\odot e^{-ts})$, $s \in S\setminus\mathrm{span}\{1\}$, is a genuine ambiguity curve for a square invertible design, so the Fisher matrix in the full (ℓ,T) parameterization is singular along its tangent, and the CRB is $+\infty$ for every functional that varies along it.

(c) Local-window corollary. For a window over which ℓ is approximately constant, the Gaussian CRB reduces to $\mathrm{Var}(\hat{\ell}_n) \geq \sigma_z^2/W_{\text{eff}}$ with $W_{\text{eff}} = W/(1+2\sum_h \rho_z(h))$ under weak dependence (equivalently, σ_z^2 replaced by the signed long-run variance σ_{LR}^2, which is the exact asymptotic constant here) — the Fisher-side counterpart of the σ_{abs}^2/W upper bound in (D.3).

b) Proof.: (a) The joint Fisher matrix of (θ, m) in the Gaussian model is the block matrix $\begin{pmatrix} U^\top\Sigma^{-1}U & U^\top\Sigma^{-1}\mathbf{1} \\ \mathbf{1}^\top\Sigma^{-1}U & \mathbf{1}^\top\Sigma^{-1}\mathbf{1}\end{pmatrix}$. Nuisance elimination is the Schur complement with respect to the m-block [40], which is exactly (D.7). The pseudoinverse CRB for singular FIM is Stoica–Marzetta [39]. The infinite-CRB claim is cleanest without CRB machinery: a direction $u \in \ker I_\theta$ that is tangent to an exact ambiguity curve leaves the data law invariant, so any estimator has *identical* distribution under θ and $\theta + tu$; unbiasedness for a functional with $Lu \neq 0$ would force $L\theta = L(\theta + tu)$, a contradiction — hence no unbiased finite-variance estimator exists.

(b) Differentiability in quadratic mean makes S_θ well defined and $I_\theta = S_\theta^* S_\theta$ the pullback metric [van der Vaart, *Asymptotic Statistics*, Ch. 7]. $I_\theta \succ 0$ iff S_θ is injective iff the differential of $[\theta] \mapsto \sqrt{dP_\theta}$ is injective, i.e. an immersion. The ambiguity-curve implication is the chain rule: $t \mapsto P_{\theta(t)}$ constant $\Rightarrow$ its L^2 derivative $S_\theta\theta'(0)$ vanishes. For Theorem A the curve is explicit (see Appendix A), so no genericity is needed. ∎

c) What this does not claim.: The converse direction — Fisher singularity implying an actual *ambiguity manifold* — requires constant-rank (or analytic) regularity so that the rank defect integrates (rank theorem); Fisher singularity by itself only defeats *local differential* identifiability. Nonsingular Fisher does not exclude isolated *global* aliases. Part (a) assumes Gaussian noise with known Σ; part (c) is a Gaussian-model computation used as an order-of-magnitude floor, not a bound proven for the actual log-carrier distribution. No efficiency claim is made for the windowed estimator beyond rate (constants open, Theorem D).

D. D.4 Theorem C (low-photon calibrated soft-log rate and the $1/(W\bar{\lambda})$ law)

a) Statement (exact finite-window identity with known carriers, weighted form).: Let counts follow the calibrated model $C_n \mid \ell_n \sim \mathrm{Pois}(\lambda_n)$, $\lambda_n = \Lambda_0 e^{\ell_n} b_n + d$, with $d \geq 0$ dark counts and $\{b_n\}$ the *known* per-frame carrier intensities ($b_n > 0$; in the Fig. 7 protocol these are *oracle* values supplied from the simulation truth — a flat-field-calibrated benchmark, not programmed design intensities — and $d = 0$ — Remark D.4.1; the unknown-random-carrier variant is Remark D.4.3). Fix $\alpha > 0$, let $\psi_\alpha(c) = \log(c+\alpha)$, and let $m_\alpha(\lambda) = \mathbb{E}[\psi_\alpha(C)]$ for $C \sim \mathrm{Pois}(\lambda)$ — the homogeneous calibration curve, strictly increasing. For a fixed window n, let $\mathcal{I}_n$ be the set of *distinct* frame indices used by the estimator and let $w_{nk} \geq 0$, $\sum_{k\in\mathcal{I}_n} w_{nk} = 1$, be its weights (for the implementation's windows, $\mathcal{I}_n = W_n$ and $w_{nk} = 1/W$). If a padding convention repeats a frame, all repeated occurrences are first aggregated into its single weight w_{nk}. Put

$$q_k(t) = m_\alpha(\Lambda_0 e^t b_k + d),$$
$$v_k(t) = \mathrm{Var}\big[\psi_\alpha(\mathrm{Pois}(\Lambda_0 e^t b_k + d))\big],$$
$$M_n(t) = \sum_{k\in\mathcal{I}_n} w_{nk} q_k(t), \qquad y_n = \sum_{k\in\mathcal{I}_n} w_{nk}\psi_\alpha(C_k),$$

and define

$$B_n = \sum_{k\in\mathcal{I}_n} w_{nk}\{q_k(\ell_k) - q_k(\ell_n)\}, \qquad V_n = \sum_{k\in\mathcal{I}_n} w_{nk}^2 v_k(\ell_k).$$

Fix a compact operating range $\Theta = [\theta_{\min}, \theta_{\max}]$ (in the implementation, the safe tabulated calibration range) and let

$$\underline{\kappa}_n = \inf_{t\in\Theta} M_n'(t) > 0, \qquad D_\Theta = \theta_{\max} - \theta_{\min}.$$

The estimator is defined *with the clamp*:

$$\hat{\theta}_n \;=\; \Pi_\Theta\big[M_n^{-1}(y_n)\big], \tag{D.8a}$$

where Π_Θ clips to Θ and M_n^{-1} is the generalized monotone inverse, returning the corresponding endpoint of Θ when y_n falls outside $M_n(\Theta)$ (an all-zero window has $y_n = \log\alpha$ and returns $\theta_{\min}$; without the clamp the inverse would diverge to $-\infty$ there — the clamp is part of the estimator, not an implementation afterthought). Assume:

1) *(Operating range; per-frame sensitivities and noise)* for every k and every $\theta \in \Theta$, $\lambda_k(\theta) = \Lambda_0 e^\theta b_k + d \in [\lambda_-, \lambda_+] \subset (0,\infty)$; consequently the per-frame sensitivity bound $\overline{\kappa}_k = \sup_{t\in\Theta} q_k'(t)$ and the per-frame noise $v_k(t)$ are finite, with $v_k(t) \leq \bar{\sigma}_\alpha^2 := \sup_{\lambda\in[\lambda_-,\lambda_+]} \mathrm{Var}[\psi_\alpha(\mathrm{Pois}(\lambda))] < \infty$, $\kappa_{\max} := \sup_k \overline{\kappa}_k < \infty$, and $\underline{\kappa}_n > 0$;
2) *(Smoothness)* $\ell \in H_\beta(L_a)$ with the *pointwise* modulus $|\ell_k - \ell_n| \leq L_a|k-n|^\beta$ and $\beta \in (0,1]$ (the paper's operative case is $\beta = 1$; the precise obstruction above one and its partial repair are Remark D.4.2);
3) *(Margin)* $\ell_n \in \Theta$ for all n, with margin $\Delta := \min_n \mathrm{dist}(\ell_n, \partial\Theta) > 0$.

Then, conditionally on the gain path and known carriers,

$$\mathrm{Var}\Bigg[\sum_{k\in\mathcal{I}_n} w_{nk}\{\psi_\alpha(C_k) - q_k(\ell_k)\}\Bigg] = V_n$$

exactly, and no mixing or stationarity assumption is required. If $\mathcal{E}_n = \{y_n \in M_n(\Theta)\}$, then

$$\mathbb{E}(\hat\theta_n - \ell_n)^2 \le \frac{B_n^2 + V_n}{\underline{\kappa}_n^2} + D_\Theta^2\,\mathbb{P}(\mathcal{E}_n^c). \tag{D.8}$$

Moreover, the pointwise modulus gives

$$|B_n| \le L_a \sum_{k\in\mathcal{I}_n} w_{nk}\overline{\kappa}_k|k-n|^\beta.$$

For an unpadded average of W distinct frames, $w_{nk} = 1/W$ and hence

$$V_n = \frac{\bar v_{n,\alpha}}{W}, \qquad \bar v_{n,\alpha} = \frac{1}{W}\sum_{k\in W_n} v_k(\ell_k) \le \bar\sigma_\alpha^2,$$

while $|B_n| \le C\,\kappa_{\max} L_a W^\beta$, so (D.8) specializes to

$$\begin{aligned}\mathbb{E}\big(\hat\theta_n - \ell_n\big)^2 \ &\le C\Big(\tfrac{\kappa_{\max}}{\underline{\kappa}_n}\Big)^2 L_a^2 W^{2\beta}\\ &\quad + \frac{\bar\sigma_\alpha^2}{\underline{\kappa}_n^2\,W} \ + \ D_\Theta^2\,\mathbb{P}(\mathcal{E}_n^c),\end{aligned}$$

with $\mathcal{E}_n$ the no-clamp event; the clamp term is bounded in Remark D.4.4 and is of the same $1/W$ order (or exponentially small under a residual margin), so for fixed D_Θ/Δ it is absorbed into the constants. Optimizing over W,

$$\mathrm{MSE}_\alpha^* \ \le \ C\,\underline{\kappa}^{-2}\left[\kappa_{\max} L_a\right]^{2/(2\beta+1)} \bar\sigma_\alpha^{4\beta/(2\beta+1)} \tag{D.9}$$

— Theorem C of the main text, with $\underline{\kappa} := \inf_n \underline{\kappa}_n$. **The loss throughout is the log-domain squared error**: $\hat\theta_n - \ell_n$ is an increment of log-gain (equivalently log-intensity), the record-mean centering fixes the additive gauge, and this is the metric in which the Fisher comparison below is stated (and which Fig. 7 reports). In the photon-starved regime the variance-to-sensitivity ratio obeys the window-average law $V_n/M_n'(\ell_n)^2 = \{1 + O_\alpha(\varepsilon)\}/(W\bar\lambda_n)$ (low-photon corollary below), which is minimax-*order* sharp *in this log-domain loss at* $d = 0$ by the gauge-invariant lower bound following the proof. Zeros only reduce Fisher information; with the clamp (D.8a) they never make the estimator diverge.

b) Proof.: Sensitivity and variance of the soft-log. The Poisson derivative identity $\frac{d}{d\lambda}\mathbb{E}f(C) = \mathbb{E}[f(C{+}1) - f(C)]$ (differentiate $\sum_c e^{-\lambda}\lambda^c f(c)/c!$ term by term) gives

$$\kappa_\alpha(\lambda) := \frac{d}{d\log\lambda}\mathbb{E}\log(C+\alpha) = \lambda\,\mathbb{E}\log\frac{C+1+\alpha}{C+\alpha},$$

with the expansions $\kappa_\alpha(\lambda) = 1 + O(1/\lambda)$ for $\lambda \gg 1$ and $\kappa_\alpha(\lambda) = c_\alpha\lambda + O(\lambda^2)$ for $\lambda \ll 1$, where $c_\alpha = \log(1+1/\alpha)$. Since $q_k'(t) = \kappa_\alpha(\lambda_k(t)) \cdot \Lambda_0 e^t b_k/\lambda_k(t)$ $(= \kappa_\alpha(\lambda_k(t))$ at $d = 0)$, hypothesis 1 holds on any compact operating range. No uniform identification $\underline{\kappa}_n \asymp c_\alpha\bar\lambda_n$ is claimed: under unrestricted heteroscedastic carriers the infimum of M_n' over Θ can sit far below $c_\alpha\bar\lambda_n$ (two-frame example $\lambda_1 = \varepsilon^2$, $\lambda_2 = 2\varepsilon - \varepsilon^2$), so $\underline{\kappa}_n$-based displays are conservative bounds only; the correct low-photon law is the window-average corollary below. The Poisson Poincaré inequality [41], [42], $\mathrm{Var}\,f(C) \le \lambda\,\mathbb{E}[(f(C{+}1) - f(C))^2]$, yields $v_k = O(1/\lambda)$ at high λ and $v_k = c_\alpha^2\lambda + O(\lambda^2)$ at low λ, whence $\bar\sigma_\alpha^2 < \infty$ on $[\lambda_-, \lambda_+]$.

Risk bound. Write

$$\begin{aligned} y_n - M_n(\ell_n) &= B_n + Z_n,\\ Z_n &= \sum_{k\in\mathcal{I}_n} w_{nk}\{\psi_\alpha(C_k) - q_k(\ell_k)\}.\end{aligned}$$

The variables in Z_n are independent and centered conditionally on the gain path and known carriers (Poisson counts are independent across frames given their intensities — the carrier is known, so no carrier randomness remains), so $\mathbb{E}Z_n = 0$ and $\mathrm{Var}(Z_n) = V_n$ *exactly* — a finite-window identity requiring no mixing and no long-run variance. On $\mathcal{E}_n$, the mean value theorem and $M_n' \ge \underline{\kappa}_n$ imply

$$|\hat\theta_n - \ell_n| \le \underline{\kappa}_n^{-1}|B_n + Z_n|,$$

and clipping to $\Theta \ni \ell_n$ can only decrease the error. On $\mathcal{E}_n^c$, both $\hat\theta_n$ and ℓ_n belong to Θ, so the error is at most D_Θ. Taking expectations and using $\mathbb{E}(B_n + Z_n)^2 = B_n^2 + V_n$ gives (D.8). The bound on B_n follows by applying the mean value theorem to each q_k and then the pointwise modulus; for equal weights the triangle inequality gives $|B_n| \le C\kappa_{\max} L_a W^\beta$ (valid for the pointwise modulus with $\beta \le 1$; no cancellation across the window is invoked — cf. Remark D.4.2), and the standard bias–variance optimization gives (D.9). ■

c) Low-photon corollary (window-average sensitivity).: For an unpadded equal-weight window, $d = 0$, fixed α, and $\max_{k\in W_n}\lambda_k(\ell_n) \le \varepsilon$, the per-frame expansions of the proof average into

$$M_n'(\ell_n) = c_\alpha\bar\lambda_n\{1 + O_\alpha(\varepsilon)\}, \qquad V_n = \frac{c_\alpha^2\bar\lambda_n}{W}\{1 + O_\alpha(\varepsilon)\},$$

$$\frac{V_n}{M_n'(\ell_n)^2} = \frac{1 + O_\alpha(\varepsilon)}{W\bar\lambda_n}, \qquad \bar\lambda_n = \frac{1}{W}\sum_{k\in W_n}\lambda_k(\ell_n),$$

with *no carrier-comparability assumption*: both the sensitivity and the variance linearize frame by frame in λ_k and are then averaged, so heteroscedastic carriers are covered. These expansions are stated for fixed α; $c_\alpha = \log(1{+}1/\alpha) \to \infty$ as $\alpha \to 0$, so the fixed-α small-λ theorem must not be read as a triangular $\alpha \to 0$ asymptotic. This window-average law is the $1/(W\bar\lambda)$ statement quoted in the main text.

d) Gauge-invariant low-photon lower bound.: The loss corresponding to the unidentifiable global scale is

$$\begin{aligned} d_{\mathrm{quot}}^2(\hat g, g) &= \inf_{a\in\mathbb{R}} \frac{1}{N}\|\log\hat g - \log g - a\mathbf{1}\|_2^2\\ &= \frac{1}{N}\|P_N(\log\hat g - \log g)\|_2^2, \quad P_N = I - \tfrac{1}{N}\mathbf{1}\mathbf{1}^\top.\end{aligned}$$

This is the centered-log-gain loss reported in Fig. 7. Multiplying either gain profile by an arbitrary positive scalar leaves the loss unchanged.

For the carrier-averaged model, let B have a parameter-independent known law and, conditionally on B, let

$$C_k \sim \mathrm{Pois}(s_k(\theta, B_k) + d), \qquad s_k(\theta, B_k) = \Lambda_0 e^\theta B_k,$$

independently across frames. The complete-data Fisher information for a common log-gain parameter is

$$I_{\mathrm{comp}}(\theta) = \sum_{k=1}^W \mathbb{E}_B\frac{s_k(\theta, B_k)^2}{s_k(\theta, B_k) + d}.$$

The observed mixed-Poisson score is the conditional expectation of the complete-data score, so $I_{\mathrm{obs}}(\theta) \le I_{\mathrm{comp}}(\theta)$. For $d = 0$ this becomes

$$I_{\mathrm{comp}}(\theta) = \sum_{k=1}^{W} \mathbb{E}_B s_k(\theta, B_k) = W\bar{\lambda}.$$

A lower bound valid for biased estimators follows directly from Le Cam's method [25]. Let $v \perp \mathbf{1}$ be a bounded contrast supported on $O(W)$ frames and compare $\ell^{(0)} = \ell_0$ with $\ell^{(1)} = \ell_0 + \delta v$. Conditional on the carrier and with $d = 0$,

$$\begin{aligned}\mathrm{KL}(P_0(\cdot \mid B), P_1(\cdot \mid B)) &= \sum_k \lambda_{k,0}\{e^{\delta v_k} - 1 - \delta v_k\} \\ &\le C\delta^2 \sum_k \lambda_{k,0} v_k^2.\end{aligned}$$

Marginalization over the unobserved carrier cannot increase KL, hence

$$\mathrm{KL}(P_0^C, P_1^C) \le C\delta^2 W\bar{\lambda}.$$

Choosing $\delta^2 \asymp \min\{\delta_0^2, (W\bar{\lambda})^{-1}\}$ and applying Le Cam's lemma gives the local quotient-risk bound

$$R_{\mathrm{minimax}} \ge c \min\left\{\delta_0^2, \frac{1}{W\bar{\lambda}}\right\}.$$

Thus the $1/(W\bar{\lambda})$ law is a local-minimax statement in the regime $W\bar{\lambda} \to \infty$; for bounded total information the risk saturates at the squared diameter of the local parameter neighborhood.

Taking a mean-zero $H_\beta(L_a)$ bump of width W and amplitude $\delta \asymp L_a W^\beta$, $0 < \beta \le 1$, gives $\mathrm{KL} \lesssim L_a^2 \bar{\lambda} W^{2\beta+1}$. The choice $W_* \asymp (L_a^2\bar{\lambda})^{-1/(2\beta+1)}$ therefore yields

$$\inf_{\hat{\ell}} \sup_{\ell \in H_\beta(L_a)} \mathbb{E}(\hat{\ell}_n^c - \ell_n^c)^2 \ge c_\beta L_a^{2/(2\beta+1)} \bar{\lambda}^{-2\beta/(2\beta+1)},$$

with the usual boundary regimes when $W_* \notin [1, N]$. This is the direct Poisson counterpart of the upper rate (D.9), and it remains valid after averaging over an unobserved carrier because marginalization only decreases KL; in particular it does *not* route through Theorem D's Gaussian bound (a Gaussian model is not a submodel of the mixed-Poisson experiment).

For fixed α, $d = 0$, and uniformly small per-frame intensities,

$$M_n'(\ell_n) = c_\alpha \bar{\lambda}_n\{1 + o(1)\}, \qquad \mathrm{Var}(y_n) = \frac{c_\alpha^2 \bar{\lambda}_n}{W}\{1 + o(1)\},$$

so the calibrated soft-log variance has leading constant $1/(W\bar{\lambda}_n)$. Sharp minimax constants beyond this local rare-event calculation are not claimed.

When $d > 0$, every occurrence of $W\bar{\lambda}$ in the information statement must instead be replaced by the effective log-gain information $\sum_{k=1}^{W} \mathbb{E}_B\, s_k^2/(s_k + d)$; the identity $I(\theta) = \lambda$ does not hold for log gain in the presence of additive dark counts.

e) Remark D.4.1 (instantiation in the Fig. 7 protocol).: There $d = 0$, the carriers b_k are *oracle* values supplied from the simulation truth — the noiseless bucket carrier of the simulated object, times the photon budget; operationally a flat-field-calibrated benchmark, *not* programmed design intensities — and Θ is the safe tabulated operating range $[\lambda_-/\min_k b_k, \lambda_+/\max_k b_k]$ with Λ_0 fixed by convention (the record-mean centering absorbs the global gauge). The implemented estimator solves $M_n(\hat{\theta}_n) = y_n$ per window by monotone bisection on the exactly tabulated homogeneous m_α — for $C \sim \mathrm{Pois}(\theta b)$, $\mathbb{E}\psi_\alpha(C) = m_\alpha(\theta b)$ with the *same* homogeneous curve, so no carrier-averaged table is needed — with the clamp of (D.8a); its windows are truncated and renormalized to their distinct in-record members at the record boundaries (the equal-weight instance of the weighted statement; no replicate padding). Calibration accuracy: between grid points the table is evaluated by linear interpolation of m_α against $\log\lambda$, and the measured interpolation error on the dense midpoint check grid is $\varepsilon_{\mathrm{cal}}$ = XXEPSCALXX (bisection resolution $\varepsilon_{\mathrm{bis}} \le$ XXEPSBISXX), entering (D.8) as a deterministic $(\varepsilon_{\mathrm{cal}} + \varepsilon_{\mathrm{bis}})^2/\underline{\kappa}_n^2$ term and shrinking the clamp margin by $\varepsilon_{\mathrm{cal}} + \varepsilon_{\mathrm{bis}}$ — an empirical measurement, honestly labeled, not a certified analytic bound. The earlier (r3) arm inverted the homogeneous curve at the window mean, $m_\alpha^{-1}(y_n)$: a *mean-Poisson calibrated proxy* that ignores the within-window carrier variation and coincides with (D.8a) only for a constant carrier. Both arms are reported in Fig. 7; at the protocol's small carrier coefficient of variation they agree numerically, but only (D.8a) is the estimator this theorem is about.

f) Remark D.4.2 (the $\beta \le 1$ theorem and the precise obstruction above one).: For $\beta \in (0, 1]$, the pointwise modulus $|\ell_k - \ell_n| \le L_a|k - n|^\beta$ and the mean value theorem give

$$|q_k(\ell_k) - q_k(\ell_n)| \le \overline{\kappa}_k L_a |k - n|^\beta,$$

so the triangle-inequality bias bound used in Theorem C is valid for arbitrary known carriers.

For $\beta \in (1, 2]$, standard smoothness has the local form

$$\ell_k - \ell_n = v_n u_k + r_{nk}, \quad u_k = \frac{k - n}{N}, \quad |r_{nk}| \le L|u_k|^\beta,$$

rather than a pointwise modulus of order β on the function itself. Let w_k be centered weights, $\sum_k w_k u_k = 0$, and write $q_k(t) = m_\alpha(\Lambda_0 e^t b_k + d)$. Taylor's theorem gives

$$\begin{aligned}|B_n| \le |v_n| &\left|\sum_k w_k q_k'(\ell_n) u_k\right| + Q_1 L \sum_k w_k |u_k|^\beta \\ &+ Q_2\left\{v_n^2 \sum_k w_k u_k^2 + L^2 \sum_k w_k |u_k|^{2\beta}\right\},\end{aligned} \tag{D.10}$$

where $Q_1 = \sup_{k, t\in\Theta} q_k'(t)$ and $Q_2 = \sup_{k,t\in\Theta} |q_k''(t)|$.

The first term in (D.10) is the load-bearing obstruction. Ordinary centering cancels $\sum_k w_k u_k$, but it need not cancel the carrier-weighted moment $\sum_k w_k q_k'(\ell_n) u_k$. With an asymmetric heteroscedastic carrier this term is generally $O(W/N)$, so an $O((W/N)^\beta)$ claim is false for $\beta > 1$ under the present assumptions.

For a constant carrier, or more generally under the sensitivity-balance condition

$$\left|\sum_k w_k q_k'(\ell_n) u_k\right| \lesssim (W/N)^\beta,$$

the first term has the desired order. The remaining nonlinear term is $O((W/N)^2)$ and is therefore no larger than

$O((W/N)^\beta)$ for $1 < \beta \le 2$, provided the local slope is bounded. Thus the existing calibrated inverse does recover the β-order deterministic bias under homogeneous or sensitivity-balanced carriers, with a constant depending also on the slope bound and on Q_2.

A second-order Taylor expansion of M_n^{-1} by itself does not remove an unbalanced first moment. For arbitrary carriers, recovery of the $\beta > 1$ rate requires carrier-adapted moment conditions, carrier-adapted local-linear weights, or a local-polynomial likelihood fit. Theorem C therefore remains restricted to $\beta \le 1$, while the homogeneous-carrier and sensitivity-balanced extensions above are valid weaker statements.

g) Remark D.4.3 (unknown random carrier; nonstationary finite-W covariance bound).: Let $\{B_k\}$ be an unobserved stationary random carrier with known law, independent of the parameter, and let

$$\bar{m}_{\alpha,k}(t) = \mathbb{E}_B\, m_\alpha(\Lambda_0 e^t B_k + d).$$

Because the gain path ℓ_k may vary, stationarity of $\{B_k\}$ does not in general make the soft-log sequence stationary. Define

$$z_k = \psi_\alpha(C_k) - \mathbb{E}\psi_\alpha(C_k), \qquad g_k(b) = m_\alpha(\Lambda_0 e^{\ell_k} b + d).$$

Conditional independence of the Poisson randomization gives, for $i \ne j$,

$$\operatorname{Cov}(z_i, z_j) = \operatorname{Cov}(g_i(B_i), g_j(B_j)).$$

Consequently the exact finite-window identity is

$$\begin{aligned}\operatorname{Var}\left(W^{-1}\sum_{i=1}^W z_i\right) &= W^{-2}\sum_{i=1}^W \operatorname{Var}(z_i)\\ &\quad + 2W^{-2}\sum_{h=1}^{W-1}\sum_{i=1}^{W-h}\operatorname{Cov}(z_i, z_{i+h}).\end{aligned} \tag{D.11}$$

Let $p = 2+\eta$, $r_\eta = \eta/(2+\eta)$, and assume

$$v_* := \sup_i \operatorname{Var}(z_i) < \infty,$$

$$G_p := \sup_i \|g_i(B_i) - \mathbb{E}g_i(B_i)\|_p < \infty.$$

If the carrier is strongly mixing with coefficient $\alpha_B(h)$, Davydov's inequality [32] gives

$$|\operatorname{Cov}(z_i, z_{i+h})| \le 8\,\alpha_B(h)^{r_\eta} G_p^2.$$

Substitution into (D.11) yields the explicit finite-W bound

$$\begin{aligned}&\operatorname{Var}\left(W^{-1}\sum_{i=1}^W z_i\right)\\ &\le \frac{1}{W}\left[v_* + 16G_p^2\sum_{h=1}^{W-1}\left(1-\frac{h}{W}\right)\alpha_B(h)^{r_\eta}\right]\\ &\le \frac{v_* + 16G_p^2\sum_{h\ge1}\alpha_B(h)^{r_\eta}}{W}.\end{aligned}$$

The factor 16 is the constant obtained from Davydov's constant 8 and the two covariance triangles. It is sharper to apply Davydov to the conditional means $g_i(B_i)$ than to z_i itself, because independent Poisson shot noise contributes only to the diagonal variance v_*.

If ℓ_k is constant and the resulting marked process is strictly stationary, write $\gamma_\psi(h) = \operatorname{Cov}(z_0, z_h)$. Then

$$\begin{aligned}\operatorname{Var}\left(W^{-1}\sum_{k=1}^W z_k\right) &= W^{-1}\sum_{|h|<W}\left(1-\frac{|h|}{W}\right)\gamma_\psi(h)\\ &\le \frac{\sum_h|\gamma_\psi(h)|}{W},\end{aligned}$$

and when $\sum_h|\gamma_\psi(h)| < \infty$,

$$\operatorname{Var}\left(W^{-1}\sum_{k=1}^W z_k\right) = \frac{\sigma^2_{\psi,\mathrm{LR}}}{W} + o(W^{-1}),$$
$$\sigma^2_{\psi,\mathrm{LR}} = \sum_h \gamma_\psi(h).$$

The multiplicative notation $\sigma^2_{\psi,\mathrm{LR}}W^{-1}\{1+o(1)\}$ is valid only when $\sigma^2_{\psi,\mathrm{LR}} > 0$. For $z_n = \epsilon_n - \epsilon_{n-1}$ the signed long-run variance is zero while the finite-window variance is $2\sigma_\epsilon^2/W^2$.

h) Remark D.4.4 (clamp probability and its risk contribution).: Use the notation $B_n, V_n, \underline{\kappa}_n$ of the finite-window proof and put

$$r_n = \min\{M_n(\ell_n) - M_n(\theta_{\min}), M_n(\theta_{\max}) - M_n(\ell_n)\}.$$

If ℓ_n has margin Δ_n from $\partial\Theta$, then $r_n \ge \underline{\kappa}_n\Delta_n$. Since $y_n - M_n(\ell_n) = B_n + Z_n$, with $\mathbb{E}Z_n = 0$ and $\operatorname{Var}(Z_n) = V_n$,

$$\mathcal{E}_n^c \subseteq \{|B_n + Z_n| \ge r_n\}.$$

Therefore

$$\mathbb{P}(\mathcal{E}_n^c) \le \min\left\{1, \frac{B_n^2 + V_n}{r_n^2}\right\} \le \min\left\{1, \frac{B_n^2+V_n}{(\underline{\kappa}_n\Delta_n)^2}\right\}.$$

If $|B_n| < r_n$, the sharper centered bound is

$$\mathbb{P}(\mathcal{E}_n^c) \le \min\left\{1, \frac{V_n}{(r_n - |B_n|)^2}\right\}.$$

An exponential Bernstein bound requires a positive residual margin $r_n - |B_n|$; it is not available merely from a positive parameter margin when the deterministic smoothing bias can reach the boundary.

On $\mathcal{E}_n^c$, both the clamped estimate and the target lie in Θ, so

$$\mathbb{E}\left[(\hat\theta_n - \ell_n)^2\mathbf{1}_{\mathcal{E}_n^c}\right] \le D_\Theta^2\,\mathbb{P}(\mathcal{E}_n^c).$$

This is the complete clamp-event contribution to the MSE. The corresponding contribution to the squared bias alone is at most $D_\Theta^2\mathbb{P}(\mathcal{E}_n^c)^2$.

For a window containing W distinct, conditionally independent Poisson counts,

$$\begin{aligned}&\mathbb{P}(C_k = 0 \text{ for all } k \in W_n)\\ &\quad = \exp\left(-\sum_{k\in W_n}\lambda_k(\ell_k)\right) \le e^{-W\lambda_-}.\end{aligned}$$

If a padding convention repeats observations, the sum and the cardinality in this formula are over distinct frames after repeated weights have been aggregated; in that case the exponent need not be $W\lambda_-$ (the first width-65 replicate-padded window of the former convention had only 33 distinct frames, so its all-zero probability was $e^{-33\lambda}$, not $e^{-65\lambda}$ — at $\lambda = 0.25$, 2.6×10^{-4} versus 8.8×10^{-8}).

Shrinkage-bias caveat (load-bearing, per Sec. 6). For $\bar\lambda < 1$ the windowed soft-log statistic is *bias-fragile*: the mass at $C = 0$ pulls the weighted mean of ψ_α toward $\log\alpha$, and the amplification factor $(\kappa_{\max}/\underline{\kappa}_n)^2$ in the specialized bias term of (D.8) blows up as the window sensitivity vanishes with $\bar\lambda_n \to 0$ (at $d = 0$; window-average form $M_n'(\ell_n) \approx c_\alpha \bar\lambda_n$). Empirically the MSE of the *uncalibrated* proxy (and of Anscombe) can then sit *below* the Fisher reference curve — that is bias, not super-efficiency, and it does not contradict the lower bound (which constrains variance-limited, locally unbiased behavior). The $1/(W\bar\lambda)$ law is asserted only in the variance-limited regime above the $\bar\lambda \sim 1$ crossover; below it, honest accounting reports the bias floor, and all Fisher comparisons are made in the log-domain loss of this theorem (Fig. 7's `log_gain_mse` metric, against the *local realized* information $I_n = \sum_{k\in W_n} \lambda_k$ at $d = 0$) — the gain-domain relMSE column retained for continuity is a different loss and is not Fisher-comparable. Recovering the rate below the crossover requires the offset-design, Anscombe [28] ($2\sqrt{C + 3/8}$), or full Poisson-mixture MLE variants; for the mixture MLE the claim $\mathrm{MSE} \approx 1/(WJ(\theta))$ with $J \sim \mathbb{E}\lambda$ at low photons is standard parametric asymptotics under regularity of the mixture likelihood and is *not* re-proven here.

i) What this does not claim.: No absolute gain scale is estimated (gauge remains). The calibration curve m_α and the carriers b_k are assumed known; in the Fig. 7 protocol the carriers are *oracle* values from the simulation truth (a flat-field-calibrated benchmark), so Fig. 7 upper-bounds what a real flat-field calibration could achieve rather than demonstrating blind carrier recovery. Miscalibration beyond the measured interpolation term $(\varepsilon_{\mathrm{cal}} + \varepsilon_{\mathrm{bis}})$ of Remark D.4.1 adds a bias not covered by (D.8). Constants in (D.8)–(D.9) are not tracked. Smoothness beyond $\beta = 1$ is claimed only in the homogeneous-carrier / sensitivity-balanced form of Remark D.4.2 (display (D.10)); for arbitrary carriers the carrier-weighted first moment obstructs every $\beta > 1$ rate. The minimax matching of (D.9) is at rate level only, *in the log-domain quotient loss and at* $d = 0$, proved by the direct Poisson Le Cam bounds above — not inherited from Theorem D, whose Gaussian model is not a submodel of the mixed-Poisson experiment; for $d > 0$ every information statement must replace $W\bar\lambda$ by $\sum_k \mathbb{E}_B\, s_k^2/(s_k + d)$. No optimality of any kind is claimed in the gain-domain (linear) metric, and no claim is made that the calibrated estimator beats bias-accepting transforms (e.g. Anscombe) in MSE below the $\bar\lambda \sim 1$ crossover — empirically it does not (Sec. 9). As $\Phi \to 0$ with no offset and no reference, Fisher information for the gain vanishes — an information limit no estimator evades (Sec. 10).

APPENDIX E
SRHT WHITENING: EXACT WALSH CONDITION AND PERMUTATION BOUNDS

This appendix proves the claims of Sec. 7: the exact covariance identity $\mathrm{Cov}_D(Z_g, Z_h) = \hat w(g+h)$, the necessary-and-sufficient Walsh-flatness conditions for sign-only whitening (pairwise, spectral-window, and window-average forms), the Bernstein–Serfling permutation bound and its union-bound corollary, the permutation-alone carrier-variance identity with its flat-object counterexample, and the Hanson–Wright window-energy bound.

A. E.0 Setting and standing assumptions

- **(E1)** $K = 2^m$ and pixels are indexed by the Walsh group $G = \mathbb{F}_2^m$, with characters $\chi_g(j) = (-1)^{g\cdot j}$, unnormalized Hadamard matrix $H_{g,j} = \chi_g(j)$, and orthonormal carrier $U = K^{-1/2}H$. For every $g \neq 0$, χ_g is balanced: exactly $K/2$ entries equal $+1$.
- **(E2)** The object $T \in \mathbb{R}^K$ is deterministic and nonzero. Write $w_j = T_j^2$; S_1, S_2, and the spread functionals K_{eff}, K_∞, K_4 are as in Appendix C. The Walsh transform of w is $\hat w(h) = \sum_j \chi_h(j)\, w_j$, so $\hat w(0) = S_2$.
- **(E3)** The SRHT coefficient vector is $x = UDPT$, where $D = \mathrm{diag}(\eta_1, \dots, \eta_K)$ with i.i.d. Rademacher signs η_j, and P is a uniformly random pixel permutation independent of D. The normalized carrier is $Z_g = \sqrt{K}\, x_g = \sum_j \chi_g(j)\, \eta_j\, (PT)_j$. Physical nonnegative patterns are the offset form $B_g = \mu S_1 + \sigma Z_g$ with positivity margin $\mu S_1 > \sigma |Z_g|$; the bucket record is $R_n = a_n B_n$ with $\ell = \log a$ as in the main text. A *window* is a subset $A \subset G$ of $W = |A|$ rows.

Permutation leaves S_1, S_2, K_{eff}, K_4, K_∞ invariant; it only reorders w, i.e. replaces $\hat w$ by $\widehat{w^P}$ with $w_j^P = T_{Pj}^2$.

B. E.1 The exact covariance identity

Lemma E.1. *Under (E1)–(E3) with P fixed (in particular $P = \mathrm{id}$), $\mathbb{E}_D Z_g = 0$ and for all $g, h \in G$*

$$\mathrm{Cov}_D(Z_g, Z_h) \;=\; \sum_j \chi_{g+h}(j)\,(PT)_j^2 \;=\; \widehat{w^P}(g+h). \tag{E.1}$$

Proof. $\mathbb{E}\eta_j = 0$ gives the mean. Since $\mathbb{E}\,\eta_i\eta_j = \delta_{ij}$,

$$\begin{aligned}\mathbb{E}_D Z_g Z_h &= \sum_{i,j} \chi_g(i)\chi_h(j)(PT)_i(PT)_j\,\mathbb{E}\eta_i\eta_j \\ &= \sum_j \chi_g(j)\chi_h(j)(PT)_j^2,\end{aligned}$$

and $\chi_g\chi_h = \chi_{g+h}$ by the character property of the Walsh group. ∎

Consequently the covariance of any window A is the pattern matrix $\Gamma_A(g,h) = \widehat{w^P}(g+h)$, and, normalized by the common marginal variance S_2, $\Gamma_A/S_2 = I_A + R_A$ with off-diagonal entries $R_A(g,h) = \widehat{w^P}(g+h)/S_2$, $g \neq h$. The entire gain–object coupling of sign-randomized acquisition is therefore carried by the non-DC Walsh spectrum of the squared (permuted) object — nothing else.

C. E.2 Sign-only whitening: necessary and sufficient conditions

Theorem E.2 (exact Walsh condition). *Under (E1)–(E3) with no permutation ($P = \mathrm{id}$):*

1) (exact pairwise) $\mathrm{Cov}_D(Z_g, Z_h) = S_2\,\delta_{g,h}$ *for all g, h* ***iff*** $\hat w(q) = 0$ *for every $q \neq 0$, equivalently* ***iff*** $T_j^2 = S_2/K$ *for all j (T^2 flat).*
2) (pairwise ε-whitening) $|\mathrm{Cov}_D(Z_g, Z_h)| \le \varepsilon S_2$ *for all $g \neq h$* ***iff*** $\max_{q\neq 0} |\hat w(q)| \le \varepsilon S_2$.
3) (spectral window) *For a window class $\mathcal{A}$, $\|\Gamma_A/S_2 - I_A\|_{\mathrm{op}} \le \varepsilon$ for all $A \in \mathcal{A}$* ***iff*** $\sup_{A\in\mathcal{A}} \|R_A\|_{\mathrm{op}} \le \varepsilon$.

4) (window average) *With* $\bar{Z}_A = W^{-1}\sum_{g\in A} Z_g$ *and multiplicity* $m_A(q) = \#\{(g,h)\in A\times A : g\neq h,\ g+h=q\}$, *the variance* $\operatorname{Var}(\bar{Z}_A)$ *lies in* $[(1\mp\varepsilon)S_2/W]$ *for every* $A\in\mathcal{A}$ ***iff*** $\sup_{A\in\mathcal{A}}\left|\sum_{q\neq 0} m_A(q)\,\hat{w}(q)\right| \le \varepsilon\, W S_2$.

Proof. (1) If whitening holds, then for any $q\neq 0$ pick g,h with $g+h=q$ (always possible in G); (E.1) forces $\hat{w}(q)=0$. Conversely, vanishing non-DC coefficients make (E.1) equal $S_2\delta_{g,h}$. The flatness equivalence is Walsh inversion: the transform is invertible, so $\hat{w}$ supported on $\{0\}$ iff w is constant, i.e. $T_j^2 = S_2/K$. (2) Immediate from (E.1) after dividing by S_2. (3) The normalized window covariance *is* $I_A + R_A$; the statement is definitional. (4) Direct expansion:

$$\begin{aligned}\operatorname{Var}(\bar{Z}_A) &= W^{-2}\sum_{g,h\in A}\hat{w}(g+h)\\ &= \frac{S_2}{W} + W^{-2}\sum_{q\neq 0} m_A(q)\,\hat{w}(q),\end{aligned}$$

and the two-sided requirement is exactly the stated inequality. ■

Two standard reductions relate the forms. If $\mathcal{A}$ contains all two-point windows, (3) implies (2), and (2) is the sharpest scalar condition. In the converse direction, since R_A has zero diagonal, Gershgorin's disc theorem gives $\|R_A\|_{\mathrm{op}} \le (W-1)\max_{q\neq 0}|\hat{w}(q)|/S_2$, so pairwise condition (2) at tolerance $\varepsilon/(W_{\max}-1)$ implies spectral condition (3) for all windows with $W\le W_{\max}$. This Gershgorin step is a (generally loose) sufficient bound, not an equivalence.

The obstruction is sharp: if T^2 is aligned with the positive support of a single character χ_q, then $|\hat{w}(q)| = S_2$ and the row pair (g,h) with $g+h=q$ is *perfectly* correlated under sign randomization. Signs cannot remove this; it is a property of the object's pixel ordering.

a) What this does not claim.: Theorem E.2 characterizes second-moment decorrelation over the sign draw for a *fixed* object and ordering; it does not assert Gaussianity of Z_g (a Rademacher sum, whose normal approximation requires $K_\infty\to\infty$ with Berry–Esseen error $\lesssim K_\infty^{-1/2}$), independence of rows, or anything about noise, gain drift, or estimation error — those enter through Secs. 5–6 and Appendix F. It is also a statement about the design/object pair, not a high-probability event: no randomness beyond D is involved.

D. E.3 Random permutation as a probabilistic Walsh-flattener

By Lemma E.1, Theorem E.2 holds verbatim for a realized permutation P with w replaced by w^P: the *deterministic* necessary-and-sufficient condition remains Walsh-flatness, now of the permuted squared object. A random P makes that condition likely.

Theorem E.3 (Bernstein–Serfling permutation bound). *Under (E1)–(E3), there exist universal constants* $c, C>0$ *such that for every fixed* $q\neq 0$ *and* $\varepsilon>0$,

$$\Pr_P\bigl(|\widehat{w^P}(q)|\ge\varepsilon S_2\bigr) \;\le\; 2\exp\bigl[-c\min(\varepsilon^2 K_4,\ \varepsilon K_\infty)\bigr], \tag{E.2}$$

and by the union bound over the $K-1$ *nonzero frequencies,*

$$\begin{aligned}&\Pr_P\bigl(\max_{q\neq 0}|\widehat{w^P}(q)|\ge\varepsilon S_2\bigr)\\ &\quad\le\; 2(K-1)\exp\bigl[-c\min(\varepsilon^2 K_4,\ \varepsilon K_\infty)\bigr].\end{aligned}\tag{E.3}$$

In particular, pairwise ε*-whitening (Theorem E.2(2) for* w^P*) holds with probability at least* $1-\delta$ *whenever*

$$\min(\varepsilon^2 K_4,\ \varepsilon K_\infty)\;\ge\; C\log(K/\delta). \tag{E.4}$$

Proof. Fix $q\neq 0$ and let $A_q = \{j : \chi_q(j)=+1\}$, so $|A_q| = K/2$ by (E1). Then

$$\widehat{w^P}(q) = \sum_{j\in A_q} w_j^P - \sum_{j\notin A_q} w_j^P = 2\sum_{j\in A_q} w_j^P - S_2,$$

a centered sum of $n=K/2$ draws *without replacement* from the finite population $\{w_1,\dots,w_K\}$ with mean $\mu_w = S_2/K$. Its exact variance is $n\sigma_w^2\frac{K-n}{K-1} \le \frac{K}{4}\sigma_w^2 \le \frac14\sum_j w_j^2 = \frac{S_2^2}{4K_4}$, where $\sigma_w^2 = K^{-1}\sum_j (w_j-\mu_w)^2$; and each summand satisfies $0\le w_j\le \|T\|_\infty^2 = S_2/K_\infty$. The Bernstein inequality for sampling without replacement (Serfling [31]; the Bernstein–Serfling form of Bardenet–Maillard [43]) then gives, for $t>0$,

$$\Pr\bigl(|\widehat{w^P}(q)|\ge t\bigr) \le 2\exp\Bigl[-c\min\Bigl(\frac{t^2}{S_2^2/K_4},\ \frac{t}{S_2/K_\infty}\Bigr)\Bigr].$$

Setting $t=\varepsilon S_2$ yields (E.2); (E.3) is the union bound; (E.4) follows by requiring the right side of (E.3) to be at most δ and absorbing the factor $2(K-1)$ into $C\log(K/\delta)$. ■

a) What this does not claim.: (E.2)–(E.4) are *sufficient* high-probability conditions with unspecified universal constants; they are not the deterministic criterion for a realized acquisition — that remains Walsh-flatness of w^P (Theorem E.2), which can and should be checked numerically for the ordering actually used. For spectral whitening of all windows up to size $W_{\max}$, the simple route substitutes $\varepsilon\to\varepsilon/(W_{\max}-1)$ in (E.4) via Gershgorin; sharper window-size dependence via matrix-Bernstein or Hanson–Wright arguments (Sec. E.5) is a concentration refinement, again sufficient rather than necessary. Both branches of the min collapse when energy concentrates on few pixels (K_4, K_∞ small): permutation cannot flatten a spike.

E. E.4 Permutation alone: exact carrier variance and the flat-object counterexample

Theorem E.4 (permutation-alone variance). *Under (E1)–(E2), with no signs and a uniformly random permutation* P, *the carrier* $Z_g(P) = \sum_j \chi_g(j) T_{Pj}$ *of any non-DC row* $g\neq 0$ *satisfies* $\mathbb{E}_P Z_g = 0$ *and*

$$\operatorname{Var}_P Z_g \;=\; \frac{KS_2 - S_1^2}{K-1} \;=\; S_2\,\frac{K-K_{\mathrm{eff}}}{K-1}. \tag{E.5}$$

Consequently, for offset physical patterns $B_g = \mu S_1 + \sigma Z_g(P)$ *with positivity margin,*

$$\frac{\operatorname{Var}_P B_g}{(\mathbb{E}_P B_g)^2} = \Bigl(\frac{\sigma}{\mu}\Bigr)^2\frac{K-K_{\mathrm{eff}}}{(K-1)\,K_{\mathrm{eff}}} \;\le\; \Bigl(\frac{\sigma}{\mu}\Bigr)^2\frac{1}{K_{\mathrm{eff}}}. \tag{E.6}$$

No matching lower bound of order $1/K_{\mathrm{eff}}$ *holds uniformly over objects: for flat* T *(*T_j *constant),* $K_{\mathrm{eff}} = K$ *while* $Z_g(P)=0$ *for* ***every*** *permutation and every* $g\neq 0$.

Proof. For $g\neq 0$, $\sum_i \chi_g(i) = 0$, so $\mathbb{E}_P Z_g = \bigl(\sum_i \chi_g(i)\bigr)\mathbb{E}_P T_{P1} = 0$. Exchangeability of $(T_{P1},\dots,T_{PK})$

gives $\mathbb{E}_P T_{Pi}^2 = S_2/K$ and, for $i \neq j$, $\mathbb{E}_P T_{Pi}T_{Pj} = (S_1^2 - S_2)/(K(K-1))$. Hence

$$\mathbb{E}_P Z_g^2 = S_2 + \frac{S_1^2 - S_2}{K(K-1)} \sum_{i \neq j} \chi_g(i)\chi_g(j),$$

and $\sum_{i\neq j} \chi_g(i)\chi_g(j) = \big(\sum_i \chi_g(i)\big)^2 - K = -K$, giving (E.5); the second form follows from $K_{\text{eff}} = S_1^2/S_2$. For (E.6), $\mathbb{E}_P B_g = \mu S_1$ and $\operatorname{Var}_P B_g = \sigma^2 \operatorname{Var}_P Z_g$; divide. The counterexample: if $T_j \equiv t$, every non-DC Walsh sum of a constant vector vanishes identically, while $S_1^2/S_2 = K$. Note (E.5) also exhibits the finite-population correction — the relative variance is $(\sigma/\mu)^2(1/K_{\text{eff}} - 1/K)\cdot K/(K-1)$. ■

a) What this does not claim.: Theorem E.4 does *not* say permutation-alone whitens: it provides finite-population stationarity (exchangeability of the carrier along the acquisition, the (★) mechanism of Sec. 5) and an $O(1/K_{\text{eff}})$ *upper* variance scale, but it is not equivalent to i.i.d. Rademacher or nonnegative random-pattern carrier noise, and it guarantees no $\Theta(1/K_{\text{eff}})$ *excitation* — a flat object has exactly zero. Two-sided coefficientwise control requires the sign flip *and* the spectral-spread condition (E.4) together (the "both" cell of the Fig. S2 ablation). An earlier claim (R1/v3) that permutation-alone "cannot attain variance $\sim 1/K_{\text{eff}}$" was too strong as an upper-scale statement and is corrected by (E.6); it is correct only in the two-sided reading.

F. E.5 Hanson–Wright window-energy bound

Whitening of window *energy* — the quantity the window estimator of Sec. 6 actually averages — admits a direct quadratic-form bound. For a window A of W rows let P_A select those rows and set

$$Q_A = \|P_A UDPT\|_2^2 = \eta^\top M \eta,$$
$$M = \operatorname{diag}(PT)\, U_A^\top U_A \operatorname{diag}(PT),$$

with η the Rademacher sign vector and U_A the selected rows of U.

Proposition E.5 (window-energy concentration). *Condition on any realized P. Then $\mathbb{E}_D Q_A = (W/K)S_2$, and there is a universal $c > 0$ with*

$$\Pr_D\Big(\big|Q_A - \tfrac{W}{K}S_2\big| \geq \varepsilon\, \tfrac{W}{K}S_2\Big) \leq 2\exp\Big[-c\min\Big(\varepsilon^2 \tfrac{WK_\infty}{K},\ \varepsilon\, \tfrac{WK_\infty}{K}\Big)\Big]. \tag{E.7}$$

Over a specified family of M windows, uniform relative energy whitening at tolerance $\varepsilon \leq 1$ holds with probability $\geq 1-\delta$ once

$$\frac{WK_\infty}{K} \geq C\varepsilon^{-2}\log(M/\delta). \tag{E.8}$$

Proof. Since U has orthonormal rows, $U_A U_A^\top = I_W$, so $U_A^\top U_A$ is an orthogonal projector and $(U_A^\top U_A)_{jj} = \sum_{g\in A} U_{gj}^2 = W/K$ (Hadamard entries are $\pm K^{-1/2}$). Hence $M \succeq 0$, and $\mathbb{E}_D Q_A = \operatorname{tr} M = \sum_j (PT)_j^2 (W/K) = (W/K)S_2$. Because $\eta_j^2 = 1$, the diagonal of M contributes deterministically and $Q_A - \mathbb{E}Q_A$ is a pure off-diagonal Rademacher chaos, to which the Hanson–Wright inequality applies (Hanson–Wright [44]; sub-Gaussian form of Rudelson–Vershynin [45]):

$$\Pr(|Q_A - \operatorname{tr} M| \geq t) \leq 2\exp\Big[-c\min\Big(\frac{t^2}{\|M\|_F^2},\ \frac{t}{\|M\|_{\text{op}}}\Big)\Big].$$

The proxies: $\|M\|_{\text{op}} \leq \|\operatorname{diag}(PT)\|_{\text{op}}^2 \|U_A^\top U_A\|_{\text{op}} \leq \|T\|_\infty^2 = S_2/K_\infty$, and for PSD M, $\|M\|_F^2 \leq \|M\|_{\text{op}} \operatorname{tr} M \leq (W/K)S_2\|T\|_\infty^2$. Substituting $t = \varepsilon(W/K)S_2$ gives $t^2/\|M\|_F^2 \geq \varepsilon^2 WK_\infty/K$ and $t/\|M\|_{\text{op}} \geq \varepsilon WK_\infty/K$, i.e. (E.7); for $\varepsilon \leq 1$ the quadratic branch is active. A union bound over M windows gives (E.8). ■

For dense objects of bounded dynamic range, $K_\infty \asymp K_4 \asymp K_{\text{eff}} \asymp K$ and (E.8) reduces to the mild $W \gtrsim \varepsilon^{-2}\log M$ — the window only needs logarithmic length. For an object supported on m comparable pixels, $K_\infty \asymp m$ and the condition hardens to $Wm/K \gtrsim \varepsilon^{-2}\log M$: this is precisely where K_{eff} alone is the wrong functional and the three spread parameters separate (Sec. 6).

a) What this does not claim.: Proposition E.5 controls window *energy* over the sign draw for a pre-specified finite window family; it is a sufficient concentration tool, not the deterministic necessary-and-sufficient condition (which remains the spectral condition of Theorem E.2(3) applied to w^P), and it does not cover all $\binom{K}{W}$ windows simultaneously without paying for the larger union. The constants c, C are universal but not optimized, and no claim is made that (E.8) is tight in its W- or K_∞-dependence — Hanson–Wright and matrix-Bernstein refinements can improve window-family dependence for structured $\mathcal{A}$, but they do not change the deterministic criterion. Finally, all of Appendix E concerns the carrier statistics of the *design*; the translation of whitening into gain-estimation rates ($1/W$ per window) and into image error (leverage $B_L = 1$ for full SRHT inversion) is carried out in Appendix D and Appendix F respectively.

G. E.6 Two randomization semantics: pixel randomization (P_{col}, D) vs. acquisition-order randomization (P_{row})

This subsection separates two operations that earlier drafts conflated under the single word "permutation". Write P_{col} for a uniformly random **pixel (column) permutation** of the object — the P of assumption (E3) — write D for the i.i.d. Rademacher **pixel sign** diagonal of (E3), and write P_{row} for a uniformly random reordering of the **acquisition (Hadamard row / time) order**. These control different claims:

- **Appendix E's theorems govern cross-row covariance.** By Lemma E.1, the sign-draw covariance $\operatorname{Cov}_D(Z_g, Z_h) = \widehat{w^P}(g+h)$ is a function of the permuted squared object w^P alone; it is therefore **invariant to the acquisition order** — P_{row} does not enter Lemma E.1 or Theorems E.2–E.3 at all. The whitening randomization of the theory is $x = UDP_{\text{col}}T$: pixel signs plus pixel permutation, in *natural* acquisition order.
- **The manuscript's temporal-stationarity screen is controlled by acquisition-order structure.** The Brown–Forsythe/Levene screen [29], [30] of Sec. 7 probes temporal homogeneity of the acquired time series; it responds to whatever breaks the coarse-to-fine sequency-energy

ordering of natural Hadamard rows — which either $P_{\rm col}$ **or** $P_{\rm row}$ accomplishes.

a) Factorized ablation (empirical attribution; `results/perm_ablation_r1/`).: All arms share the same ordered natural-Hadamard baseline ($K = 4096$, 64×64, 8192 paired frames, noiseless carrier), the same 10-object panel, and the same Levene protocol (8 chunks, reject at $p < 10^{-3}$); 32 draws per randomized arm. Levene rejection rates:

arm	$P_{\rm col}$	D	$P_{\rm row}$	rate
ordered (baseline)	–	–	–	**0.700**
$P_{\rm col}$ only (theory's P)	Y	–	–	**0.056**
$P_{\rm row}$ only	–	–	Y	**0.122**
D only	–	Y	–	**0.434**
$P_{\rm col} + D$ (theory)	Y	Y	–	**0.119**
$P_{\rm row} + D$ (code's `srht_paired`)	–	Y	Y	**0.059**
$P_{\rm col} + P_{\rm row}$ (old Exp.-B arm)	Y	–	Y	**0.059**

The $P_{\rm col} + P_{\rm row}$ arm reproduces the historical headline ($19/320 = 5.9\%$) exactly, confirming the seed-compatible reconstruction of the old experiment; that historical 5.9% is attributable to *either* permutation alone ($P_{\rm col}$: 5.6%; $P_{\rm row}$: 12.2%).

b) Code caveat (stated plainly).: The implemented basis `make_srht_paired_basis` (`src/basis.py`, `permute_rows=True`) applies pixel signs D but permutes **Hadamard rows** (acquisition order), i.e. it implements $P_{\rm row} + D$ — **not** the theorem's $P_{\rm col} + D$. Both combinations are empirically sufficient for the temporal-stationarity screen (5.9% and 11.9%, versus 70% ordered), while D **alone is not** (43.4%): sign flips whiten carrier *amplitudes* (chunk-std CV collapses from 0.76 to 0.03) but leave the temporal sequency ordering intact, so the chunk-variance test still rejects. Text and captions must therefore attach each claim to its own arm: the covariance-whitening claims of E.1–E.3 attach to $P_{\rm col}$ (+D); the temporal-screen results attach to the arm actually acquired ($P_{\rm row} + D$ for the code's SRHT).

c) What this does not claim.: The table is an empirical attribution on one object panel and one protocol, not a theorem; no claim is made that $P_{\rm row}$ enjoys the concentration guarantees (E.2)–(E.4), which are proved for $P_{\rm col}$. The small increase from adding D on top of $P_{\rm col}$ (5.6% → 11.9%) is a property of the variance-homogeneity test on near-white sequences, not a whitening failure. Nothing here alters the theorems of E.1–E.5; it fixes which experimental arm may cite them.

Appendix F
The master finite-noise identity and the flip boundary

This appendix proves Theorem 1 (the master finite-noise relMSE identity) and Prop 3 (the finite-N flip boundary) of Sec. 8, together with the noise plug-ins and the three basis specializations quoted there. Throughout, $T \in \mathbb{R}^K$ is the object, with S_1, S_2, $K_{\rm eff}$ as in Appendix C; $A \in \mathbb{R}^{N\times K}$ is the design, $b = AT$ the ideal bucket vector; the raw bucket is $R_n = a_n B_n$ with gain $a_n = e^{\ell_n}$, $\ell \in S$ (the drift class of Sec. 4, $p = \dim S$). After the pipeline applies a gain estimate $\hat a_n = e^{\hat\ell_n}$, the corrected bucket is

$$\begin{aligned} z_n \;&=\; R_n/\hat a_n \;=\; (1+\delta_n)\, b_n + \xi_n, \\ \delta_n &= \frac{a_n}{\hat a_n} - 1 = e^{\ell_n - \hat\ell_n} - 1, \end{aligned} \tag{F.1}$$

where δ_n is the *residual* multiplicative gain error and ξ_n is additive bucket-domain noise after correction, with $\mathbb{E}\,\delta = m_\delta$, $\mathrm{Cov}(\delta) = V_\delta$, $\mathbb{E}\,\xi = 0$, $\mathrm{Cov}(\xi) = \Sigma_\xi$, and $\delta \perp \xi$ (uncorrelated, $C_{\delta\xi} = \mathrm{Cov}(\delta, \xi) = 0$ — a hypothesis, revisited in the cross-term note below). The reconstruction is $\hat T = Lz$ for any fixed linear operator $L \in \mathbb{R}^{K\times N}$ (exact inverse, pseudoinverse, DGI correlator, or a regularized estimator linearized around a fixed design).

A. F.1 The master identity

Theorem 1 (exact second-moment bridge). *Assume model (F.1) with A, T, L fixed (all expectations conditional on them), and assume — as part of the hypothesis, not as background convention — that:*

(H1) $\mathbb{E}\,\xi = 0$;

(H2) δ and ξ are uncorrelated, $C_{\delta\xi} := \mathrm{Cov}(\delta, \xi) = 0$;

*(H3) the gain estimate $\hat a$ is **exogenous** with respect to the reconstruction record — computed independently of the noise realization ξ (injected/synthetic residuals, an independent calibration record, or a cross-fitted estimate from held-out frames) — so that (H1)–(H2) hold conditionally on the record being reconstructed;*

(H4) the second moments $m_\delta, V_\delta, \Sigma_\xi$ are finite.

Then, exactly and with no Gaussian or small-noise approximation,

$$\frac{\mathbb{E}\|\hat T - T\|_2^2}{S_2} = \underbrace{\frac{\|(LA-I)T + L\,\mathrm{diag}(b)\, m_\delta\|_2^2}{S_2}}_{bias} + \underbrace{\frac{\mathrm{tr}\big(L\,\mathrm{diag}(b)\, V_\delta\,\mathrm{diag}(b)\, L^\top\big)}{S_2}}_{residual\ gain} + \underbrace{\frac{\mathrm{tr}\big(L\,\Sigma_\xi L^\top\big)}{S_2}}_{additive\ noise}. \tag{F.2}$$

*(a) If moreover $\mathbb{E}\,\delta_n = 0$ and the δ_n are uncorrelated with common variance v, the residual-gain term equals $v\, B_L(A,T)$ with the **leverage***

$$B_L(A,T) = \frac{1}{S_2}\sum_{n=1}^{N} b_n^2\,\|Le_n\|_2^2. \tag{F.3}$$

(b) If instead the residual gains are stationary with $\mathrm{Cov}(\delta_n,\delta_m) = v\, r_\delta(n-m)$, the residual-gain term equals

$$R_{\rm gain} = \frac{v}{S_2}\sum_{n,m} r_\delta(n-m)\, b_n b_m\,\langle Le_n, Le_m\rangle. \tag{F.4}$$

Proof. Write $z = b + \mathrm{diag}(b)\,\delta + \xi$, so $\hat T - T = (LA - I)T + L\,\mathrm{diag}(b)\,\delta + L\xi$. Decompose $\delta = m_\delta + (\delta - m_\delta)$. The mean of $\hat T - T$ is $(LA - I)T + L\,\mathrm{diag}(b)\, m_\delta$ (using

(H1)); its squared norm is the bias term. By the bias–variance decomposition $\mathbb{E}\|X\|^2 = \|\mathbb{E}X\|^2 + \operatorname{tr}\operatorname{Cov}(X)$ and (H2), the covariance of $\hat{T}$ is $L\operatorname{diag}(b)\,V_\delta\operatorname{diag}(b)\,L^\top + L\,\Sigma_\xi L^\top$; taking traces and dividing by S_2 gives (F.2). For (a), $V_\delta = vI$ and $\operatorname{tr}(L\operatorname{diag}(b)\operatorname{diag}(b)\,L^\top) = \sum_n b_n^2\|Le_n\|^2$, giving (F.3). Case (b) is the coordinate expansion of the same trace with $(V_\delta)_{nm} = v\,r_\delta(n-m)$. ■

a) Remark F.1.1 (same-record gain estimates violate the hypotheses; scope of the exact identity).: Hypothesis (H3) is not decorative. When $\hat{a}$ is estimated from the *same* noisy record that carries ξ — the generic situation for a blind pipeline — the conditional mean $\mathbb{E}[\xi \mid \text{record}]$ is generically nonzero and δ is correlated with ξ, so (H1)–(H2) fail: the mean of $\hat{T} - T$ acquires an additional bias term $L\,m_\xi$ (with $m_\xi = \mathbb{E}\,\xi \neq 0$) inside the bias norm, and the covariance acquires the cross-term $(2/S_2)\,\operatorname{tr}\big[L\operatorname{diag}(b)\,C_{\delta\xi}\,L^\top\big]$. Neither term is included in (F.2). The exact identity therefore covers the **exogenous / injected-residual case** — which is precisely what the injected-residual protocol of Sec. 9 (experiment E4, Fig. S3) validates — and cross-fitted estimates that restore (H1)–(H2) by construction; for same-record blind pipelines, (F.2) is exact only up to the two extra terms above.

b) What this does not claim.: With (H1)–(H3) now stated *in* the theorem, the caveat is Remark F.1.1: (F.2) is not claimed for same-record blind gain estimation, where the $L\,m_\xi$ bias and the $C_{\delta\xi}$ cross-term appear. Beyond that, (F.2) is a second-moment accounting identity for a *linear* reconstructor with the design held fixed; it does not bound nonlinear or data-adaptive L, and for random-pattern pipelines the constants below are obtained by additionally averaging over the pattern draw. It says nothing about how small v can be made — that is the estimation-rate question of Sec. 6. Finally, the scalar collapse $v\,B_L$ *requires* uncorrelated residuals: for smooth (coherent) residual gain — the generic output of a windowed blind estimator — only the matrix form (F.4) is correct. In particular, the simple summary "orthogonal inversion gives v" holds only for independent coefficient-wise residuals under exact inversion.

B. F.2 Noise plug-ins

Proposition F.2 (read and Poisson noise, exact). (a) Gaussian read. *If detector-domain noise* $\eta_n \sim \mathcal{N}(0, \sigma^2_{\text{read},n})$ *(independent across frames) precedes gain correction, then* $\xi_n = \eta_n/\hat{a}_n$ *and, exactly,* $\operatorname{Var}(\xi_n) = \sigma^2_{\text{read},n}/\hat{a}_n^2 = \big(\sigma^2_{\text{read},n}/a_n^2\big)\big(a_n/\hat{a}_n\big)^2$*; for calibrated gains* $\hat{a}_n \approx a_n$ *the factor* $(a_n/\hat{a}_n)^2 \to 1$ *and* $\operatorname{Var}(\xi_n) = \tau^2_{G,n} := \sigma^2_{\text{read},n}/a_n^2$. (b) Poisson. *If counts obey* $C_n \mid T, a \sim \operatorname{Pois}(\Phi_n)$ *with* $\Phi_n = \gamma_n a_n b_n$ *and* $z_n = C_n/(\gamma_n\hat{a}_n)$*, then exactly*

$$\mathbb{E}[z_n \mid T, a] = \frac{a_n}{\hat{a}_n} b_n, \qquad \operatorname{Var}(z_n \mid T, a) = \frac{b_n^2}{\Phi_n}\Big(\frac{a_n}{\hat{a}_n}\Big)^2. \tag{F.5}$$

Proof. (a) is scaling of a Gaussian. (b): a Poisson variable has mean = variance = Φ_n; dividing by $\gamma_n\hat{a}_n$ scales the mean by $(\gamma_n\hat{a}_n)^{-1}$ and the variance by $(\gamma_n\hat{a}_n)^{-2}$, and $\Phi_n/(\gamma_n\hat{a}_n) = (a_n/\hat{a}_n)b_n$, $\Phi_n/(\gamma_n\hat{a}_n)^2 = (b_n^2/\Phi_n)(a_n/\hat{a}_n)^2$. ■

Hence — and only after the post-calibration simplification $\hat{a}_n \to a_n$, which sets the exact $(a_n/\hat{a}_n)^2$ factors of (a) and (F.5) to one — $\Sigma_\xi = \operatorname{diag}(\tau^2_{G,n} + b_n^2/\Phi_n)$ plugs into (F.2), and the Poisson term is **exact at all photon counts**, including $\Phi_n \ll 1$; no log or Gaussian approximation enters the reconstruction layer. Under a total budget $\Phi_n = \omega_n\Phi_{\text{tot}}$, $R_{\text{Pois}} = (S_2\Phi_{\text{tot}})^{-1}\sum_n(b_n^2/\omega_n)\|Le_n\|^2$.

a) What this does not claim.: Exactness holds *conditionally on the gains*; the difficulty of low photons lives entirely in estimating $\hat{a}_n$ (the calibrated soft-log analysis; see Appendix D), not in propagating noise through L. The optimal allocation $\omega_n \propto |b_n|\,\|Le_n\|$ requires knowing T and is not claimed to be achievable.

C. F.3 Specializations

a) F.3.1 Orthogonal / full-SRHT inversion ($B_L = 1$).: If $A = U$ is orthonormal ($N = K$, signed coefficients available) and $L = U^\top$, then $LA = I$ and $\|Le_n\| = 1$, so $B_L = (1/S_2)\sum_n b_n^2 = 1$ by Parseval. With independent residuals,

$$\text{relMSE}_{\text{orth}} = v + \frac{1}{S_2}\sum_n\big(\tau^2_{G,n} + b_n^2/\Phi_n\big). \tag{F.6}$$

Exact orthogonal inversion transmits independent coefficient-wise residual gain at unit leverage. A full SRHT acquisition $A = P_{\text{row}}UDP_{\text{col}}$, $L = A^\top$ has identically the same propagation; SRHT's role is *not* a better inverse but temporal stationarization of the carrier so that the blind estimator can shrink v (Sec. 5/7).

b) F.3.2 Pairwise Hadamard: exact perturbation law.: Let $c_k = h_k^\top T$, $h_k \in \{\pm1\}^K$, with physical masks $B_k^\pm = (S_1 \pm c_k)/2$ and reconstruction $T = K^{-1}H^\top c$, so coefficient errors e_k give $\text{relMSE} = (KS_2)^{-1}\sum_k\mathbb{E}e_k^2$. If the paired masks carry gains a_k^+ and $a_k^- = a_k^+(1+\Delta_k)$, the pairwise-normalized estimator satisfies the **exact** algebraic law

$$\hat{c}_k - c_k = -\frac{S_1\,\Delta_k\,(1-x_k^2)}{2+\Delta_k(1-x_k)}, \qquad x_k = c_k/S_1, \tag{F.7}$$

whence, exactly, $R_{\text{pair,gain}} = \dfrac{S_1^2}{KS_2}\displaystyle\sum_k\mathbb{E}\Big[\frac{\Delta_k^2(1-x_k^2)^2}{(2+\Delta_k(1-x_k))^2}\Big]$. For small mismatch with Δ_k independent of the coefficient index,

$$R_{\text{pair,gain}} = \frac{K_{\text{eff}}}{4}\,D_H(T)\operatorname{Var}(\Delta) + O\big(K_{\text{eff}}\,\mathbb{E}|\Delta|^3\big),$$
$$D_H(T) = \frac{1}{K}\sum_k(1-x_k^2)^2, \tag{F.8}$$

by Taylor expansion of (F.7) in Δ_k. For most non-DC Hadamard coefficients of a nonnegative object $|x_k| \ll 1$, so $D_H \approx 1$; D_H is nevertheless a *pipeline constant to be measured*, not assumed. The K_{eff} amplification is structural: pairwise normalization converts a differential gain error into a coefficient error of order S_1, and Parseval then divides by S_2. Noise terms: read noise gives exactly $R_{\text{pair,read}} = 2\sigma^2_{\text{read}}/S_2$; for Poisson counts with pair budget Φ_{pair}, using $B_k^+ + B_k^- = S_1$, the coefficient shot variance is $\operatorname{Var}(e_k \mid c_k) \approx S_1^2/\Phi_{\text{pair},k}$ — an approximation, exact only up to the c_k-dependent split between the two buckets — giving $R_{\text{pair,Pois}} = K_{\text{eff}}/\Phi_{\text{pair}}$ (equal-pair budget) or $(K_{\text{eff}}+1)/(2\Phi_{\text{mask}})$ (equal-mask budget); these differ only by allocation convention.

c) What F.3.2 does not claim.: (F.7)–(F.8) presuppose a stable DC/S_1 normalization; at low photons the denominator bucket sum can be noisy or zero, and a robust implementation needs offset/clipping or a likelihood-ratio estimator. The exact law (F.7) is algebra; only the moment step (F.8) invokes small Δ and index-independence.

d) F.3.3 Random patterns / DGI: the one-sample constant C_0.: Let $I_n = \mu\mathbf{1} + \sigma x_n$ with iid coordinates, $\mathbb{E}x = 0$, $\mathbb{E}x^2 = 1$, $\mathbb{E}x^3 = \gamma_3$, $\mathbb{E}x^4 = \beta_4$, and the ideal correlator $\hat{T}_{\mathrm{DGI}} = (N\sigma)^{-1}\sum_n x_n z_n$. Define the one-sample vector $Z = \sigma^{-1}x\,(\mu S_1 + \sigma x^\top T)$; then $\mathbb{E}Z = T$ and, with known gain and no detector noise,

$$\frac{\mathbb{E}\|\hat{T}_{\mathrm{DGI}} - T\|^2}{S_2} = \frac{C_0}{N}, \qquad C_0 := \frac{\mathbb{E}\|Z - T\|^2}{S_2}, \tag{F.9}$$

which is the implementation-independent *definition* of C_0. **The evaluation of C_0 for iid patterns is a moment calculation**: expanding $\mathbb{E}\|Z\|^2$ with $Z = (\mu S_1/\sigma)x + xx^\top T$ gives the three terms $K(\mu/\sigma)^2 S_1^2$, $2(\mu/\sigma)\gamma_3 S_1^2$, and $(K + \beta_4 - 1)S_2$; subtracting $\|\mathbb{E}Z\|^2 = S_2$,

$$C_0 = K + \beta_4 - 2 + K_{\mathrm{eff}}\left[K(\mu/\sigma)^2 + 2\gamma_3(\mu/\sigma)\right] \xrightarrow[\mu=0,\ \gamma_3=0]{} K + \beta_4 - 2. \tag{F.10}$$

The $K_{\mathrm{eff}}K(\mu/\sigma)^2$ term is the familiar nonnegative-background penalty. Residual gain, read, and Poisson terms (equal-count convention $\Phi_n = \Phi_{\mathrm{frame}}$):

$$\mathrm{relMSE}_{\mathrm{DGI}} = \frac{C_0}{N} + \frac{(1 + C_0)v}{N} + \frac{K\tau_G^2}{N\sigma^2 S_2} + \frac{1 + C_0}{N\,\Phi_{\mathrm{frame}}}. \tag{F.11}$$

The gain term follows from Theorem 1(a) with the *leverage $\times N$ framing*: for the correlator, $\|Le_n\|^2 = \|x_n\|^2/(N^2\sigma^2)$, so averaging over pattern draws, $N\,\mathbb{E}B_L = \mathbb{E}\|Z\|^2/S_2 = 1 + C_0$ — i.e., $R_{\mathrm{DGI,gain}} = (1 + C_0)v/N$, and in residual-injection experiments the measurable slope is $NB_L \to 1 + C_0$. **Status (labeled).** This pattern-averaging step treats the residuals δ as independent of the pattern draw — exact for injected residuals (the Fig. S3 protocol), a heuristic decoupling for blind pipelines, where $\hat{a}$ is computed from the same record the patterns generated. Correlated residuals multiply v by $\chi_\delta = 1 + 2\sum_{h\geq 1}\mathrm{corr}(\delta_0, \delta_h)$ — a heuristic long-run-variance correction valid only when correlations are weak enough that pattern cross-terms still average; the rigorous object remains (F.4).

e) Correction (drift-leverage constant of the gain term; measured, no fitted parameters).: As previously written, the gain term $(1+C_0)v/N$ of (F.11) — with the *pipeline* constant C_0 of definition (F.9) — is **wrong for mean-subtracted DGI pipelines** under time-structured (drift-like) residual gain: on the published grid it underpredicts the measured random-arm drift excess by a median factor $\approx 1.4\times 10^3$ (1364 / 1527 / 1521 across the none / AGC / SCGI-proxy corrections; white-drift diagnostic, `results/prop3_nofreeparam_r1/`). The correct leverage carries the **raw-correlator constant** $1+C_0^{\mathrm{raw}}$, where C_0^{raw} is (F.10) evaluated for the ideal correlator at the measured raw pattern moments (at the grid's operating point $C_0^{\mathrm{raw}} \approx 0.7$–$1.2 \times 10^6$, versus pipeline $C_0 \approx 6.2$–7.2×10^2); with it the measured excess is reproduced to 0.3–4.7% (median ratios 1.003 / 1.047 / 1.039), still with zero fitted parameters. Mechanism: mean subtraction removes the $K_{\mathrm{eff}}K(\mu/\sigma)^2$ background penalty from the **sampling floor** — hence the small pipeline C_0 in the C_0/N term, confirmed exactly by the flat oracle arm at C_0/N — but **not** from the **drift leverage**: a time-varying gain modulates the large DC background coherently, and the correlator pays the full raw-correlator constant. For mean-subtracted pipelines the gain term of (F.11) must therefore read $(1 + C_0^{\mathrm{raw}})\,v/N$, while the sampling-floor term keeps the pipeline C_0. This vindicates — and extends to the gain term — the "no universal C_0" warning below, which earlier versions applied to the floor term only.

f) What F.3.3 does not claim.: There is **no universal** C_0: (F.10) is exact only for the ideal known-(μ, σ) correlator with iid coordinates. Mean subtraction, reference normalization, and photon allocation all change the constant (not the C_0/N structure), so C_0 must be measured from the pipeline via definition (F.9). Moreover — per the correction above — the sampling-floor constant and the drift-leverage constant of the *same* pipeline are different measured numbers whenever the pipeline subtracts a background; quoting one for the other is a $\sim 10^3$ error at nonnegative-background operating points. The flux-limited convention $\Phi_n = \gamma b_n$ replaces the last term of (F.11) by $(N\sigma^2 S_2)^{-1}\mathbb{E}[\|x\|^2 b_n/\gamma]$.

D. F.4 Prop 3: the finite-N flip boundary

a) Drift model and assumptions.: Let s^2 be the variance scale of the log-gain process ℓ and let ρ_{pair} be the **adjacent-pair decorrelation parameter**, meaning the paired-mask gain increment obeys $\mathrm{Var}(\Delta_k) \approx \mathrm{Var}(\ell_k^- - \ell_k^+) = 2s^2\,r(\rho_{\mathrm{pair}})$ with $r(\rho) = \rho + O(\rho^2)$ small-ρ; for an Ornstein–Uhlenbeck log-gain, $r(\rho) = 1 - e^{-\rho}$. Assume the small-mismatch regime of (F.8) (so $R_{\mathrm{pair,gain}}(\rho_{\mathrm{pair}}) \approx \frac{1}{2}K_{\mathrm{eff}}D_H(T)\,s^2\,r(\rho_{\mathrm{pair}})$) and the DGI decomposition (F.11) with blind residual variance $v_{\mathrm{blind}}(\rho_{\mathrm{pair}}, N)$ — the window-optimized output of the Sec.-6 rate theorem, $v_{\mathrm{blind}} \lesssim \min_{W\in[1,N]}[\sigma_{\mathrm{eff}}^2/W + C(s\rho_{\mathrm{pair}}W)^2] \lesssim C\,\sigma_{\mathrm{eff}}^{4/3}(s\rho_{\mathrm{pair}})^{2/3}$ when the optimizer is interior.

Prop 3 (implicit flip boundary). *Define $\rho^* = \inf\{\rho_{\mathrm{pair}} \geq 0 : R_{\mathrm{pair}}(\rho_{\mathrm{pair}}) \geq R_{\mathrm{rand}}(\rho_{\mathrm{pair}}, N)\}$, where $R_{\mathrm{pair}} = R_{\mathrm{pair,gain}} + R_{\mathrm{pair,read}} + R_{\mathrm{pair,Pois}}$ and R_{rand} is (F.11) with $v = v_{\mathrm{blind}}(\rho_{\mathrm{pair}}, N)$.* ***Assume single crossing:*** *since $v_{\mathrm{blind}}(\rho_{\mathrm{pair}}, N)$ itself varies with ρ_{pair}, both sides of the comparison move with ρ_{pair}, and we assume the difference $R_{\mathrm{pair}} - R_{\mathrm{rand}}$ changes sign exactly once on the range considered — existence and uniqueness of the crossing is an assumption at the same heuristic level as the small-drift expansion, not a proven property. Under the small-drift expansion and this assumption, ρ^* solves the implicit equation*

$$r(\rho^*) = \frac{2}{K_{\mathrm{eff}}\,D_H(T)\,s^2}\Big[\frac{C_0}{N} + \frac{(1 + C_0^{\mathrm{lev}})\,v_{\mathrm{blind}}(\rho^*, N)}{N} + R_{\mathrm{DGI,noise}} - R_{\mathrm{pair,noise}}\Big], \tag{F.12}$$

where C_0^{lev} is the pipeline's ***drift-leverage*** *constant: $C_0^{\mathrm{lev}} = C_0^{\mathrm{raw}}$ of (F.10) for mean-subtracted DGI pipelines (the F.3.3* *correction), and $C_0^{\mathrm{lev}} = C_0$ only for pipelines without background subtraction. For the OU convention this is explicit: $\rho^* = -\log(1-Q)$ with $Q = 2\Delta_R/(K_{\mathrm{eff}} D_H s^2)$, Δ_R the bracket of (F.12). If $Q \le 0$, the random/DGI arm is already at least as good as the pairwise arm at $\rho = 0$ (the bracket $\Delta_R \le 0$: the pairwise arm's noise overhead exceeds the random arm's sampling/noise floor), so $\rho^* = 0$ — the flip is immediate, not absent; if $Q \ge 1$, pairwise never flips within the OU small-to-moderate range.*

Proof. At the boundary, $R_{\mathrm{pair}}(\rho^*) = R_{\mathrm{rand}}(\rho^*, N)$. Substituting $R_{\mathrm{pair,gain}} = \frac{1}{2} K_{\mathrm{eff}} D_H s^2\, r(\rho^*)$ (from F.8 with $\mathrm{Var}\Delta = 2s^2 r$) and (F.11) with the corrected leverage constant, and solving for $r(\rho^*)$, gives (F.12); monotonicity of r makes the crossing the infimum. The OU form inverts $r(\rho) = 1 - e^{-\rho}$. Note (F.12) is a fixed-point equation because v_{blind} is evaluated at ρ^*; the single-crossing assumption stated in the proposition is what makes the crossing exist and be unique. The monotonicity intuition — $R_{\mathrm{pair,gain}}$ increasing in ρ_{pair} while the right side varies slowly ($v_{\mathrm{blind}} \propto \rho_{\mathrm{pair}}^{2/3}$) — motivates the assumption but does not prove it. ■

Empirical status of Prop 3 (two regimes; corrected constant; `results/prop3_nofreeparam_r1/`). The published equal-frame grid ($N = 2048$, $K = 1024$, OU drift, 10 objects, all constants measured, zero fitted parameters) splits the proposition's status in two:

(a) The $v = 0$ skeleton is VALIDATED, no-free-parameter. For a gain-known random arm ($v = 0$: oracle gains, so only the C_0/N floor competes with the pair term), the predicted boundary $\rho^* = -\log\bigl(1 - 2(C_0/N)/(K_{\mathrm{eff}} D_H s^2)\bigr)$ lands within a **median factor 1.54** of the 42 observed crossings; restricted to $\sigma_a \ge 0.1$ (small-drift expansion intact), **40/40 cells within factor 2** (max 1.73). The empirical σ-scaling exponent is -2.11 ($R^2 = 0.976$) versus -2.07 predicted — the leading-order σ^{-2} law is correct. The prediction is systematically early (median $\log_{10}$ ratio -0.187), consistent with (F.8)'s $O(\Delta^3)$ corrections. Ingredients check out separately: the pair law (F.8) is essentially exact (median measured/predicted 0.997 over 639 cells), and the oracle random arm sits flat at C_0/N (0.3286) across the whole grid.

(b) The blind-arm flip prediction, as previously stated, is WITHDRAWN — the corrected theory predicts no flip, and none occurs. With the erroneous pipeline-C_0 leverage, (F.12) predicted flips at $\rho^* = 0.007$–0.33 for $\sigma_a \ge 0.1$; empirically ordered pairwise Hadamard stays strictly better than blind random+DGI in **all 100** (source × correction × σ × object) cells over the entire scanned range $\rho \in [10^{-3}, 20]$. With the corrected constant $C_0^{\mathrm{lev}} = C_0^{\mathrm{raw}}$, the bracket satisfies $Q(\rho) > r(\rho)$ for every ρ at every grid (σ_a, object) for both blind corrections — the theory then predicts **no flip in range**, exactly matching the 100/100 empirical no-flip. The failure of the original prediction was localized to one constant (the F.3.3 leverage correction, verified to 0.3–4.7% at white drift), not to the functional form of (F.12); the boundary curves previously shown in Fig. 4 are SRHT-paired-versus-ordered-Hadamard crossovers — a comparison Prop 3 explicitly does not rank — summarized by free two-parameter power-law fits, and are not a test of (F.12).

b) Leading-order heuristic (labeled as such).: If $r(\rho) = \rho$, $D_H = 1$, noise differences are negligible, v_{blind} is negligible (or absorbed into C_0), and ρ *means* adjacent-pair decorrelation, (F.12) collapses to the engineering rule

$$\rho^* \approx \frac{2C_0}{N\, K_{\mathrm{eff}}\, s^2}, \tag{F.13}$$

with first corrections $\rho^* \approx 2\bigl(C_0 + (1 + C_0^{\mathrm{lev}}) v_{\mathrm{blind}}\bigr)/[N K_{\mathrm{eff}} D_H s^2] + 2(R_{\mathrm{DGI,noise}} - R_{\mathrm{pair,noise}})/(K_{\mathrm{eff}} D_H s^2)$ and the nonlinear replacement $\rho \mapsto r^{-1}(\cdot)$ when adjacent increments are not infinitesimal. Note that for mean-subtracted pipelines "negligible v_{blind}" means $(1 + C_0^{\mathrm{raw}}) v_{\mathrm{blind}} \ll C_0$ — a far stronger requirement than $v_{\mathrm{blind}} \ll C_0$, and the reason the blind-arm flip vanished on the published grid (empirical-status point (b) above). (F.13) is a *leading-order heuristic*, not a theorem; the theorem is (F.12).

c) ρ-convention warning.: Throughout, ρ_{pair} (adjacent-pair decorrelation) is the variable of (F.12)–(F.13); it is **not** the normalized bandwidth ρ_{bw} of the tall-design thresholds ($p \approx \rho_{\mathrm{bw}} N$, Sec. 4). If a bandwidth parameter with adjacent-increment variance $\sim 2s^2\rho_{\mathrm{bw}}/N$ is substituted, the explicit $1/N$ in (F.13) cancels and the boundary reads $\rho_{\mathrm{bw}}^* \approx 2C_0/(K_{\mathrm{eff}} s^2)$ — same physics, different symbol. Every phase-diagram axis must declare its convention.

d) What Prop 3 does not claim.: The boundary compares two *specific* pipelines (pairwise-normalized Hadamard vs. blind random+DGI with the ideal correlator); it does not rank SRHT-paired pipelines, externally calibrated systems, or prior-regularized reconstructions, each of which shifts Δ_R. It inherits every assumption of its ingredients: small mismatch (F.8), measured constants C_0, C_0^{lev}, and D_H (with the floor/leverage distinction of the F.3.3 correction), a stationary drift with a declared $r(\rho)$, and a v_{blind} from the window rate theorem whose constants are implementation-specific. It is a mean-square crossing, not a guarantee about any single realization; the single-crossing property itself is assumed, not derived; and at ρ_{pair} beyond the small-drift expansion only the defining infimum — not (F.12) — is meaningful. Finally, the validated regime is the $v = 0$ skeleton (empirical-status point (a)); no flip prediction is currently claimed for the blind arm — the corrected constant predicts, and the data confirm, no flip on the published grid.

References

[1] T. B. Pittman, Y. H. Shih, D. V. Strekalov, and A. V. Sergienko, "Optical imaging by means of two-photon quantum entanglement," *Phys. Rev. A* **52**, R3429–R3432 (1995).

[2] R. S. Bennink, S. J. Bentley, and R. W. Boyd, " 'Two-Photon' Coincidence Imaging with a Classical Source," *Phys. Rev. Lett.* **89**, 113601 (2002).

[3] J. H. Shapiro, "Computational ghost imaging," *Phys. Rev. A* **78**, 061802(R) (2008).

[4] Y. Bromberg, O. Katz, and Y. Silberberg, "Ghost imaging with a single detector," *Phys. Rev. A* **79**, 053840 (2009).

[5] M. Bina *et al.*, "Backscattering Differential Ghost Imaging in Turbid Media," *Phys. Rev. Lett.* **110**, 083901 (2013).

[6] J. Cheng, "Ghost imaging through turbulent atmosphere," *Opt. Express* **17**(10), 7916–7921 (2009).

[7] Y. Peng and W. Chen, "Deep learning-enhanced ghost imaging through dynamic and complex scattering media with supervised corrections of dynamic scaling factors," *Appl. Phys. Lett.* **124**, 181104 (2024).

[8] Y. Peng, Y. Xiao, and W. Chen, "High-fidelity optical wireless transmission in complex environments around a corner using the design of a single-layer neural network for data encoding," *Opt. Express* **33**(14), 30123–30135 (2025).

[9] M. F. Duarte *et al.*, "Single-Pixel Imaging via Compressive Sampling," *IEEE Signal Process. Mag.* **25**(2), 83–91 (2008).

[10] M. P. Edgar, G. M. Gibson, and M. J. Padgett, "Principles and prospects for single-pixel imaging," *Nat. Photonics* **13**, 13–20 (2019).

[11] Z. Zhang, X. Wang, G. Zheng, and J. Zhong, "Hadamard single-pixel imaging versus Fourier single-pixel imaging," *Opt. Express* **25**(16), 19619–19639 (2017).

[12] W.-K. Yu, "Super Sub-Nyquist Single-Pixel Imaging by Means of Cake-Cutting Hadamard Basis Sort," *Sensors* **19**(19), 4122 (2019).

[13] M. Kech and F. Krahmer, "Optimal Injectivity Conditions for Bilinear Inverse Problems with Applications to Identifiability of Deconvolution Problems," *SIAM J. Appl. Algebra Geom.* **1**(1), 20–37 (2017).

[14] Y. Li, K. Lee, and Y. Bresler, "Identifiability in Bilinear Inverse Problems With Applications to Subspace or Sparsity-Constrained Blind Gain and Phase Calibration," *IEEE Trans. Inf. Theory* **63**(2), 822–842 (2017).

[15] A. Ahmed, B. Recht, and J. Romberg, "Blind Deconvolution Using Convex Programming," *IEEE Trans. Inf. Theory* **60**(3), 1711–1732 (2014).

[16] S. Choudhary and U. Mitra, "Identifiability Scaling Laws in Bilinear Inverse Problems," arXiv:1402.2637 (2014).

[17] R. Kueng, H. Rauhut, and U. Terstiege, "Low rank matrix recovery from rank one measurements," *Appl. Comput. Harmon. Anal.* **42**(1), 88–116 (2017).

[18] F. Ferri, D. Magatti, L. A. Lugiato, and A. Gatti, "Differential Ghost Imaging," *Phys. Rev. Lett.* **104**, 253603 (2010).

[19] B. Sun *et al.*, "Normalized ghost imaging," *Opt. Express* **20**(15), 16892–16901 (2012).

[20] A. J. Weiss and B. Friedlander, "Eigenstructure methods for direction finding with sensor gain and phase uncertainties," *Circuits Syst. Signal Process.* **9**(3), 271–300 (1990).

[21] L. Balzano and R. Nowak, "Blind calibration of sensor networks," in *Proc. IPSN*, 2007, pp. 79–88.

[22] S. Ling and T. Strohmer, "Self-calibration and biconvex compressive sensing," *Inverse Problems* **31**(11), 115002 (2015).

[23] X. Nie *et al.*, "Noise-robust computational ghost imaging with pink noise speckle patterns," *Phys. Rev. A* **104**, 013513 (2021).

[24] Y. Han *et al.*, "Generalized sparse Hadamard single-pixel imaging," *Opt. Express* **33**(20), 43324–43341 (2025).

[25] A. B. Tsybakov, *Introduction to Nonparametric Estimation*, Springer Series in Statistics. New York: Springer, 2009.

[26] R. D. Gill and B. Y. Levit, "Applications of the van Trees inequality: a Bayesian Cramér–Rao bound," *Bernoulli* **1**(1–2), 59–79 (1995).

[27] M. S. Pinsker, "Optimal filtering of square-integrable signals in Gaussian noise," *Probl. Inf. Transm.* **16**(2), 120–133 (1980).

[28] F. J. Anscombe, "The transformation of Poisson, binomial and negative-binomial data," *Biometrika* **35**(3–4), 246–254 (1948).

[29] M. B. Brown and A. B. Forsythe, "Robust tests for the equality of variances," *J. Amer. Statist. Assoc.* **69**(346), 364–367 (1974).

[30] H. Levene, "Robust tests for equality of variances," in *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*, I. Olkin *et al.*, Eds. Stanford, CA: Stanford Univ. Press, 1960, pp. 278–292.

[31] R. J. Serfling, "Probability inequalities for the sum in sampling without replacement," *Ann. Statist.* **2**(1), 39–48 (1974).

[32] Yu. A. Davydov, "Convergence of distributions generated by stationary stochastic processes," *Theory Probab. Appl.* **13**(4), 691–696 (1968).

[33] F. Merlevède, M. Peligrad, and E. Rio, "A Bernstein type inequality and moderate deviations for weakly dependent sequences," *Probab. Theory Relat. Fields* **151**(3–4), 435–474 (2011).

[34] H. C. P. Berbee, *Random Walks with Stationary Increments and Renewal Theory*, Mathematical Centre Tracts, vol. 112. Amsterdam: Mathematisch Centrum, 1979.

[35] E. Rio, *Asymptotic Theory of Weakly Dependent Random Processes*, Probability Theory and Stochastic Modelling, vol. 80. Berlin: Springer, 2017.

[36] R. Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, Cambridge Series in Statistical and Probabilistic Mathematics, vol. 47. Cambridge: Cambridge Univ. Press, 2018.

[37] L. D. Brown and M. G. Low, "Asymptotic equivalence of nonparametric regression and white noise," *Ann. Statist.* **24**(6), 2384–2398 (1996).

[38] C. R. Rao and S. K. Mitra, *Generalized Inverse of Matrices and Its Applications*. New York: Wiley, 1971.

[39] P. Stoica and T. L. Marzetta, "Parameter estimation problems with singular information matrices," *IEEE Trans. Signal Process.* **49**(1), 87–90 (2001).

[40] E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed., Springer Texts in Statistics. New York: Springer, 1998.

[41] C. A. J. Klaassen, "On an inequality of Chernoff," *Ann. Probab.* **13**(3), 966–974 (1985).

[42] S. G. Bobkov and M. Ledoux, "On modified logarithmic Sobolev inequalities for Bernoulli and Poisson measures," *J. Funct. Anal.* **156**(2), 347–365 (1998).

[43] R. Bardenet and O.-A. Maillard, "Concentration inequalities for sampling without replacement," *Bernoulli* **21**(3), 1361–1385 (2015).

[44] D. L. Hanson and F. T. Wright, "A bound on tail probabilities for quadratic forms in independent random variables," *Ann. Math. Statist.* **42**(3), 1079–1083 (1971).

[45] M. Rudelson and R. Vershynin, "Hanson–Wright inequality and sub-Gaussian concentration," *Electron. Commun. Probab.* **18**, paper no. 82, 1–9 (2013).